

UNIVERSITY OF MINES AND TECHNOLOGY

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SCHOOL OF RAILWAY AND INFRASTRUCTURE DEVELOPMENT

DEPARTMENT OF COMPUTING AND DATA ANALYTICS



LECTURE NOTES ON

PROBABILITY AND STATISTICS

(CE/EL/ME/GL/GM/ES 361)

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Preface

Probability and statistics is the mathematics of uncertainty. Engineers spend their working lives making decisions from measurements — an assay from a drill core, a strain gauge reading, a set of rainfall records, the failure times of a batch of components — and every one of those measurements carries error and variation. An engineer who cannot quantify that uncertainty cannot say how much to trust the number in front of them. Probability gives us a language for uncertainty before the data arrive; statistics gives us a method for reasoning once they have.

This course, MC/EL/CE/CV/GL/ES/GM 361, introduces both. You will learn to summarise data graphically and numerically, to calculate probabilities of events using the rules of probability and the techniques of counting, to work with random variables and their distributions, to recognise the special distributions that model common engineering situations, to compute expectations and variances, and to draw conclusions about a population from a sample through estimation, hypothesis testing, and regression. The emphasis throughout is practical: each result is followed by worked examples, because the difficulty in this subject is rarely the algebra — it is recognising which model applies, and what the answer means once obtained.

Objective. To introduce the basic concepts and methods of probability and statistics, and to enable students to apply them to engineering data.

Aim. At the end of the course, students should be able to summarise and present data, compute probabilities of simple and compound events, work with discrete and continuous random variables and their distributions, compute expectations, variances, and covariances, construct confidence intervals, test hypotheses about means and proportions, and fit and interpret a simple linear regression.

Prerequisites. A first course in differential and integral calculus, and basic mathematical maturity. Statistical tables for the standard normal, t , and χ^2 distributions are assumed to be available.

Readers are urged to attempt each example before reading its solution, and to pay as much attention to the statement of the assumptions as to the arithmetic. A result derived under the assumption of independence, or of a known variance, or of a large sample, is worth nothing when applied where that assumption fails.

Dr Abdulzeid Yen Anafo

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Chapter 1

Introduction to Statistics

Learning Objectives

- Discuss the reasons for studying statistics as an engineer.
- Identify basic statistical concepts.
- Identify the levels of measurements
- Discuss the sampling procedures
- Understand data visualization using several graphical devices

1.0 Introduction to Statistics

Statistics is a way to get information from data. Statistics is a discipline which is concerned with:

- summarizing information to aid understanding,
- drawing conclusions from data,
- estimating the present or predicting the future, and
- designing experiments and other data collection.

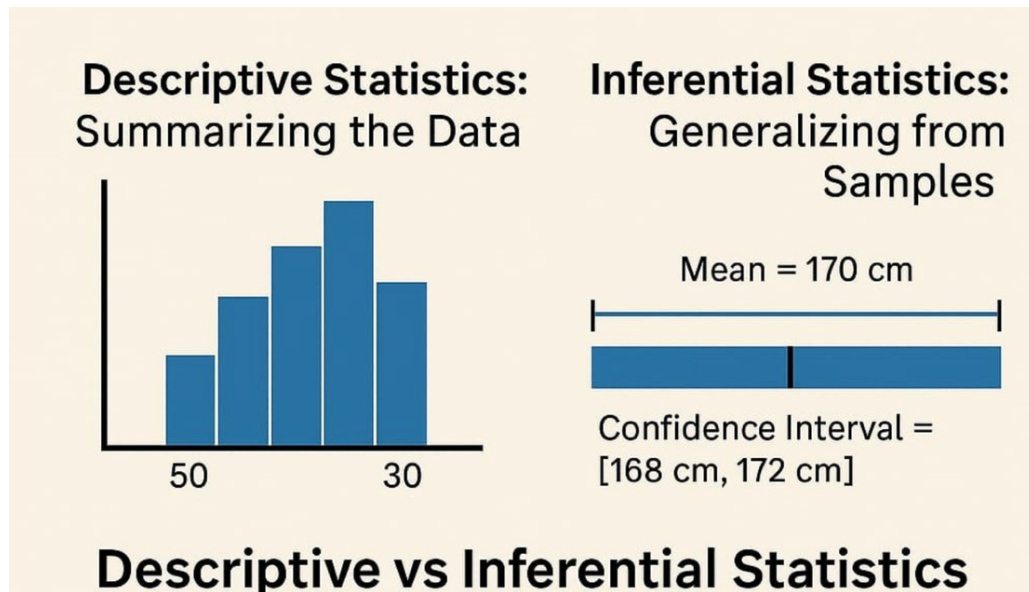
In making predictions, Statistics uses the concept of probability, which models chance mathematically and enables calculations of chance in complicated cases.

1.1 Why Statistics?

Engineering problems are rarely deterministic. Measurements vary, materials behave inconsistently, and environmental conditions change. Statistics equips engineers with the ability to analyze variability, assess reliability, and make decisions under uncertainty. Statistical methods are essential for designing new products, improving existing systems, optimizing processes, and evaluating performance. Through statistical analysis, engineers can distinguish meaningful signals from random noise, assess the quality of data, and quantify uncertainty in their conclusions. Beyond technical applications, statistical thinking supports objectivity in scientific inquiry. Conclusions drawn from data are strengthened when they are supported by formal statistical reasoning rather than subjective judgment. As a result, statistics plays a central role in research, quality control, risk assessment, and evidence-based decision-making across many disciplines. Even when engineers do not directly perform statistical analyses, understanding statistical concepts allows them to interpret results critically and make informed professional judgments.

1.2 Branches of Statistics

Statistics is traditionally divided into two interconnected branches: descriptive statistics and inferential statistics.



1.1.1 Descriptive Statistics

Descriptive statistics focuses on organizing, summarizing, and presenting data in a form that is easy to understand. This includes representing data visually through graphs and tables, as well as numerically through summary measures such as averages and measures of variability. The goal of descriptive statistics is to provide a clear and concise picture of the data at hand. At this stage, no generalizations are made beyond the observed dataset. Instead, descriptive statistics answers the question of what the data reveal about the observed phenomenon.

1.1.2 Inferential Statistics

Inferential statistics extends beyond the observed data by using information from a sample to draw conclusions about a larger population. Because it is often impractical or impossible to measure an entire population, inferential methods allow us to make informed statements about population characteristics based on limited data.

Probability theory underpins inferential statistics by providing a framework for quantifying uncertainty. Through estimation and hypothesis testing, inferential statistics enables researchers and engineers to assess the strength of evidence, evaluate competing explanations, and make predictions with known levels of confidence.

1.3 Variables and Types of Data

A variable is any characteristic or attribute that can take different values across individuals, objects, or observations. Variables form the basic building blocks of statistical analysis.

Variables can be broadly classified as qualitative or quantitative. Qualitative variables describe characteristics or categories and are typically expressed as labels or names. Examples include material type, defect category, or color classification. Quantitative variables, on the other hand, represent numerical measurements and express quantities such as weight, temperature, pressure, or time.

Quantitative variables can be further classified based on how they take values. Discrete variables arise from counting processes and take distinct, separate values, while continuous variables arise

from measurement processes and can take any value within a specified range. This distinction is important because it influences the choice of statistical methods and graphical representations.

	Quantitative Variables	Qualitative Variables
Definition	<i>Take on numeric values</i>	<i>Take on names or labels</i>
Examples	Number of students in a class	Eye color
	Number of square feet in a house	Gender
	Population size of a city	Breed of dog
	Age of an individual	Level of Education
	Height of an individual	Marital status

1.4 Univariate and Bivariate Data

Statistical data can also be classified based on the number of variables involved in the analysis. When data consist of measurements on a single variable, the data are referred to as univariate. Univariate analysis focuses on understanding the distribution, central tendency, and variability of that variable.

When two variables are measured simultaneously for each observation, the data are described as bivariate. Bivariate analysis explores relationships between variables, such as associations, trends, or dependencies. Understanding whether and how two variables are related is a key objective in many engineering and scientific studies.

1.5 Populations and Samples

In statistics, a population refers to the complete set of all individuals, objects, or measurements of interest in a study. A sample is a subset of the population selected for analysis. Because studying an entire population is often impractical, samples are used to represent populations.

The manner in which a sample is selected is critical. Well-designed sampling procedures aim to ensure that samples are representative of the population, thereby allowing valid inference. A commonly used approach is simple random sampling, in which every member of the population has an equal chance of being selected.

Numerical characteristics that describe a population are called parameters, while numerical characteristics computed from sample data are called statistics. For example, the population mean is denoted by the symbol μ , whereas the sample mean is denoted by x . One of the central goals of statistics is to use sample statistics to estimate unknown population parameters.



1.6 Graphical and Numerical Summaries of Data

Graphical displays play a crucial role in understanding data. For qualitative data, graphs such as bar charts and pie charts are commonly used to display frequencies and proportions. For quantitative data, graphical tools such as histograms, box plots, dot plots, and stem-and-leaf displays reveal important features of the distribution, including its center, spread, and shape.

In addition to graphical summaries, numerical summary statistics provide concise descriptions of data. Measures of location, such as the mean, median, and mode, describe where the data tend to cluster. Measures of variability, including the range, variance, and standard deviation, describe how spread out the data are around the center. Together, graphical and numerical summaries provide complementary perspectives on the same dataset.

1.7 Describing Data Patterns

When analyzing data, statisticians commonly describe distributions in terms of several key features. These include the center of the distribution, the degree of variability or spread, and the overall shape of the distribution. Shape is often discussed in terms of symmetry, skewness, and the presence of multiple peaks.

Modern data analysis emphasizes not just computing summary measures but also interpreting what they reveal about the underlying process that generated the data.

1.8 Unusual Features in Data

Real-world data often contain unusual features that merit special attention. Gaps in a distribution occur when certain ranges of values have no observations, potentially indicating structural constraints or data collection issues. Outliers are extreme values that differ markedly from the rest of the data and may arise from measurement errors, rare events, or meaningful deviations.

A commonly used rule for identifying outliers is based on the interquartile range. Observations that lie more than 1.5 times the interquartile range below the first quartile or above the third quartile are often classified as outliers. Identifying and interpreting unusual features is an important step in responsible data analysis.

1.9 Comparing Data Sets

Comparing two or more datasets is a common task in engineering and scientific studies. Graphical tools such as side-by-side box plots, comparative histograms, and grouped bar charts allow analysts to assess similarities and differences across datasets.

When comparing distributions, attention is typically given to differences in center, variability, shape, and the presence of unusual features. A systematic comparison helps ensure that conclusions are based on evidence rather than visual impressions alone.

1.5 Summarizing data graphically

Selected graphs for qualitative data

- Pie chart
- Bar Chart
- (Also frequency distribution)

Selected graphs for Numerical data

- Box plot
- Dot plot
- Stem-and-leaf
- Histogram

1.5 Summarizing Data Graphically

One of the first and most important steps in statistical analysis is visualizing the data. Graphical summaries allow us to explore data quickly, detect patterns, identify unusual features, and guide further analysis. Before computing numerical summaries such as means or standard deviations, statisticians almost always begin by examining appropriate graphs.

The choice of a graphical display depends fundamentally on the type of data being analyzed. In general, different graphical tools are used for qualitative (categorical) data and quantitative (numerical) data. Using an inappropriate graph can obscure important information or lead to misleading interpretations.

1.4.1 Graphical Displays for Qualitative (Categorical) Data

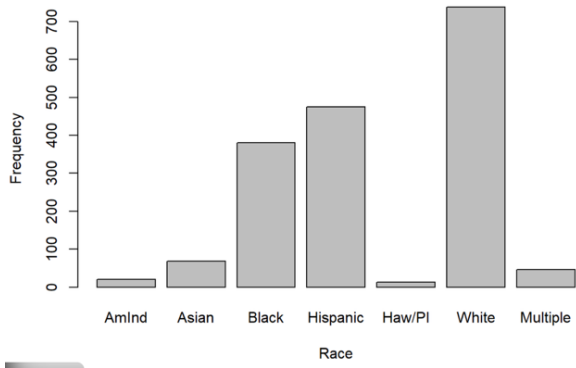
Qualitative data describe attributes or categories rather than numerical magnitudes. Graphs for qualitative data emphasize counts, proportions, and comparisons between categories.

The bar chart remains the most effective and widely used graphical tool for qualitative data. Each category is represented by a bar whose height corresponds to its frequency or relative frequency. Bar charts allow for easy comparison across categories and are particularly useful in engineering quality assessments, such as defect classification.

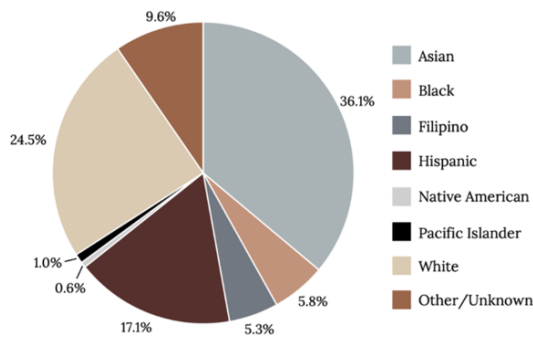
The pie chart displays each category as a sector of a circle, with sector sizes proportional to category percentages. Pie charts are useful for illustrating how categories contribute to a whole, although they are less effective when precise comparisons are required or when many categories are present.

A stacked bar chart extends the basic bar chart by showing subcategories within each bar. This type of graph is useful when comparing compositions across groups, such as defect types across multiple production lines.

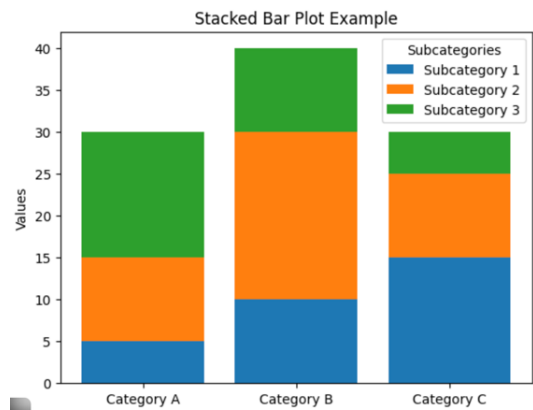
A Pareto chart is a specialized bar chart in which categories are ordered from most frequent to least frequent. Often used in quality control and industrial engineering, Pareto charts help identify the most significant contributors to a problem by visually reinforcing the “vital few versus trivial many” principle.

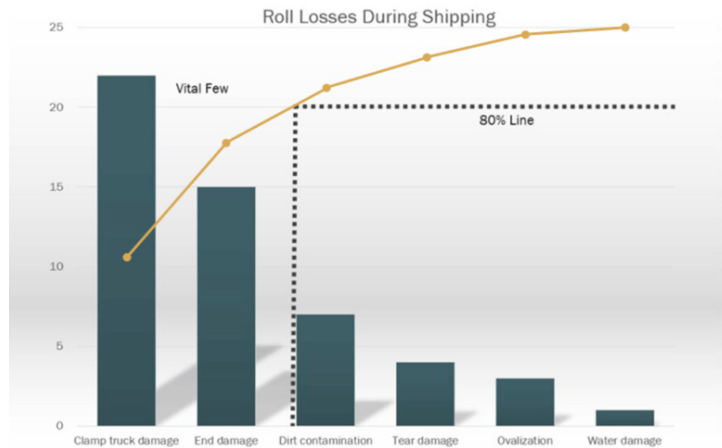


Ethnicity of Students



Pie chartPie chartBar chartBar chart



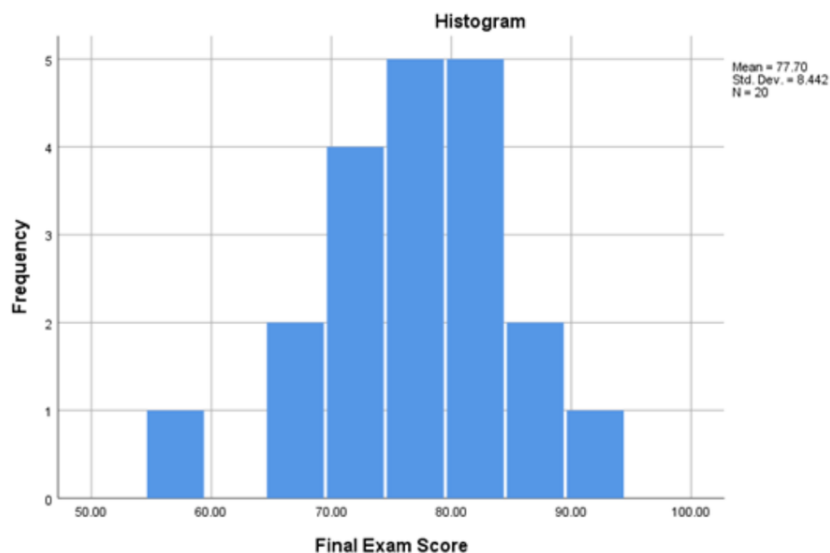


Pareto chart Pareto chart Stacked bar chart Stacked bar chart

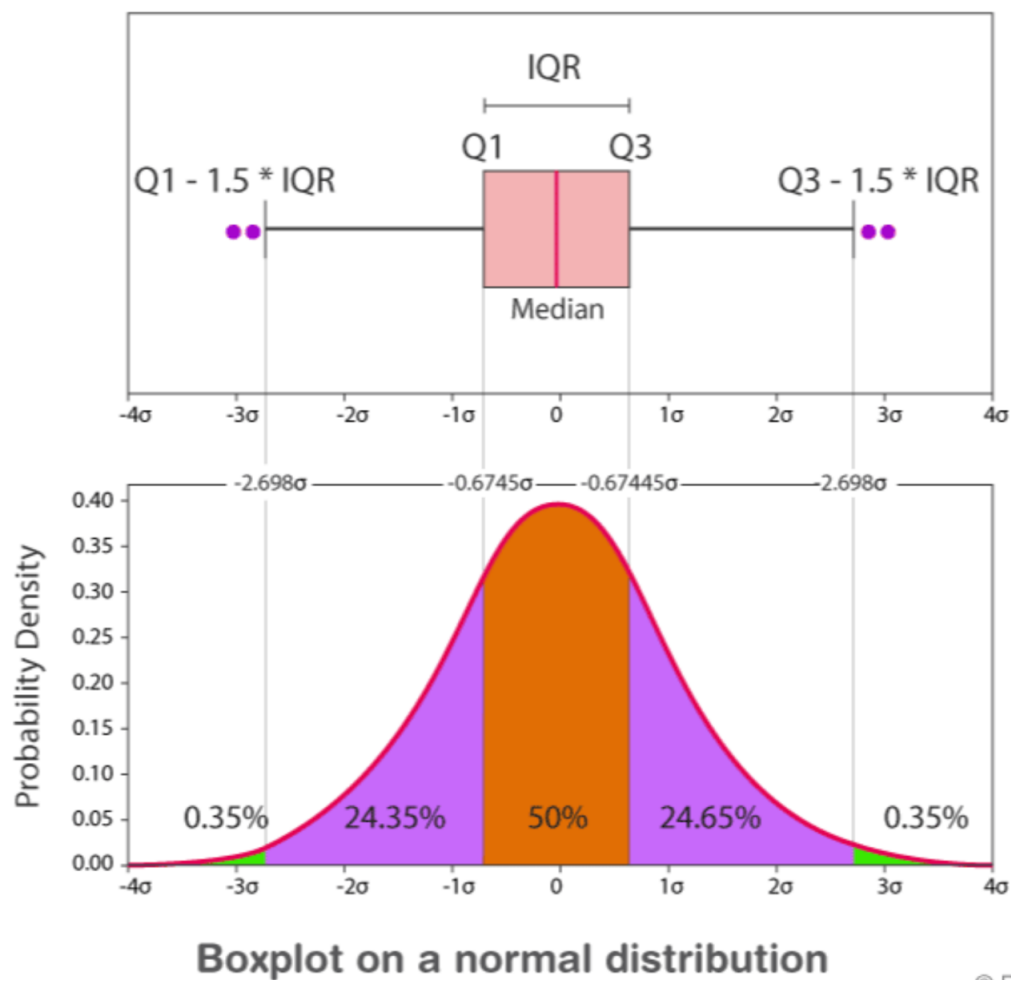
1.4.2 Graphical Displays for Quantitative (Numerical) Data

Quantitative data consist of numerical measurements, and their graphical summaries focus on revealing the distribution of values. Distributional graphs help answer questions about where data are centered, how spread out they are, and whether they exhibit skewness or extreme values.

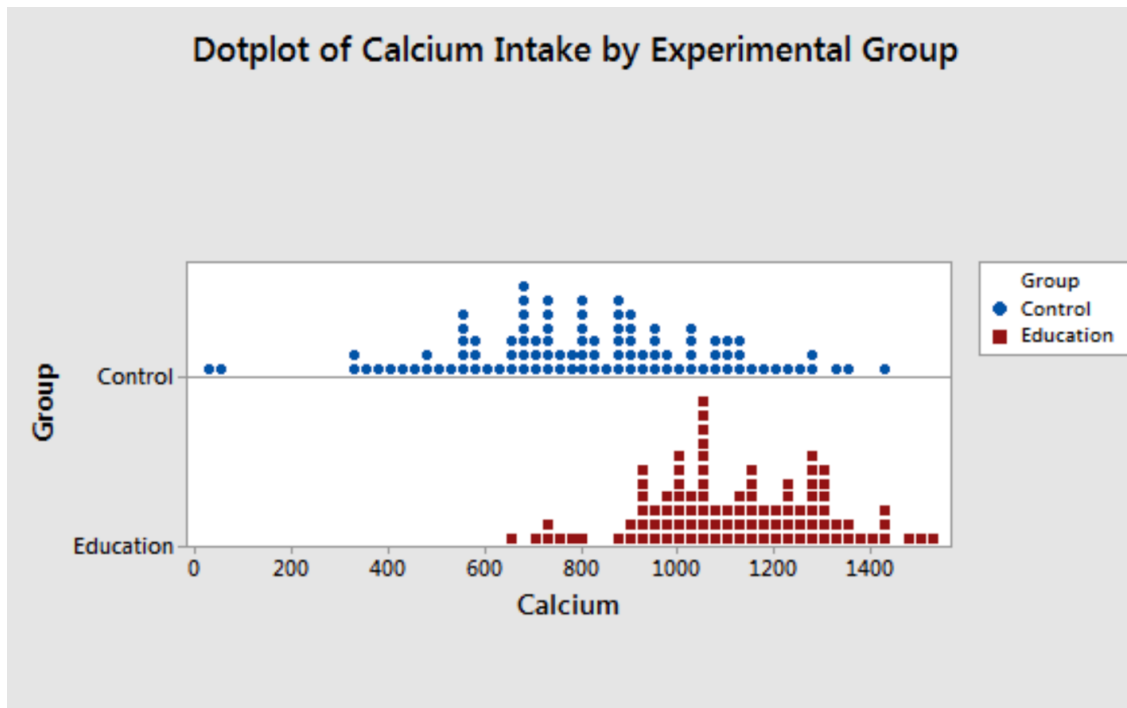
The histogram is one of the most important tools for exploring quantitative data. By grouping data into intervals, histograms show the overall shape of the distribution, including symmetry, skewness, and the presence of multiple modes. The choice of bin width can influence interpretation, making thoughtful construction essential.



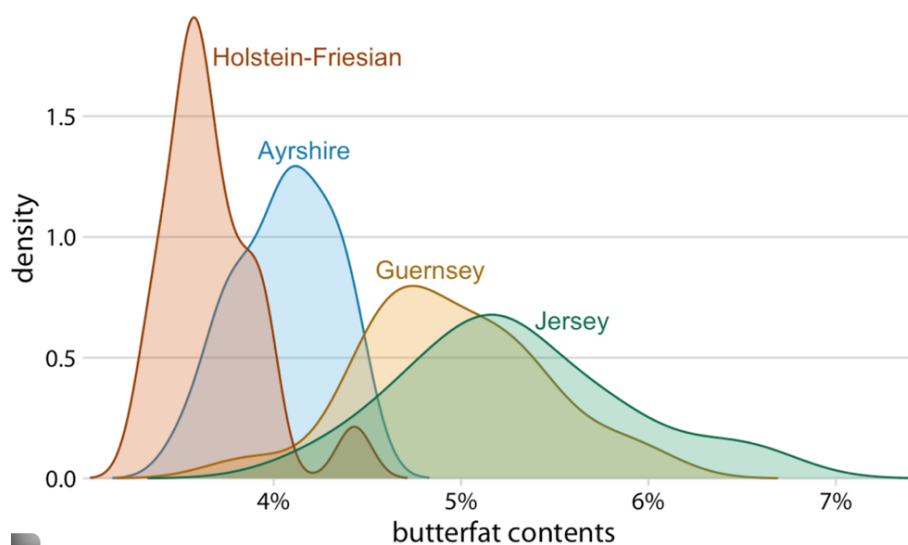
A box plot provides a concise summary of a distribution using the median, quartiles, and interquartile range. Box plots are especially valuable for comparing multiple datasets side by side and for identifying potential outliers.



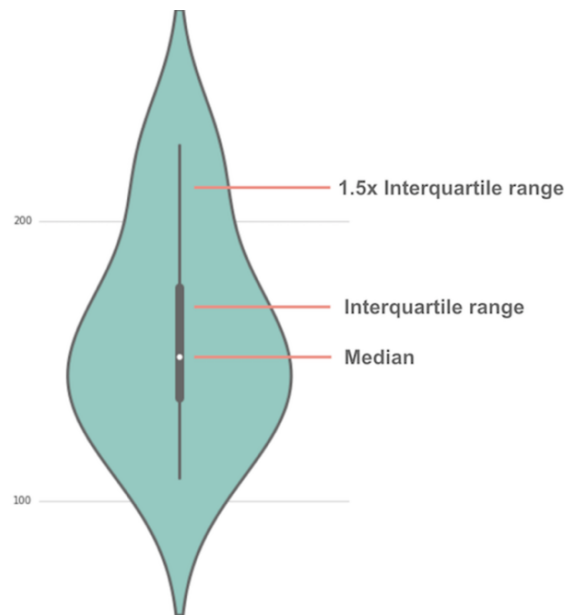
A dot plot displays individual observations along a number line. Because every data point is visible, dot plots are particularly useful for small to moderate datasets and for teaching purposes, as they preserve the raw data while still revealing patterns.



A density plot is a smoothed version of a histogram that estimates the probability density function of the data. Density plots are especially useful for comparing distributions and are widely used in modern statistical software.



A violin plot combines features of box plots and density plots. It displays the distribution's shape while also showing summary statistics. Violin plots are particularly effective when comparing distributions across groups.



Conceptual Takeaway for Students Students should understand that graphs are analytical tools, not decorations. A well-chosen graph can reveal insights that numerical summaries alone cannot. Developing the habit of visualizing data early in the analysis process is a defining characteristic of strong statistical reasoning and effective engineering practice.

1.6 Summary Statistics

While graphical displays allow us to visualize data, summary statistics provide concise numerical descriptions of key features of a dataset. Together, graphical and numerical summaries form the foundation of exploratory data analysis. Summary statistics are commonly grouped into measures of location and measures of variability.

Measures of Location (Central Tendency)

Measures of location describe where the data tend to cluster, or what might be considered a “typical” value in the dataset. The most commonly used measures of location are the mean, median, and mode.

The mean is the arithmetic average of the data and is widely used in engineering and scientific applications. However, it is sensitive to extreme values. The median is the middle value when the data are ordered and provides a measure of center that is resistant to outliers. The mode is the most frequently occurring value and is particularly useful for qualitative data or for identifying common outcomes.

In engineering practice, the choice of an appropriate measure of location depends on the nature of the data and the presence of unusual values. For example, median lifetimes are often preferred over means when analyzing component failure times due to skewness.

1.5.1 Measures of Spread (Variability)

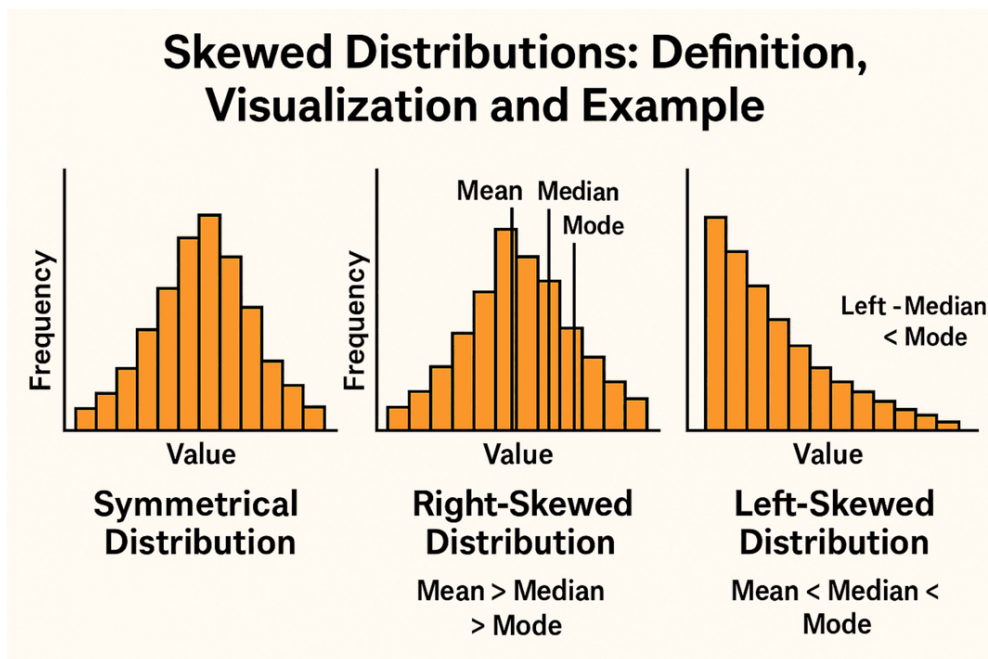
Measures of spread describe how dispersed the data are around the center. Two datasets may have the same mean but very different variability, leading to different engineering implications.

The range is the difference between the maximum and minimum values and provides a simple measure of spread. The variance measures the average squared deviation from the mean and forms the basis for many statistical methods. The standard deviation, which is the square root of the variance, is the most commonly used measure of variability because it is expressed in the same units as the original data.

In manufacturing and quality control, low variability is often as important as achieving a target mean, since excessive variation can lead to defects and system failure.

1.5.2 Describing Data Patterns

Beyond center and spread, statisticians describe data using additional features that capture the overall pattern of the distribution. These features are most easily recognized through graphical displays



The shape of a distribution refers to its general appearance. A distribution may be symmetric or skewed to the left or right. Skewness indicates the direction and degree of asymmetry, while kurtosis describes the heaviness of the tails and the peakedness of the distribution. Understanding these features helps engineers interpret the underlying processes generating the data. For example, right-skewed distributions are common in waiting times and failure data, while symmetric distributions often arise from measurement error around a stable process.

1.7 Unusual Features in Data

Real-world data rarely follow idealized patterns perfectly. As a result, statisticians pay close attention to unusual features that may indicate important phenomena, data collection issues, or rare but critical events.

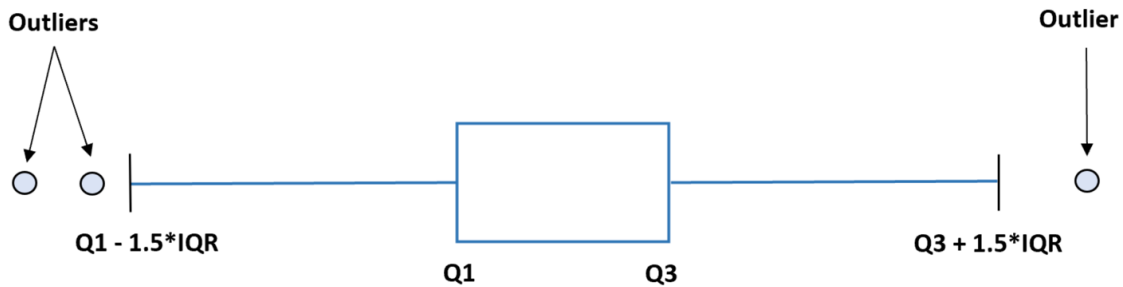
The two most common unusual features are gaps and outliers.

1.6.1 Gaps

A gap refers to a region within the range of the data where no observations occur. Gaps may arise due to physical constraints, changes in operating conditions, or data recording errors. In some cases, a gap may indicate the presence of distinct subgroups within the data, suggesting that a single distribution may not adequately describe the dataset.

1.6.2 Outliers

An outlier is an observation that lies unusually far from the rest of the data.



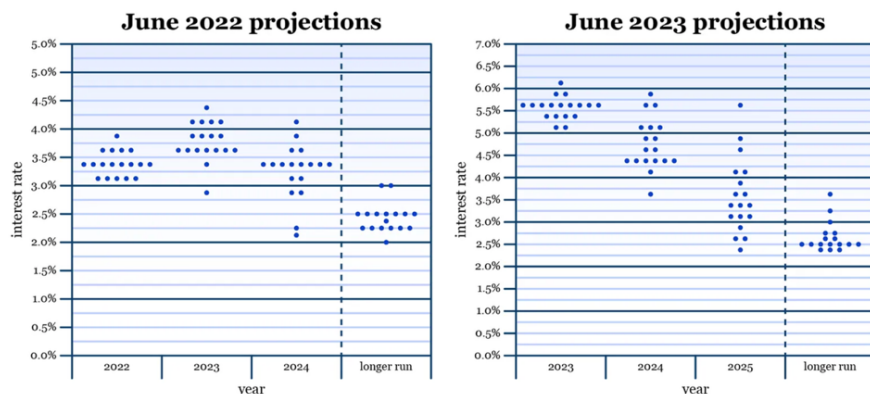
Outliers can result from measurement error, data entry mistakes, or genuinely extreme events. A commonly used rule for identifying outliers is based on the interquartile range (IQR). Observations lying more than 1.5 IQRs below the first quartile or above the third quartile are often flagged as potential outliers.

In engineering applications, outliers should never be removed automatically. Instead, they should be investigated carefully, as they may signal system faults, safety risks, or rare but important operating conditions.

1.8 Comparing Data Sets

In many practical situations, the goal is not to analyze a single dataset but to compare two or more datasets. Graphical displays are especially powerful for this purpose, as they allow direct visual comparison across groups or conditions.

Dot plot comparison: June 2022 vs. June 2023



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Common graphical tools for comparison include side-by-side box plots, comparative histograms,

dot plots, and bar charts. These displays help reveal differences and similarities that may not be apparent from numerical summaries alone.

1.7.1 Key Features for Comparing Data Sets

When comparing multiple datasets, statisticians focus on four fundamental features. The center indicates typical values, the spread reflects variability, the shape reveals distributional characteristics such as skewness or multimodality, and unusual features highlight gaps or outliers. A systematic comparison across these features ensures that conclusions are grounded in evidence rather than visual impression or single summary measures.

1.9 Sampling Procedures

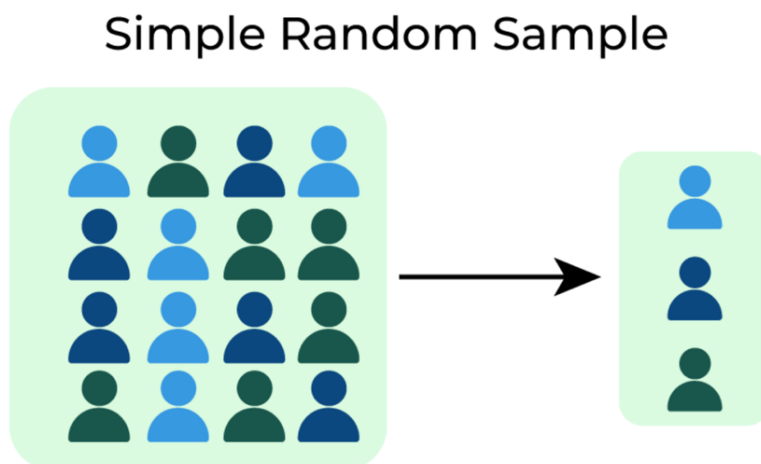
In statistical practice, it is rarely feasible to observe every element of a population. Instead, statisticians rely on sampling procedures to select a subset of observations, called a sample, from which conclusions about the population can be drawn. The fundamental goal of any sampling procedure is to obtain a sample that is representative of the population, meaning it accurately reflects the population's key characteristics.

A sampling procedure is said to be random if every possible sample of a given size has a known and non-zero probability of selection. Random sampling protects against systematic bias and provides the theoretical foundation for statistical inference. Without randomness, probability-based statements about uncertainty and confidence are no longer valid.

Different sampling procedures are used depending on the structure of the population, the objectives of the study, and practical constraints such as cost and accessibility.

1.8.1 Simple Random Sampling

Simple random sampling is the most fundamental sampling method and serves as the conceptual benchmark against which other methods are compared. In simple random sampling, every possible sample of size n drawn from a population of size N has equal probability of being selected.



For a finite population of size N , there are

$$\binom{N}{n}$$

distinct samples of size n . A sample is considered a simple random sample if each of these possible samples has probability $1/\binom{N}{n}$ of being chosen.

For an infinite population, or for populations that are very large relative to the sample size, a sample is considered random if each observation is selected independently from the others and according to the same probability distribution. In practice, many large populations can be treated as infinite when the sample size is small relative to the population size.

Example 1

There are 26 letters in the English alphabet. The total number of distinct samples of 5 letters that can be drawn is

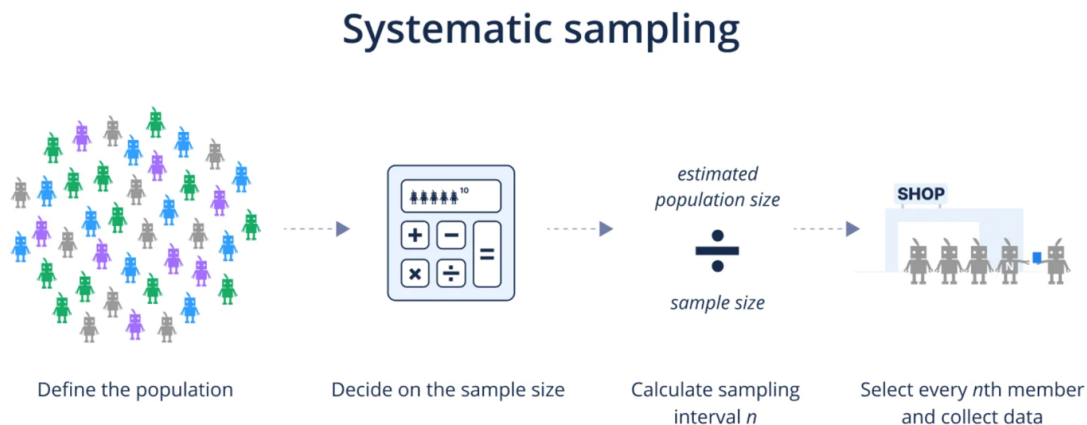
$$\binom{26}{5} = 65,780.$$

If a procedure is designed so that each of these 65,780 possible samples has an equal probability of being selected, then the resulting sample is a simple random sample. No letter or combination of letters is favoured over another.

Simple random sampling is conceptually ideal, but in practice it can be difficult to implement when populations are large, geographically dispersed, or poorly defined.

1.8.2 Systematic (Regular) Sampling

Systematic sampling is a structured alternative to simple random sampling that is often easier to implement in industrial and engineering settings. In this method, elements are selected at regular intervals from an ordered population.



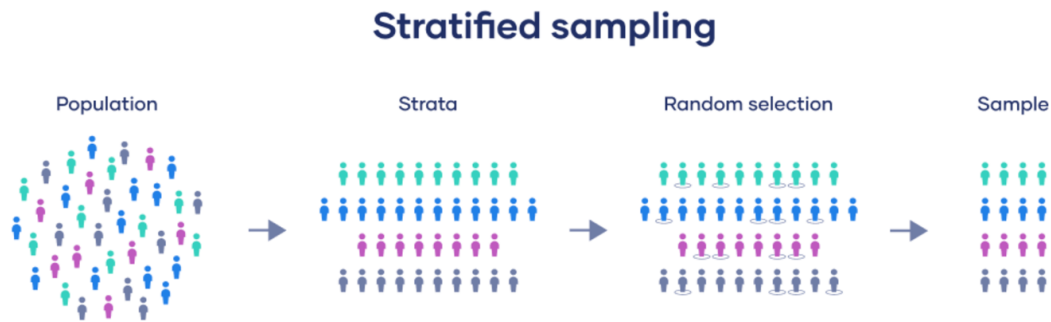
The procedure begins by randomly selecting a starting point from the first k elements of the population. Thereafter, every k -th element is selected for inclusion in the sample. The value of k , known as the sampling interval, is typically determined by dividing the population size by the desired sample size.

Systematic sampling has the advantage that the selected observations are spread evenly across the population, which often leads to samples that are highly representative. However, care must be taken to ensure that the population does not contain hidden periodic patterns that coincide with the sampling interval.

Example 2

In inspecting a batch of 1,000 pipes for defects, an engineer may choose to inspect every 10th pipe. The inspected pipes would then be the 10th, 20th, 30th, and so on, up to the 1,000th pipe.

This method is efficient and easy to implement. However, if every 10th pipe happens to be produced by a specific machine or process stage, the sample may be biased and fail to represent the entire batch accurately. Thus, systematic sampling requires careful consideration of the production process.

1.8.3 Stratified Random Sampling

Stratified random sampling is designed to improve representativeness when a population consists of distinct subgroups that differ in important ways. In this approach, the population is divided into mutually exclusive and collectively exhaustive sub-populations, called strata, and random samples are drawn independently from each stratum.

The strata are chosen so that observations within each stratum are relatively homogeneous with respect to the parameter of interest, while differences between strata are meaningful. This structure often leads to more precise estimates than simple random sampling.

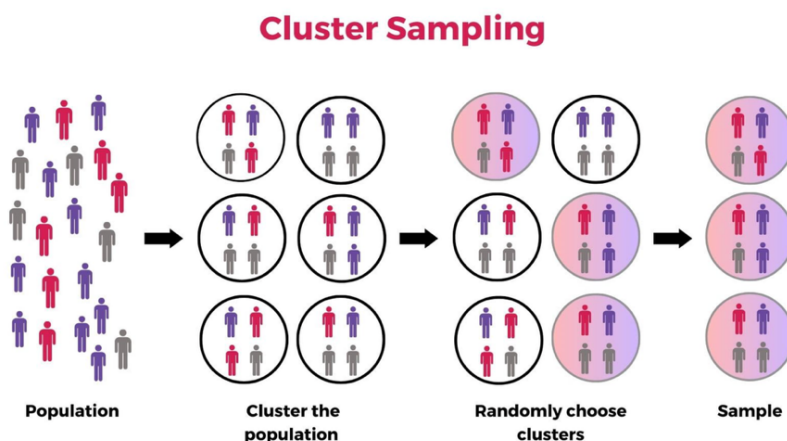
Example 3

To study income distribution among engineers in the Bay Area, the population can be divided into strata based on engineering discipline, such as electrical, mechanical, civil, chemical, and industrial engineering. Random samples are then selected from each discipline.

This approach is justified by the reasonable assumption that income levels are influenced by specialization. Stratified sampling ensures that all major engineering fields are adequately represented in the final sample.

1.8.4 Cluster Sampling

Cluster sampling is a sampling procedure in which the population is divided into groups, or clusters, and a random selection of these clusters is made. All elements within the selected clusters, or a random subset of them, are then included in the sample.



Unlike stratified sampling, clusters are typically chosen so that each cluster is a miniature representation of the entire population, rather than being internally homogeneous. Cluster sampling is particularly useful when populations are geographically dispersed or when obtaining a complete sampling frame is difficult.

Example 4

In conducting a statewide voter preference poll, researchers may randomly select congressional districts and survey all voters, or a random selection of voters, within those districts. When clusters are chosen carefully, the resulting sample can provide a reliable picture of statewide voter preferences at a fraction of the cost of simple random sampling.

1.10 Levels of Measurement (Types of Data)

Variables can be classified on the basis of their level of measurement. The way we classify variables greatly affects how we can use them in our analysis. Variables can be (1) nominal, (2) ordinal, (3) interval, or (4) ratio.

A nominal measurement is created when names are used to establish categories into which variables can be exclusively recorded. For example, sex can be classified as "male" or "female." You could also code it with a "1" or a "2"; oxide and carbonate ores may also be coded 1 and 2, but the numbers would serve only to indicate the categories and would carry no numerical significance; mathematical calculations using these codes would be meaningless. Soft drinks may be classified as Coke, Pepsi, 7-Up, or Ale 8. Each drink could be recorded in one of these categories to the exclusion of the others.

Nominal Measurements Names or classifications are used to divide data into separate and distinct categories.

It is important to remember that a nominal measurement carries no indication of order of preference, but merely establishes a categorical arrangement into which each observation can be placed.

Unlike a nominal measurement, an ordinal scale produces a distinct ordering or arrangement of the data. That is, the observations are ranked on the basis of some criterion. A retail company may rank its products as "good," "better," and "best." Opinion polls often use an ordinal scale such as "strongly agree," "agree," "no opinion," "disagree," and "strongly disagree."

As with nominal data, numbers can be used to order the rankings. Like nominal data, the magnitude of the numbers is not important; the ranking depends only on the order of the values. The retailer could have used the rankings of "1," "2," and "3," or "1," "3," and "12," for that matter. The arithmetic differences between the values are meaningless. A product ranked "2" is

not twice as good as one with a ranking of "1."

Ordinal Measurements Measurements that rank observations into categories with a meaningful order.

Variables on an interval scale are measured numerically, and, like ordinal data, carry an inherent ranking or ordering. However, unlike the ordinal rankings, the differences between the values is important. Thus, the arithmetic operations of addition and subtraction are meaningful. The Fahrenheit scale for temperatures is an example of an interval scale. Not only is 70 degrees hotter than 60 degrees, but also the same difference of 10 degrees exists as between 90 and 100 degrees Fahrenheit.

The value of zero is arbitrarily chosen in an interval scale. There is nothing sacrosanct about the temperature of zero; it is merely an arbitrary reference point. The Fahrenheit scale could have been created so that zero was set at a much warmer (or colder) temperature. No specific meaning is attached to zero other than to say it is 10 degrees colder than 10 degrees Fahrenheit. Thus, 80 degrees is not twice as hot as 40 degrees and the ratio $80/40$ has no meaning.

Interval Measurements Measurements on a numerical scale in which the value of zero is arbitrary but the difference between values is important.

Of all four levels of measurement, only the ratio scale is based on a numbering system in which zero is meaningful. Therefore, the arithmetic operations of multiplication and division also take on a rational interpretation. A ratio scale is used to measure many types of data found in business and geoscientific analyses. Variables such as costs, profits, inventory levels and grades are expressed as ratio measures. The value of zero dollars to measure revenues, for example, can be logically interpreted to mean that no sales have occurred. Furthermore, a firm with a 40 percent market share has twice as much of the market as a firm with a 20 percent market share. Measurements such as weight, time, and distance are also measured on a ratio scale since zero is meaningful, and an item that weighs 100 pounds is one-half as heavy as an item weighing 200 pounds.

Ratio Measurements Numerical measurements in which zero is a meaningful value and the difference between values is important.

You may notice that the four levels of measurement increase in sophistication, progressing from the crude nominal scale to the more refined ratio scale. Each measurement offers more information about the variable than did the previous one. This distinction among the various degrees of refinement is important, since different statistical techniques require different levels of measurements. While most statistical tests require interval or ratio measurements, other tests, called nonparametric tests (which will be examined later in this text), are designed to use nominal or ordinal data.

1.11 Measures of Location and Dispersion

In addition to graphical summaries such as histograms and frequency curves, the information contained in a dataset can be summarized numerically. Two numerical characteristics are of primary importance: the **location** of the data and the **dispersion** of the observations.

Measures of location, also called **measures of central tendency**, describe a value that is typical or representative of the dataset. They indicate where the observations tend to cluster. Measures of dispersion describe how widely the observations are spread around this central value. A meaningful statistical description of data must consider both aspects together.

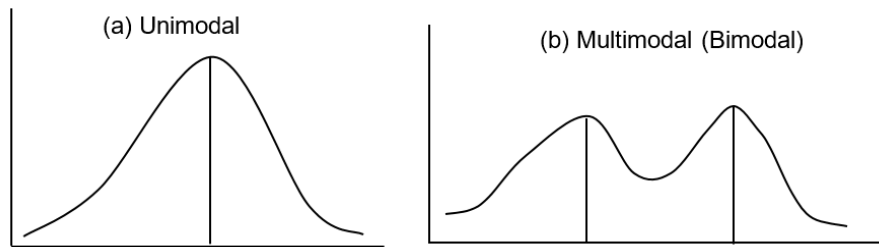
This section focuses on **measures of location**.

1.10.1 Measures of Location

Different measures of location capture different ideas of what is meant by a “typical” value. The most commonly used measures of location are the mode, the mean, and the median.

(a) The Mode

The mode is the value that occurs most frequently in a dataset. A dataset with a single mode is unimodal, while a dataset with multiple modes is multimodal. The mode is especially useful for qualitative data and for identifying the most common outcome in a set of observations.



Example 1

Consider the number of defects observed on identical manufactured components in a production batch:

2, 3, 1, 2, 0, 2, 1, 3, 2

The value 2 occurs most frequently. Therefore, the mode of the dataset is 2 defects per component. If two or more values occur with the same highest frequency, the dataset is multimodal, indicating that there is no single most typical value.

(b) The Mean

The mean is the most widely used measure of location. In general terms, the mean represents the “balance point” of the data. Several types of means are used in statistics, each appropriate for different types of data and applications.

(i) Arithmetic Mean (AM)

The arithmetic mean is defined as the sum of all observations divided by the number of observations. If a variable x takes values $x_1, x_2, \dots, x_n$, the arithmetic mean is denoted by $\bar{x}$.

The arithmetic mean uses all observations in the dataset, but it is sensitive to extreme values.

For a set of observations $x_1, x_2, \dots, x_n$, the arithmetic mean is defined as:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

The arithmetic mean is the most widely used measure of location and incorporates all observations. However, it is sensitive to extreme values.

Example 2

Suppose the measured lengths (in mm) of five machined parts are:

98, 100, 101, 99, 102

The arithmetic mean length is:

$$\bar{x} = \frac{98 + 100 + 101 + 99 + 102}{5} = 100\text{mm}$$

This value represents the average length of the parts. For large datasets, data are often grouped

into classes. In such cases, the arithmetic mean is computed using class marks and frequencies, which provides an efficient approximation for large volumes of data.

(ii) Geometric Mean

The geometric mean is defined only for positive, non-zero values. It is obtained by taking the n th root of the product of the observations. The geometric mean is particularly appropriate when dealing with percentage changes, growth rates, and multiplicative processes. The geometric mean answers the question:

“What constant growth rate would produce the same overall change?”

The geometric mean is defined for positive, non-zero values and is given by:

$G = (\prod_{i=1}^n x_i)^{1/n}$ The geometric mean is appropriate for multiplicative processes, such as growth rates and percentage changes.

Example 3 (Growth rate application)

Suppose the annual growth rates of a company's output over three years are:

+10%, +15%, and +5%

Expressed as growth factors, these are:

1.10, 1.15, and 1.05

The geometric mean growth factor is:

$$G = \sqrt[3]{1.10 \times 1.15 \times 1.05}$$

This value represents the average annual growth rate over the three years. Because growth compounds multiplicatively, the geometric mean provides a more meaningful summary than the arithmetic mean in such situations. It is always true that the geometric mean is less than or equal to the arithmetic mean, with equality only when all observations are equal.

(iii) Harmonic Mean

The harmonic mean is primarily used to compute average rates, particularly when a fixed quantity, such as distance or workload, is involved. It is defined as the reciprocal of the arithmetic mean of the reciprocals of the observations. The harmonic mean is appropriate when averaging quantities expressed as “per unit,” such as speed or efficiency. The harmonic mean is used to compute average rates when a fixed quantity such as distance or workload is involved.

It is defined as:

$$H = \frac{n}{\sum \frac{1}{x_i}}$$

Example 4

An engineer drives a test vehicle over the same distance at two different speeds:

60 km/h on the first segment and 40 km/h on the second segment.

The average speed over the entire trip is not the arithmetic mean of 60 and 40. Instead, it is given by the harmonic mean:

$$H = \frac{2}{\frac{1}{60} + \frac{1}{40}} = 48 \text{ km/h}$$

This example illustrates why the harmonic mean is the correct measure when distances are fixed. For any set of positive numbers, the following inequality always holds:

$$\text{Harmonic Mean} \leq \text{Geometric Mean} \leq \text{Arithmetic Mean}$$

(c) The Median

The median is the middle value of an ordered dataset. If the number of observations is odd, the median is the central value. If the number of observations is even, the median is the average of the two central values. Unlike the mean, the median is resistant to extreme values, making it particularly useful for skewed data.

Median for Ungrouped Data

Let $x_1, x_2, \dots, x_n$ denote a random sample of size n , arranged in increasing order of magnitude.

If n is odd, the median is the observation in position

$$\frac{n+1}{2}$$

If n is even, the median is the average of the observations in positions

$$\frac{n}{2} \text{ and } \frac{n}{2} + 1$$

That is,

$$\text{Median} = \begin{cases} x_{\frac{n+1}{2}}, & \text{if } n \text{ is odd} \\ \frac{1}{2} \left(x_{\frac{n}{2}} + x_{\frac{n}{2}+1} \right), & \text{if } n \text{ is even} \end{cases}$$

Median for Grouped Data

When data are grouped into class intervals, the exact median cannot be determined directly. Instead, it is estimated using the median class, which is the class interval containing the median position.

The general formula for the median of grouped data is:

$$\text{Median} = b_L + \left(\frac{\frac{n}{2} - f_{m-1}}{f_m} c \right)$$

where:

- b_L = lower boundary of the median class
- n = total number of observations
- f_{m-1} = cumulative frequency of the class preceding the median class
- f_m = frequency of the median class
- c = class width

This formula assumes that observations are uniformly distributed within the median class.

Example 5 (Ungrouped data)

Consider the annual salaries (in thousands) of five engineers:

45, 48, 50, 52, 120

The arithmetic mean is inflated by the extremely high salary, while the median (50) better represents the typical salary in the group.

Example 6 (Grouped data)

Given:

- $b_L = 199.5$
- $n = 300$
- $f_{m-1} = 116$
- $f_m = 73$
- $c = 50$

Substituting into the formula:

$$\text{Median} = 199.5 + \left(\frac{150 - 116}{73} \right) 50$$

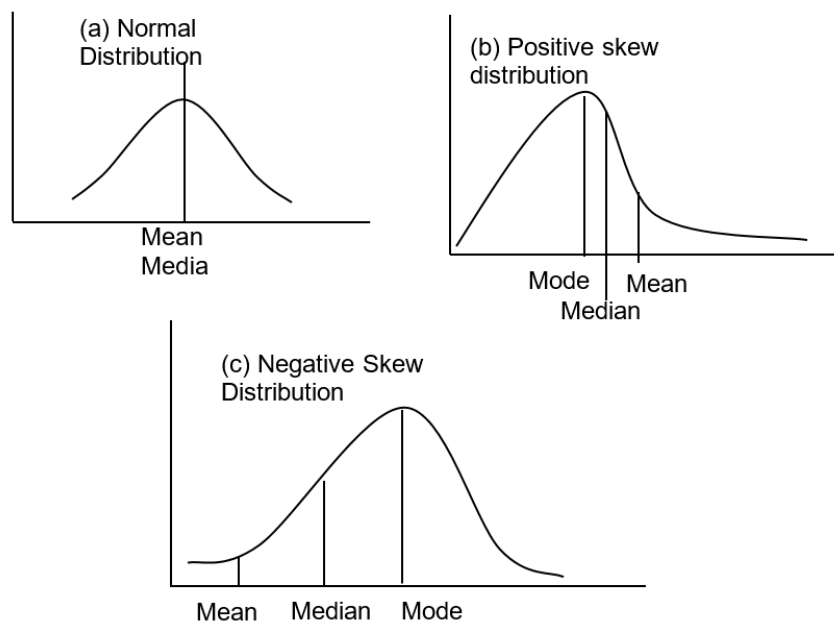
$$\text{Median} = 199.5 + (0.4658)(50)$$

$$\text{Median} \approx 222.8$$

(d) Relationship Between Measures of Location and Distribution Shape

The relative positions of the mode, median, and mean depend on the shape of the frequency distribution. In symmetrical distributions, all three measures coincide. In positively skewed distributions, the mean is greater than the median, which is greater than the mode. In negatively skewed distributions, the mean is less than the median, which is less than the mode.

Understanding this relationship is essential for selecting an appropriate measure of location and for interpreting data correctly in engineering applications.



1.11 Measure of Location and Dispersion

In addition to the histogram, the information in the frequency distribution can be further summarised by means of just two numbers. The first is the location of the data, and the various numbers that provide information about this are known as ‘measures of location’ or ‘measures of central tendency’. ‘Location of the data’ refers to a value that is typical of all the sample observations.

The second important aspect of the data is the dispersion of the observations. This implies how the data are scattered (dispersed). This is also called ‘measure of variation’.

1.12 Measures of Dispersion

Measures of location describe the central or typical value of a dataset, but they do not reveal how the data are distributed around that central value. Two datasets may have the same mean or median yet differ greatly in their degree of variability. Measures of dispersion quantify this variability and describe how widely the observations are spread.

In engineering and scientific applications, dispersion is critically important. A process may achieve the correct average output, but if the variability is too high, the process may still be unreliable or unsafe. Measures of dispersion therefore provide insight into consistency, precision, stability, and risk.

Measures of dispersion can be broadly classified into three categories:

1. Measures based on extreme or selected values,
2. Measures based on deviations from a central value,
3. Measures based on squared deviations.

1.11.1 Measures Based on Extreme or Selected Values

(a) The Range

The range is the simplest measure of dispersion and is defined as the difference between the largest and smallest observations in the dataset:

$$\text{Range} = x_{\max} - x_{\min}$$

Example 1

The tensile strengths (MPa) of five steel samples are:
410, 415, 420, 418, 412

$$\text{Range} = 420 - 410 = 10 \text{ MPa}$$

The range gives a quick indication of spread but depends only on two values and ignores the behavior of the remaining observations.

(b) The Interquartile Range (IQR)

The interquartile range (IQR) measures the spread of the central 50% of the data and is defined as:

$$\text{IQR} = Q_3 - Q_1$$

where Q_1 and Q_3 are the first and third quartiles, respectively.

Example 2

If the 25th percentile of a dataset is 52 and the 75th percentile is 68:

$$\text{IQR} = 68 - 52 = 16$$

The IQR is resistant to extreme values and is commonly used in box plots and outlier detection.

(c) The Interdecile Range

The interdecile range measures the spread of the central 80% of the data and is defined as:

$$InterdecileRange = D_9 - D_1$$

This measure is mainly used in descriptive statistics because it is less convenient for theoretical analysis.

1.11.2 Measures Based on Deviations from a Central Value

(a) Mean Deviation

The mean deviation is defined as the average of the deviations of observations from a central value (usually the mean or median)

$$MeanDeviation = \frac{1}{n} \sum (x_i - x)$$

Because positive and negative deviations cancel out, this measure can be misleading and is rarely used in practice.

(b) Mean Absolute Deviation (MAD)

To avoid cancellation of deviations, the mean absolute deviation (MAD) uses absolute values:

$$MAD = \frac{1}{n} \sum |x_i - x|$$

Example 3

Consider the dataset:

6, 9, 10, 13, 17, 19, 21, 25

The absolute deviations from the mean are calculated and averaged to obtain the MAD.

The MAD is intuitive and uses all observations, but it is difficult to manipulate mathematically in advanced statistical analysis.

1.11.3 Variance and Standard Deviation

To overcome the limitations of absolute deviations, statisticians square the deviations from the mean. This leads to the variance, which is the foundation of many statistical methods.

(a) Variance

For a sample, the variance is defined as:

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

The denominator $n - 1$ is used to obtain an unbiased estimate of the population variance.

For a population, the variance is:

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{n}$$

(b) Standard Deviation

The standard deviation is the positive square root of the variance:

$$s = \sqrt{s^2}$$

The standard deviation is expressed in the same units as the original data, making it the most interpretable measure of dispersion.

Types of Standard Deviation

(i) Population Standard Deviation

Used when data include every member of the population:

$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{n}}$$

(ii) Sample Standard Deviation

Used when data represent a sample from a larger population:

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}}$$

This is the most common form used in engineering and scientific analysis.

(iii) Standard Deviation for Grouped Data

When data are grouped into classes, the standard deviation is computed using class midpoints:

$$s = \sqrt{\frac{\sum f_i (x'_i - \bar{x})^2}{n - 1}}$$

where:

- x'_i is the class midpoint,
- f_i is the class frequency,
- n is the total number of observations.

Example 4 (Sample Standard Deviation)

Consider the measured voltages (V):

98, 100, 102, 100, 100

Mean:

$$\bar{x} = 100$$

Squared deviations:

$$(98 - 100)^2 = 4, (100 - 100)^2 = 0, (102 - 100)^2 = 4$$

$$s^2 = \frac{4 + 0 + 4 + 0 + 0}{4} = 2$$

$$s = \sqrt{2} \approx 1.41V$$

This indicates that the voltages typically deviate from the mean by about 1.4 V.

Interpretation of Standard Deviation

- Small standard deviation $\rightarrow$ data tightly clustered $\rightarrow$ high precision
- Large standard deviation $\rightarrow$ data widely spread $\rightarrow$ low consistency

In quality control and reliability engineering, reducing standard deviation is often more important than shifting the mean.

1.11.4 Why Dispersion Matters in Engineering

Dispersion affects:

- product quality,
- system reliability,
- safety margins,
- risk of failure.

Two processes with identical means may perform very differently if their variability differs. Engineers must therefore evaluate dispersion before making decisions based on averages.

Key Insight for Students Measures of dispersion quantify uncertainty and reliability. In engineering practice, controlling variability is often more important than controlling the average. Key Insight for Students Measures of dispersion quantify uncertainty and reliability. In engineering practice, controlling variability is often more important than controlling the average.

1.14 Measures of Dispersion, Skewness

It should be clear that a measure of central tendency by itself can exhibit only one of the

PRACTISE QUESTIONS

Question 1

Two manufacturing machines produce steel rods intended to have a length of 100 mm.

Machine A produces rods with lengths (mm):

98, 99, 100, 101, 102

Machine B produces rods with lengths (mm):

90, 95, 100, 105, 110

- Compute the arithmetic mean for both machines.
- Compute the range and standard deviation for both machines.
- Which machine would you recommend for mass production?
- Which measure would you report to a non-technical manager, and why?

Question 2

Test scores from an engineering statistics course are grouped into classes and summarized as follows:

Class Interval	Frequency
50–59	5
60–69	8
70–79	14
80–89	18
90–99	6

- Estimate the mean and standard deviation using grouped-data formulas.

- (ii) Explain why the grouped-data standard deviation is only an approximation.
- (iii) Under what circumstances is grouping unavoidable in engineering practice?

Question 3

A dataset describing waiting times for system failure is strongly right-skewed.

- (i) Would the standard deviation still be an appropriate measure of dispersion?
- (ii) Which combination of location and dispersion measures would you recommend?
- (iii) Explain how skewness affects risk assessment.

THEORITICAL QUESTIONS

1. Explain the difference between descriptive statistics and inferential statistics. Give one engineering example where each is used.
2. Why is probability essential in inferential statistics? Explain in the context of making decisions from sample data.
3. Distinguish clearly between:
 - a population and a sample
 - a parameter and a statistic

Provide one example of each from engineering practice.

4. Explain why a sample can never perfectly represent a population, even when random sampling is used.
5. Why is it dangerous to rely on a single numerical summary (such as the mean) when describing a dataset?
6. Explain why a histogram is more appropriate than a bar chart for numerical data.
7. Under what circumstances would a box plot be preferred over a histogram?
8. Describe one situation where a Pareto chart would be more informative than a pie chart.

PROJECT ASSIGNMENT (100 %)

Project Purpose

This project is designed to help students apply the concepts learned in Chapter 1: Introduction to Statistics to a real-world situation. The emphasis is on data collection, descriptive analysis, and statistical interpretation, rather than on probability or inference. By completing this project, students will demonstrate their ability to:

- collect and organize data,
- summarize data graphically and numerically,
- assess variability and distribution shape,
- communicate statistical findings clearly and logically.

Project Overview

Each student (or approved group) must identify a real-world variable, collect original data, and conduct a complete descriptive statistical analysis using only the methods covered in Chapter 1. No inferential methods (confidence intervals, hypothesis tests, probability models) should be used.

Step 1: Selection of Variable and Context

Students must select multiple primary variables that is measurable and relevant to engineering, science, or everyday systems.

The chosen variables must be:

- quantitative,
- clearly defined,
- measurable using a consistent method.

Examples of acceptable variables include:

- dimensions of manufactured items,
- waiting or processing times,
- energy or water consumption,
- environmental measurements (temperature, noise level, illumination),
- test scores or laboratory results,
- number of defects per unit.

Students must clearly state:

- what is being measured,
- the unit of measurement,
- the context in which the data arise.

Step 2: Data Collection Plan

Before collecting data, students must describe their data collection procedure, including:

- where the data will be collected,
- how measurements will be taken,
- the tool or method used for measurement,
- how consistency will be ensured.

Data Requirements

- Minimum of 30 observations,
- Data must be collected personally or obtained from a clearly identified reliable source,
- All observations must relate to the same variable and context.

Any potential sources of measurement error or bias should be briefly discussed.

Step 3: Data Organization and Classification

Students should present the raw data in a clearly labeled table and arrange the data in ascending order. The variable should be classified as discrete or continuous, and the level of measurement (nominal, ordinal, interval, or ratio) should be identified. If appropriate, the data may be grouped into class intervals, with justification provided for the chosen class width.

Step 4: Graphical Representation

Students must construct at least three appropriate graphical displays selected from those studied in Chapter 1. These may include histograms, box plots, stem-and-leaf plots, dot plots, or bar or Pareto charts where appropriate. Each graph must include a clear title, labeled axes with units, and a short written interpretation describing the observed patterns.

Step 5: Measures of Location

Students must compute and interpret the mode, arithmetic mean, and median. Where appropriate, the geometric mean or harmonic mean may also be computed. For each measure, students should explain what the value represents in context, whether it is appropriate as a typical value, and how the shape of the distribution affects its interpretation.

Step 6: Measures of Dispersion

Students must compute and interpret the range, interquartile range, variance, and standard deviation. Where relevant, the coefficient of variation should also be computed. Interpretation should focus on the degree of variability, the consistency of the data, and the practical implications of high or low dispersion.

Step 7: Distribution Shape and Unusual Features

Students must analyze the shape of the distribution by discussing symmetry or skewness, the relative positions of the mean, median, and mode, and the presence of gaps or outliers. Kurtosis should be discussed qualitatively in terms of peakedness or tail behavior. The relationship between graphical features and numerical summaries should be clearly explained.

Step 8: Discussion and Interpretation

A discussion section should address the main patterns observed in the data, possible reasons for variability, limitations of the data collection process, and practical implications of the findings. The emphasis should be on interpretation rather than computation.

Step 9: Report Structure

The project report should be organized as follows:

Title and objective

Description of variable and data collection method

Data table

Graphical summaries

Measures of location

Measures of dispersion

Distribution shape analysis

Discussion and conclusions

Reports must be clearly written, logically organized, and neatly presented.

NOTE!!! Academic Integrity

All data must be original or properly referenced. Fabricated or copied datasets will not be accepted.

Chapter 2

Introduction to Probability

Learning Objectives

- Interpret probabilities and use probabilities of outcomes to calculate probabilities of events in discrete sample spaces.
- Interpret and calculate conditional probabilities of events
- Use Bayes' theorem to calculate conditional probabilities
- Discuss random variables.
- Counting techniques in calculating probabilities of events

2.0 Introduction to Probability

Probability is an important and complex field of study. Fortunately, only a few basic issues in probability theory are essential for understanding statistics at the level covered in this book. These basic issues are covered in this chapter. The introductory section discusses the definitions of probability. This is not as simple as it may seem. The section on basic concepts covers how to compute probabilities in a variety of simple situations. The section on base rates discusses an important but often-ignored factor in determining probabilities.

Artifacts from ancient Egypt, Mesopotamia, Greece, China, and India show that humans interacted with random outcomes thousands of years before probability was formalized. Games such as Senet and early dice-based activities reveal practical encounters with uncertainty, even though numerical probability concepts were not yet developed.







The modern theory of probability is generally traced to the mid-1600s, when mathematicians began analyzing games of chance using logical and numerical reasoning. A pivotal moment occurred through correspondence between **Blaise Pascal** and **Pierre de Fermat**.

They were asked to solve the “problem of points,” which concerned how to fairly divide the stakes of an unfinished gambling game. Their solution required counting possible future outcomes and assigning equal likelihood to each. This work introduced the idea that probability could be calculated as a ratio of favorable outcomes to total possible outcomes, laying the foundation for the classical definition of probability.

2.1 Definitions

- (a) **Experiment:** An experiment is any process that generates a set of data or well-defined outcomes. There are two types of experiments, namely Deterministic and Random (or Chance) Experiments. In the deterministic experiments the observed results are not subject to chance while the outcomes of random experiments cannot be predicted with certainty. A random experiment could be as simple as tossing a coin or die and observing an outcome or complex as choosing 50 people from a population and testing them for the AIDS disease.
 - (b) **Trial:** Each repetition of an experiment is called a trial. That is, a trial is a single performance of an experiment.
 - (c) **Outcome:** The possible result of each trial of an experiment is called an outcome. When an outcome of an experiment has equal chance of occurring as the others the outcomes are said to be equally likely. For example, the toss of a coin and a die yield the possible outcomes in the sets, {H, T} and {1, 2, 3, 4, 5, 6} and a play of a football match yields {win (W), loss (L), draw (D)}.
 - (d) **Sample Space:** Sample space is the collection of all possible outcomes at a probability experiment. We use the notation S for sample space. Each element or outcome of the experiment is called a sample point. For example,
- (i) The results of two and three tosses of a coin give the following sample spaces:

$$S = \{HH, HT, TH, TT\}$$

$$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

- (ii) A toss of a die and a coin simultaneously give the results.

$$S = \{H1, H2, H3, H4, H5, H6, T1, T2, T3, T4, T5, T6\}$$

- (iii) The outcomes of two tosses of a die are as shown in the table

T1 \ T2	1	2	3	4	5	6
1	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
2	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
3	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
4	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
5	(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
6	(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)

where T1 and T2 represent the first and second tosses respectively.

- (i) Drawing a card from a packet of playing cards has sample space with 52 cards made up of 13 Heart, 13 Spade, 13 Diamond and 13 Club cards.

(e) Event: An event is a collection of one or more outcomes from an experiment.

That is, it is a subset of a sample space. It is denoted by a capital letter. For example we may have:

- (i) The event of observing a head (H) in three tosses of a coin,

$$A = \{HTT, TTH\}$$

- (i) The event of obtaining a total score of 8 on two tosses of a die,

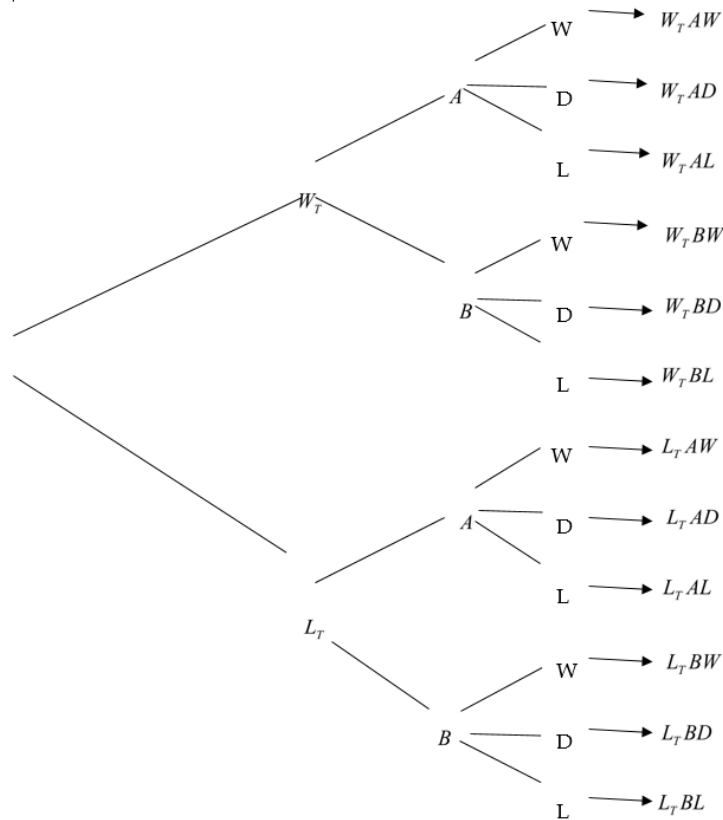
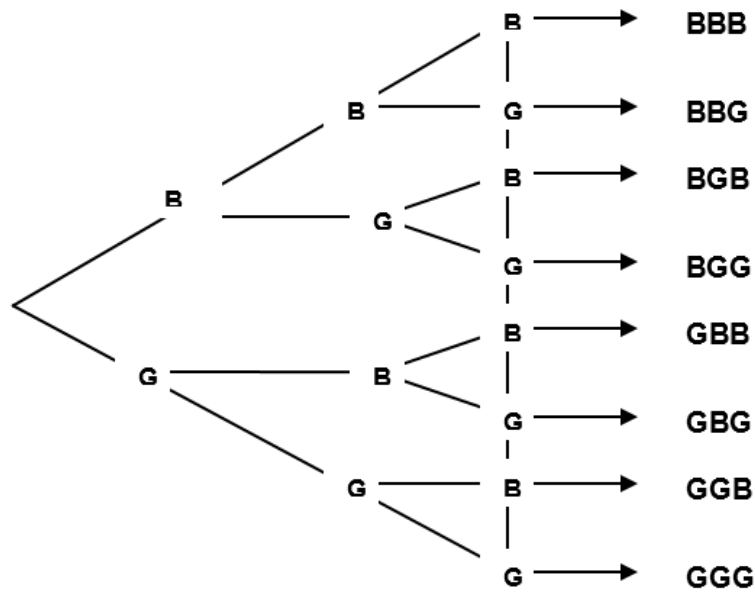
$$B = \{2, 6\}, \{3, 5\}, \{4, 4\}, \{5, 3\}, \{6, 2\}$$

- (i) Consider a newly married couple planning to have three children. The event of the family having two girls is:

$$D = \{BGG, GBG, GGB\}$$

(f) Tree Diagram: The tree diagram represents pictorially the outcomes of random experiment. The probability of an outcome which is a sequence of trials, is represented by any path of the tree. For example,

- (i) Consider a couple planning to have three children, assuming each child born is equally likely to be a boy (B) or girl (G).
- (ii) A soccer team on winning (WT) or losing (LT) a toss can defend either post A or B. It plays the match and either win (W), draw (D) or lose (L). We illustrate the experiment on a diagram as follows



2.2 Determination of Probability of an Event

The probability of an event A , denoted, $P(A)$, gives the numerical measure of the likelihood of the occurrence of event A which is such that $0 \leq P(A) \leq 1$. If $P(A) = 0$, the event A is said to be impossible to occur and if $P(A) = 1$, A is said to be certain. If A^c is the complement of the event A , then $P(A^c) = 1 - P(A)$, called the probability that event A will not occur. There are three main schools of thought in defining and interpreting the probability of an event. These are the Classical Definition, Empirical Concept and the Subjective Approach. The first two are

referred to as the Objective Approach.

- (a) **The Classical Definition:** This is based on the assumption that the outcomes of an experiment are equally likely. For example, if an experiment can lead to n mutually exclusive and equally likely outcomes, then the probability of the event A is defined by

$$P(A) = \frac{n(A)}{n(S)} = \frac{\text{Number of successful outcomes}}{\text{Number of possible outcomes}}$$

The classical definition of probability of event A is referred to as a priori probability because it is determined before any experiment is performed to observe the outcomes of event A .

- (a) **The Empirical Concept:** This concept uses the relative frequencies of past occurrences to develop probabilities for future. The probability of an event A happening in future is determined by observing what fraction of the time similar events happened in the past. That is,

$$P(A) = \frac{\text{Number of times } A \text{ occurred in the past}}{\text{Total number of observations}}$$

The relative frequency of the occurrence of the event A used to estimate $P(A)$ becomes more accurate if trials are largely repeated. The relative frequency approach of defining $P(A)$ is sometimes called posteriori probability because $P(A)$ is determined only after event A is observed.

- (a) **The Subjective Definition:** The subjective concept of probability is based on the degree of belief through the evidence available. The probability of an event A may therefore be assessed through experience, intuitiveness, judgment or expertise. For example, determining the probability of getting a cure of a disease or going to rain today. This approach to probability has been developed relatively recently and is related to *Bayesian Decision Analysis*. Although the subjective view of probability has enjoyed increased attention over the years, it has not been fully accepted by statisticians who have traditional orientations.

Example 9

Consider the problem of a couple planning to have three children, assuming each child born is equally likely to be a boy (B) or a girl (G).

- (a) List the possible outcomes in this experiment
 (b) What is the probability of the couple having exactly two girls?

Solution

- (a) The sample space for this experiment is

$$S = \{BBB, BBG, BGB, BGG, GBG, GGB, GGG\}$$

- (b) Let A be the event of the couple having exactly two girls. Then,

$$A = \{BGG, GBG, GGB\}$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{3}{8}$$

Example 10

Suppose a card is randomly selected from a packet of 52 playing cards.

- (i) What is the probability that it is a “Heart”?
 (ii) What is the probability that the card bears the number 5 or a picture of a queen?

(a) A box contains 4 red, 2 black and 3 white balls. What is the probability of drawing a red ball?

Solution

(a) Let the sample space be the set, $S = \{\text{playing cards}\}$, $A = \{\text{Heart cards}\}$, $B = \{\text{Cards numbered 5}\}$, $Q = \{\text{Cards with a picture of queen}\}$. Then

$$n(S) = 52, n(A) = 13, n(B) = 4 \text{ and } n(Q) = 4$$

(i)

$$P(A) = \frac{n(A)}{n(S)} = \frac{13}{52} = \frac{1}{4}$$

(ii) $P(B \text{ or } Q) = P(B) + P(Q)$

=

$$\frac{n(B)}{n(S)} + \frac{n(Q)}{n(S)} = \frac{4}{52} + \frac{4}{52} = \frac{2}{13}$$

(b) The sample space, $S = \{4R, 2B, 3W\text{-balls}\}$ and let $R = \{\text{red balls}\}$. Then

$$P(R) = \frac{n(R)}{n(S)} = \frac{4}{9}$$

Example 11

A die is tossed twice. List all the outcomes in each of the following events and compute the probability of each event.

- (a) The sum of the scores is less than 4
- (b) Each toss results in the same score
- (c) The sum of scores on both tosses is a prime number
- (d) The product of the scores is at least 20

Solution

The sample space for the experiment is the set of ordered paired (m, n) , where m and n each takes the values 1, 2, 3, 4, 5 and 6. Thus,

$$S = \{(1, 1), (1, 2), (1, 3), \dots, (6, 6)\}, \text{ where } n(S) = 36$$

$$(a) A = \{(1, 1), (1, 2), (2, 1)\}$$

$$P(A) = \frac{3}{36} = \frac{1}{12}$$

(b) $B = \{\text{each toss results in the same score}\}$

$$= \{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6)\}$$

$$P(B) = \frac{6}{36} = \frac{1}{6}$$

(c) $D = \{\text{sum of scores on both tosses is prime}\}$

$$D = \{(1, 1), (1, 2), (1, 4), (1, 6), (2, 1), (2, 3), (2, 5), (3, 2), (3, 4), (4, 1), (4, 3), (5, 2), (5, 6), (6, 1), (6, 5)\}$$

$$P(D) = \frac{15}{36} = \frac{5}{12}$$

(d) $E = \{\text{product of the scores is at least 20}\}$

$$= \{(4, 5), (4, 6), (5, 4), (5, 5), (5, 6), (6, 4), (6, 5), (6, 6)\}$$

$$P(E) = \frac{8}{36} = \frac{2}{9}$$

2.3 Probability of Compound Events

Two or more events are combined to form a single event using the set operations, $\cup$ and $\cap$. The event

- (i) $(A \cup B)$ occurs if either A or B both occur(s).
- (ii) $(A \cap B)$ occurs if both A and B occur.

2.2.1 Definitions:

(a) **Mutually Exclusive Events:** Two or more events which have no common outcome(s) (i.e. never occur at the same time) are said to be mutually exclusive. If A and B are mutually exclusive events of an experiment, then $A \cap B = \emptyset$ and $P(A \cup B) = P(A) + P(B)$, since $P(A \cap B) = 0$.

(b) **Independent Events:** Two or more events are said to be independent if the probability of occurrence of one is not influenced by the occurrence or non-occurrence of the other(s). Mathematically, the two events, A and B are said to be independent, if and only if $P(A \cap B) = P(A) \times P(B)$. However, if A and B are such that, $P(A \cap B) = P(A) \times P(B | A)$, they are said to be conditionally independent.

(c) **Conditional Probability:** Let A and B be two events in the sample space, S with $P(B) > 0$. The probability that an event A occurs given that event B has already occurred, denoted $P(A/B)$, is called the conditional probability of A given B. The conditional probability of A given B is defined as.

$$P(A | B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0$$

In particular, if S is a finite equiprobable space, then

$$P(A \cap B) = \frac{n(A \cap B)}{n(S)}, \quad P(B) = \frac{n(B)}{n(S)} \text{ and } P(A | B) = \frac{n(A \cap B)}{n(B)}$$

(d). **Exhaustive Events:** Two or more events defined on the same sample space are said to be exhaustive if their union is equal to the sample space S (thus, if they partition the sample space mutually exclusively). Eg: if $A_1, A_2, A_3 \in S$, $A_1 \cup A_2 \cup A_3 = S$.

Definition (partition of sample space): The events $A_1, A_2, A_3, \dots, A_n$ form a partition of the same sample space S if the following hold:

- (a) $A_i \neq \emptyset$ for all $i = 1, 2, 3, \dots, n$
- (b) $A_i \cap A_j = \emptyset$ for all $i \neq j$; $i, j = 1, 2, 3, \dots, n$
- (c)

$$\bigcup_{i=1}^n A_i = S$$

In other words, the n - events $A_1, A_2, A_3, \dots, A_n$ form a partition of the sample space S if the n - events are (a) nonempty, (b) mutually exclusive and (c) collectively exhaustive.

Example 12

(a) In a certain population of women, 40% have had breast cancer, 20% are smokers and 13% are smokers and have had breast cancer. If a woman is selected at random from the population, what is the probability that she had breast cancer, smokes or both?

(b) Let A and B be events such that $P(A) = 0.6$, $P(B) = 0.5$ and $(A \cup B) = 0.8$

(i) Find $P(A | B)$

(ii) Are A and B independent?

Solution

(a) Let B be the event of women with breast cancer and W the event of women who smoke. Then,

$P(B) = 0.4$, $P(W) = 0.2$ and $(B \cap W) = 0.13$

$$P(B \cup W) = P(B) + P(W) - P(B \cap W)$$

$$= 0.4 + 0.20 - 0.13$$

$$= 0.47$$

(b) Given that $P(A) = 0.6$, $P(B) = 0.5$ and $(A \cup B) = 0.8$

(i) $P(A \cap B) = P(A) + P(B) - P(A \cup B)$

$$= 0.6 + 0.5 - 0.8 = 0.3$$

$$P(A | B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0$$

$$= \frac{0.3}{0.5} = \frac{3}{5} = 0.6$$

(i) A and B are independent if $P(A) \times P(B) = P(A \cap B)$

$$P(A) \times P(B) = (0.6)(0.5) = 0.3 = P(A \cap B)$$

Which means that A and B are independent.

2.2.2 Example on Conditional Probability

Example 13

Complex components are assembled in a plant that uses two different assembly lines, A and A' . Line A uses older equipment than A' , so it is somewhat slower and less reliable. Suppose on a given day line A has assembled 8 components, of which 2 have been identified as defective (B) and 6 as nondefective (B'), whereas A' has produced 1 defective and 9 nondefective components. This information is summarized in the accompanying table.

	Condition	Total		
	B	B'		
Line	A	2	6	8
	A'	1	9	10
	3	15	18	

Unaware of this information, the sales manager randomly selects 1 of these 18 components for a demonstration. Prior to the demonstration

$P(\text{line A component selected}) = P(A) =$

$$\frac{N(A)}{N} = \frac{8}{18} = 0.44$$

However, if the chosen component turns out to be defective, then the event B has occurred, so the component must have been 1 of the 3 in the B column of the table. Since these 3 components are equally likely among themselves after B has occurred,

$$P(A | B) = \frac{P(A \cap B)}{P(B)} = \frac{2/18}{3/18} = \frac{2}{3}$$

Exercise:

Suppose that of all individuals buying a certain digital camera, 60% include an optional memory card in their purchase, 40% include an extra battery, and 30% include both a card and battery. Given that the selected individual purchased an extra battery, what is the probability that an optional card was also purchased?

The Multiplication Rule for $P(A \cap B)$

The definition of conditional probability yields the following result, obtained by multiplying both sides of the conditional probability equation by $P(B)$.

$$\begin{aligned} P(A | B) &= \frac{P(A \cap B)}{P(B)} \\ P(A | B) \times P(B) &= \frac{P(A \cap B)}{P(B)} \times P(B) \\ P(A | B) \times P(B) &= P(A \cap B) \end{aligned}$$

This rule is important because it is often the case that $P(A \cap B)$ is desired, whereas both $P(B)$ and $P(A | B)$ can be specified from the problem description.

The Law of Total Probability

Let $A_1, \dots, A_k$ be mutually exclusive and exhaustive events. Then for any other event B ,

$$\begin{aligned} P(B) &= P(B | A_1) \times P(A_1) + \dots + P(B | A_k) \times P(A_k) \\ &= \sum_{i=1}^k P(B | A_i) \times P(A_i) \end{aligned}$$

2.4 Bayes' Rule

The power of Bayes' rule is that in many situations where we want to compute $P(A/B)$ it turns out that it is difficult to do so directly, yet we might have direct information about $P(B/A)$. Bayes' rule enables us to compute $P(A/B)$ in terms of $P(B/A)$.

$$P(A | B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B | A)P(A)}{P(B)}$$

2.3.1 Bayes' Theorem

Let A and A^c constitute a partition of the sample space S such that with $P(A) > 0$ and $P(A^c) > 0$, then for any event B in S such that $P(B) > 0$,

$$P(A | B) = \frac{P(B | A)P(A)}{P(B | A)P(A) + P(B | A^c)P(A^c)}$$

The denominator $P(B)$ in the equation can be computed,

$$\begin{aligned} P(B) &= P[(A \cap B) \cup (A^c \cap B)] \\ &= P(A \cap B) + P(A^c \cap B) \\ &= P(A)P(B | A) + P(A^c)P(B | A^c) \quad (\text{by multiplicative rule}) \end{aligned}$$

Example 14

A paint-store chain produces and sells latex and semigloss paint. Based on long-range sales, the probability that a customer will purchase latex paint is 0.75. Of those that purchase latex paint, 60% also purchase rollers. But only 30% of semi gloss pain buyers purchase rollers. A randomly selected buyer purchases a roller and a can of paint. What is the probability that the paint is latex?

Solution

$L = \{\text{The customer purchases latex paint.}\}$, $P(L) = 0.75$

$S = \{\text{The customer purchases semigloss paint.}\}$, $P(S) = 0.25$

$R = \{\text{The customer purchases roller.}\}$

$$P(R|L) = 0.6$$

$$P(R|S) = 0.3$$

$$P(R) = P(R|L)P(L) + P(R|S)P(S) = 0.6 \times 0.75 + 0.3 \times 0.25 = 0.525$$

$$P(L | R) = \frac{P(L \cap R)}{P(R)} = \frac{P(R | L)P(L)}{P(R)} = \frac{0.6 \times 0.75}{0.6 \times 0.75 + 0.3 \times 0.25} \approx 0.857$$

2.5 Axioms of Probability

Given an experiment and a sample space, S , the objective of probability is to assign to each event A a number $P(A)$, called the probability of the event A , which will give a precise measure of the chance that A will occur. To ensure that the probability assignments will be consistent with our intuitive notions of probability, all assignments should satisfy the following axioms (basic properties) of probability.

A.1: For every event A , $0 \leq P(A) \leq 1$

$$A.2 : P(S) = 1$$

A.3: If A and B are mutually exclusive events, i.e $A \cap B = \emptyset$ then

$$P(A \cup B) = P(A) + P(B).$$

A.4: If $A_1, A_2, A_3, \dots, A_n$ is a sequence of n mutually exclusive events, then,

$$P(A_1 \cup A_2 \cup A_3 \cup \dots \cup A_n) = P(A_1) + P(A_2) + P(A_3) + \dots + P(A_n) \text{ or}$$

(

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P(A_i)$$

.

The following theorems arise directly from the above axioms:

(i) **Theorem 1:** If $\emptyset$ is the empty set, then $P(\emptyset) = 0$.

Proof:

Let A be any event, then A and $\emptyset$ are mutually exclusive and $A = A \cup \emptyset$

Then by A.3 $P(A) = P(A \cup \emptyset) = P(A) + P(\emptyset)$ and $P(\emptyset) = 0$

(ii) **Theorem 2:** If A^c is the complement of an event A , then

$$P(A^c) = 1 - P(A)$$

Proof

$$S = A \cup A^c$$

$$P(S) = P(A) + P(A^c)$$

$$1 = P(A) + P(A^c), \text{ by A.2}$$

$$P(A^c) = 1 - P(A)$$

2.4.1 Some Rules of Probability

(a) **The Addition Rule:**

Let $A_1, A_2, A_3, \dots, A_n$ be events of the sample space, S . Then

$$(i) P(A_1 \cup A_2) = P(A_1) + P(A_2) - P(A_1 \cap A_2)$$

$$(ii) P(A_1 \cup A_2 \cup A_3) = P(A_1) + P(A_2) + P(A_3) - P(A_1 \cap A_2) - P(A_1 \cap A_3) - P(A_2 \cap A_3) + P(A_1 \cap A_2 \cap A_3).$$

If the events $A_1, A_2, A_3, \dots, A_n$ are mutually exclusive, then

$$(i) P(A_1 \cup A_2) = P(A_1) + P(A_2)$$

$$(ii) P(A_1 \cup A_2 \cup A_3) = P(A_1) + P(A_2) + P(A_3)$$

$$(iii) P(A_1 \cup A_2 \cup \dots \cup A_n) = P(A_1) + P(A_2) + P(A_3) + \dots + P(A_n)$$

(b) **The Multiplication Theorem:**

If $A_1, A_2, A_3, \dots, A_n$ are events of the same sample space, S , then

$$(i) P(A_1 \cap A_2) = P(A_1) \times P(A_2 | A_1)$$

$$(ii) P(A_1 \cap A_2 \cap A_3) = P(A_1) \times P(A_2 | A_1) \times P(A_3 | A_1 \cap A_2)$$

2.6 Application of Counting Techniques

The classical definition of probability of an event A , $P(A)$ requires the knowledge of the number of outcomes of A and the total possible outcomes of the experiment, S . To find these outcomes we list such outcomes explicitly, which may be impossible if they are too many. Counting Techniques may be useful to determine the number of outcomes and compute $P(A)$. We shall examine three basic counting techniques, namely the *Multiplication Principle*, *Permutation* and *Combination*

2.5.1 The Multiplication Principle

The *Multiplication Principle*, also known as the *Basic Counting Principle* states that:

- If an operation can be performed in n_1 ways, and a second operation can be performed in n_2 ways and so on for k th operation which can be performed in n_k ways, then the combined experiment or operations can be performed in $n_1 \times n_2 \times n_3 \times \cdots \times n_k$ ways.

For example: A homeowner doing some remodeling requires the services of both a plumbing contractor and an electrical contractor. If there are 12 plumbing contractors and 9 electrical contractors available in the area, in how many ways can the contractors be chosen? If we denote the plumbers by $P_1, \dots, P_{12}$ and the electricians by $Q_1, \dots, Q_9$, then we wish the number of pairs of the form (P_i, Q_j) . With $n_1=12$ and $n_2=9$, the product rule yields $N = (12)(9) = 108$ possible ways of choosing the two types of contractors.

Example 15

1. Tossing a coin has two possible outcomes and tossing a die has six possible outcomes. Then the combined experiment, tossing the coin and die together will result in $(2 \times 6 = 12)$ twelve possible outcomes provided below:

$H1, H2, H3, H4, H5, H6, T1, T2, T3, T4, T5, T6$

(2) Another example is the number of different ways for a man to get dressed if he has 8 different shirts and 6 different pairs of trousers. The combination of the 8 different shirts and the six different pairs of trousers results in $(8 \times 6 = 48)$ possible ways.

(3) In a certain examination paper, students are required to answer 5 out of 10 questions from Section A another 3 out of 5 questions from Section B and 2 out of 5 questions from Section C. In how many ways can the students answer the examination paper?

Solution

The number of ways of answering the questions in Section A

$$\begin{aligned} &= 10 \times 9 \times 8 \times 7 \times 6 \\ &= 30240 \end{aligned}$$

The number of ways of answering the questions in Section B

$$= 5 \times 4 \times 3 = 60$$

The number of ways of answering the question in Section C

$$= 5 \times 4 = 20$$

Hence the students can answer the question in the three sections in

$$\begin{aligned} &= 30240 \times 60 \times 20 \\ &= 36,288,000 \end{aligned}$$

Application of the multiplication principle results in the other two counting techniques: *Permutation* and *Combination*, used to find the number of possible ways when a fixed number of items are to be picked from a lot without replacement.

2.5.2 Permutation of Objects

An ordered arrangement of objects is called a permutation. For example, the possible permutations of the letters A , B and C are as follows: ABC , ACB , BAC , BCA , CAB , CBA .

Definitions:

- (a) The number of permutations of n distinct objects, taken all together is:

$$n! = n(n-1)(n-2)(n-3)\dots 3 \times 2 \times 1 \quad (\text{or } {}^nP_n)$$

- (a) The number of permutations of n distinct objects taken k at a time is:

$${}^nP_k \text{ or } P(n, k) = \frac{n!}{(n-k)!}, \text{ where } k < n$$

- (a) The number of permutations of n objects consisting of groups of which n_1 of the first group are alike, n_2 of the second group are alike and so on for the k th group with n_k objects which are alike is:

$$\frac{n!}{n_1! n_2! n_3! \dots n_k!}$$

where

$$n_1 + n_2 + n_3 + \dots + n_k = n$$

- (a) *Circular Permutations*: Permutations that occur when objects are arranged in a circle are called circular permutations. The number of ways of arranging n different objects in a circle is given by:

$$\frac{n!}{n} = (n-1)!$$

Example 16

1. (i) The number of permutations of 10 distinct digits taken two at a time is

$${}^{10}P_2 = \frac{10!}{(10-2)!} = 10 \times 9 = 90$$

- (ii) A company codes its customers by giving each customer an eight character code. The first 3 characters are the letters A, B and C in any order and the remaining 5 are the digits 1, 2, 3, 4 and 5 also in any order. If each letter and digit can appear only once then the number of customers the company can code is obtained as follows:

The first 3 letters can be filled in $3!$

The next 5 digits can be filled in $5!$

Then the required number $= 3! \times 5! = 720$

- 2.(a) The number of permutations of the letters of the word, **POSSIBILITY**, which contains 3I's and 2S's is

$$\begin{aligned} \frac{11!}{3! \times 2! \times 1! \times 1! \times 1! \times 1! \times 1! \times 1!} &= 11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 2 \\ &= 3,326,400 \end{aligned}$$

(b) The number of arrangements of the letters of the word, **ADDING**, if the two letters D and D are together (**ADDING**)

$$= 5! = 120$$

(c) The number of circular permutations of 6 persons sitting around a circular table

$$= \frac{n!}{n} = (n-1)! = 5! = 120$$

(d) In how many ways can 4 boys and 2 girls seat themselves in a row if

(i) the 2 girls are to sit next to each other?

(ii) the 2 girls are not to sit next to each other?

Solution

(i) If we regard the 2 girls as separate persons ($B_1 B_2 B_3 B_4 G_1 G_2$), then the number of arrangements of 5 different persons, taken all at a time $= 5!$

The 2 girls can exchange places and so the required number of ways the can seat themselves

$$= 5! \times 2! = 240$$

(ii) The number of ways the boys can arrange themselves $= 4!$

The number of ways the 2 girls can occupy the arrowed places in their mix with the boys is as shown below:

$$\uparrow B_1 \uparrow B_2 \uparrow B_3 \uparrow B_4 \uparrow = {}^5P_2 = 5 \times 4$$

The required number of permutations (with the 2 girls not sitting next to each other) $= 4! \times 5 \times 4 = 480$

2.5.3 Combination of Objects

A Combination is a selection of objects in which the order of selection does not matter.

Definition:

The number of ways in which k objects can be selected from n distinct objects, irrespective of their order is defined by:

$${}^nC_k \text{ or } \binom{n}{k} = \frac{n!}{(n-k)!k!} = \frac{{}^nP_k}{k!}, \text{ where } k < n$$

Example 17

1.(a) (i) The number of ways a committee of 5 people can be chosen out of 9 is

$${}^9C_5 = \frac{9!}{4!5!} = 126$$

(ii) The number of combinations of the letters a, b, c, d and e, taken three at a time is $\binom{5}{3} = 10$ which are listed below:

$$abc, abd, abe, acd, ace, ade, bcd, bce, bde, cde$$

(b) Find the number of ways in which a committee of 4 can be chosen from 6 boys and 5 girls if it must

(i) consist of 2 boys and 2 girls

(ii) consist of at least 1 boy and 1 girl.

Solution

(i) The number of ways of choosing 2 boys from 6 and 2 girls from 5 is as follows:

$$\begin{aligned}\binom{6}{2} \times \binom{5}{2} &= \frac{6!}{4!2!} \times \frac{5!}{3!2!} \\ &= 15 \times 10 = 150\end{aligned}$$

(ii) For the committee to contain at least 1 boy and 1 girl will involve the following:

$$1B3G \quad \text{or} \quad 2B2G \quad \text{or} \quad 3B1G$$

The required number of ways

$$\begin{aligned}&= \binom{6}{1} \times \binom{5}{3} + \binom{6}{2} \times \binom{5}{2} + \binom{6}{3} \times \binom{5}{1} \\ &= \left(\frac{6!}{5!1!} \times \frac{5!}{2!3!} \right) + \left(\frac{6!}{4!2!} \times \frac{5!}{2!3!} \right) + \left(\frac{6!}{3!3!} \times \frac{5!}{4!1!} \right) \\ &= (6 \times 10) + (15 \times 10) + (20 \times 5) = 310\end{aligned}$$

Example 18

1. A box contains 6 red, 3 white and 5 blue balls. If three balls are drawn at random, one after the other without replacement, find the probability that

- (i) all are red
- (ii) 2 are red and 1 is white
- (iii) at least 1 is red
- (iv) 1 of each colour

Solution

(i)

$$\begin{aligned}\text{Pr}(\text{all the 3 are red balls}) &= \frac{\text{No. of selection of 3 from 6}}{\text{No. of selection of 3 from 14}} \\ &= \frac{\binom{6}{3}}{\binom{14}{3}} = \frac{6 \times 5 \times 4}{14 \times 13 \times 12} = \frac{5}{91}\end{aligned}$$

(ii)

$$\begin{aligned}\text{Pr}(2 \text{ red and 1 white ball}) &= \frac{\binom{6}{2} \times \binom{3}{1}}{\binom{14}{3}} \\ &= \frac{15 \times 3}{14 \times 13 \times 12} = \frac{45}{364}\end{aligned}$$

(iii) $\text{Pr}(\text{at least 1 red ball}) = 1 - \text{Pr}(\text{none is a red ball})$

$$\begin{aligned}&= 1 - \frac{\binom{8}{3}}{\binom{14}{3}} = 1 - \frac{8 \times 7}{14 \times 13 \times 12} \\ &= 1 - \frac{2}{13} = \frac{11}{13}\end{aligned}$$

(iv)

$$\text{Pr}(1 \text{ of each colour}) = \frac{\binom{6}{1} \times \binom{3}{1} \times \binom{5}{1}}{\binom{14}{3}}$$

$$= \frac{6 \times 3 \times 5}{14 \times 13 \times 2} = \frac{45}{182}$$

2. A board consists of 12 men and 8 women. If a committee of 3 members is to be formed, what is the probability that

- (i) It includes at least one woman?
- (ii) It includes more women than men?

Solution

The number of ways of forming the committee of 3 from twelve men and 8 women ($12M+8W$) = $\binom{20}{3} = 1140$

- (i) The probability that it includes at least 1 woman

$$\begin{aligned} &= \Pr(1W \ 2M) + \Pr(2W \ 1M) + \Pr(3W) \\ &= \frac{\binom{8}{1} \times \binom{12}{2} + \binom{8}{2} \times \binom{12}{1} + \binom{8}{3}}{\binom{20}{3}} \\ &= \frac{(8 \times 66) + (28 \times 12) + (56)}{1140} \\ &= \frac{528 + 336 + 56}{1140} = \frac{46}{57} \end{aligned}$$

- (ii) The probability that it includes more women than men

$$\begin{aligned} &= \Pr(2W \ 1M) + \Pr(3W) \\ &= \frac{\binom{8}{2} \times \binom{12}{1} + \binom{8}{3}}{\binom{20}{3}} \\ &= \frac{(28 \times 12) + (56)}{1140} \end{aligned}$$

2.7 Supplementary Questions

Example 19

If the probability of achieving monthly production targets at Goldfields Ghana Limited, (A), and Ashanti (Obuasi), (B), are 0.8 and 0.9 respectively, what is $P(A \cap B)$?

Solution

$$P(A \cap B) = P(A) \times P(B|A)$$

But production in GGL and production at Ashanti are independent. Hence,

$$P(B|A) = P(B)$$

Thus, $P(A \cap B) = P(A) \times P(B) = 0.8 \times 0.9 = 0.72$.

Since the mines are independent of each other, their productions are assumed to be independent of each other.

- (a) For dependent events. Here the first event is considered in determining the probability of the second. The principle of conditional probability is required. Thus, the probability of the joint events A and B is:

$$P(A \cap B) = P(A) \times P(B|A).$$

Example 20

The Credit Manager at SSB collects data on 100 of her customers. Of the 60 men, 40 have credit cards (C). Of the 40 women, 30 have credit cards (C). Ten of the men with credit cards have balances (B), whilst 15 of the women have balances (B). The Credit Manager wants to determine the probability that a customer selected at random is:

- (i) A woman with credit card
- (ii) A man with a balance.

Solution

$$(i) P(W \cap C) = P(W) \times P(C|W)$$

$$P(W) = \frac{40}{100}; P(C|W) = \frac{30}{40}$$

$$\therefore P(W \cap C) = \frac{40}{100} \times \frac{30}{40} = \frac{30}{100} = 0.3$$

$$(i) P(M \cap B) = P(M) \times P(B|M)$$

$$P(M) = \frac{60}{100}; P(B|M) = \frac{10}{60}$$

$$\therefore P(M \cap B) = P(M) \times P(B|M) = \frac{60}{100} \times \frac{10}{60} = \frac{10}{100} = 0.10$$

OR

$$P(M \cap B) = P(B) \times P(M|B) = \frac{25}{100} \times \frac{10}{25} = \frac{10}{100}$$

Example 21

The probability that a mining company will make profit at an annual production rate of 5000t/yr is 0.7 if the gold price is \$660/oz. If the gold price goes below \$660/oz the probability will fall to 0.40. The current world politics indicates that there is a 50% probability that the dollar will

be strong and gold price will fall below \$660/oz. If:

A: Gold price falls below \$660/oz

B: The mine is profitable.

1. What is the probability that both A and B occur?
2. What is the probability that either A or B will occur?

Solution

$$P(A) = 0.5, P(B) = 0.7, P(B|A) = 0.4$$

$$a) P(A \cap B) = P(A) \times P(B|A) = 0.5 \times 0.4 = 0.2$$

$$\begin{aligned} b) P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= 0.5 + 0.7 - 0.2 \\ &= 1 \end{aligned}$$

Example 22

2:

A coin is tossed twice. What is the probability that at least one head occurs?

Solution

The sample space for this experiment is: $S = \{HH, HT, TH, TT\}$

HTHT = = = HTHTHT = = = HT

$$A = \{HH, HT, TH\}$$

$$\therefore P(A) = 1/4 + 1/4 + 1/4 = 3/4$$

In general if an experiment can result in any one of N different equally likely outcomes, and if exactly n of these outcomes correspond to event A , then the probability of event A is: $P(A) = \frac{n}{N}$

Example 23

3:

If a player picks 5 cards, find the probability of holding 2 aces and 3 jacks.

Solution

Number of ways being dealt 2 aces from 4 is:

$$C_2^4 = \frac{4!}{2!2!} = 6$$

Number of ways of being dealt 3 jacks from 4 is: $C_3^4 = \frac{4!}{3!1!} = 4$

For each combination of 2 aces there are the number of combinations of 3 jacks. Thus there are $n = (6) \cdot (4) = 24$ hands with 2 aces and 3 jacks. The total number of 5 cards all of which are equally likely, is: $N = C_5^{52} = \frac{52!}{5!47!} = 2598960$

$$P(C) = \frac{n}{N} = \frac{24}{2598960} = 0.9 \times 10^{-6}$$

Example 24**Example 24:**

The probability of a certain lecturer arriving to lectures on time is $P(A) = 0.82$. The probability of closing on time is $P(D) = 0.83$. The probability that he arrives and departs on time is $P(A \cap D) = 0.78$. Find the probability that he will depart on time if he arrives on time $P(D|A)$.

Solution

$$P(D|A) = \frac{P(D \cap A)}{P(A)} = \frac{0.78}{0.82} = 0.95$$

The notion of conditional probability provides the capacity of re-evaluating the idea of probability information that is when it is known that another event has occurred. The probability of $P(AB)$ is an “updating” of $P(A)$ based on the knowledge that event B has occurred.

Chapter 3

Random Variables and Distributions

Learning Objectives

- Discuss random variables
- Determine probabilities from probability density functions.
- Determine probabilities from cumulative distribution functions and cumulative distribution functions from probability density functions, and the reverse.

3.0 Random Variables and Distribution

3.1 Introduction

Experiments are conducted with results that are subject to chance. Suppose that to each point of a sample space we assign a number, we then have a function defined on the sample space. This function is called a random variable (or stochastic variable). In other words, in a particular experiment a random variable X would be some function that assigns a real number $X(s)$ to each possible outcome $s \in S$.

Definition: Let S be the sample space associated with some experiment ε . A random variable X is a function that assigns a real number $X(s)$ to each sample point $s \in S$.

There are two basic types of random variables: the Discrete random variable and the Continuous random variable.

3.2 Discrete Random Variable

A **discrete** random variable is a random variable whose possible values either constitute a finite set or else can be listed in an infinite sequence in which there is a first element, a second element, and so on (“countably” infinite).

3.1.1 Probability Distribution of Discrete Random Variable

For a random variable X of the discrete type, it is necessary and sufficient to know the following:

- (a) the list of all possible values x_i , $i = 1, 2, 3, \dots$
- (b) the corresponding probability $p(X = x_i)$ of each of these values.

When such information is available we say that the distribution law or the probability distribution or simply the distribution of the random variable X is known.

Definition: A probability distribution of a discrete random variable X is a sequence of the values of X , together with the probability assigned to each point x_i , ($i = 1, 2, 3, \dots$) x_i , $i = 1, 2, 3, \dots$.

The probability distribution of a discrete random variable is more often called probability function ($p.f.$) or a probability mass function ($p.m.f.$) and is frequently denoted by $p(x_i)$ and $P(X = x_i)$.

Representation of Probability Distribution of the Discrete Random Variable

The probability distribution of a discrete random variable X can be represented by a table, a graph or a function.

Tabular form

It is convenient to specify the probability distribution of a random variable by means of a table having two rows: the upper row contains the possible values the random variable assumes and the lower row contains the corresponding probabilities of the values. This is represented as in the table 1 below:

Table 1: Probability distribution of X

x_i	x_1	x_2	$\dots$	x_n
$p(x_i)$	$p(x_1)$	$p(x_2)$	$\dots$	$p(x_n)$

Graphical form

The probability distribution may also be given graphically. The graph represents chance and not data.

Definition: A graph of $p(x_i)$ against x_i is called a probability graph.

To obtain a probability graph, vertical lines or bars are drawn above the possible values x_i of the random variable X on the horizontal axis. The height of each line or bar (rectangle) equals the probabilities of the corresponding values of the random variables.

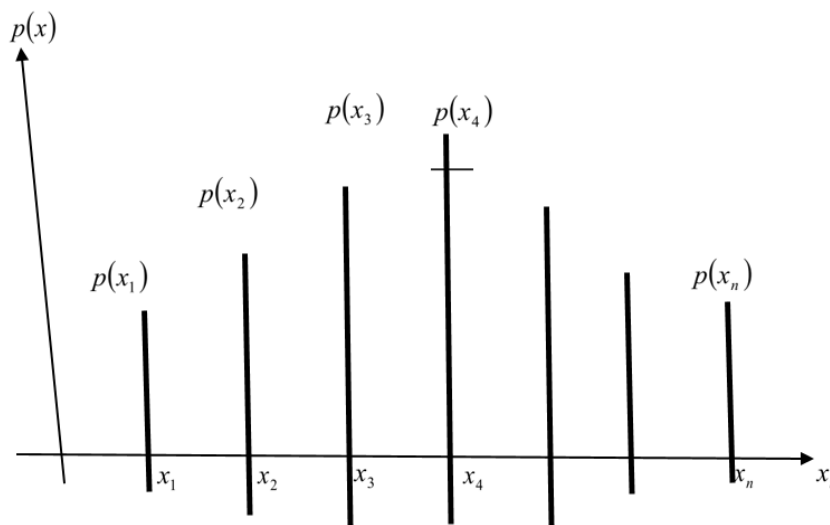


Figure 3.1: Figure 1: A graph of a probability distribution

Formula:

Representing a discrete probability by means of a formula is the most useful method.

For example:

$$p(x) = P(X = x) = \frac{1}{n}, \quad x = 1, 2, 3, \dots, n$$

defines a probability distribution of a random variable X . Not all functions of a discrete random variable qualify to be called probability mass function. We shall now give a formal definition of a probability mass function.

Definition: Probability mass function: Any function $p(x)$ defined on all possible values of the discrete random variable $X = x_i$, $i = 1, 2, 3, \dots$ is called a probability mass function if it satisfies the following properties:

$$\begin{aligned} \text{property 1} \quad & p(x_i) \geq 0, \quad i = 1, 2, 3, \dots \\ \text{property 2} \quad & \sum_i p(x_i) = 1 \end{aligned}$$

where the summation is over all possible values of the random variable X .

A random variable X in one experiment takes only one of its n possible values with a corresponding probability which is non-negative. Thus, in an experiment, one and only one of the numerous possible mutually exclusive events will happen. Consequently, the sum of the probabilities of these events:

$$p(x_1) + p(x_2) + p(x_3) + p(x_4) + \dots + p(x_n) = 1$$

Example 25

A fair coin is tossed three times. Let the random variable X represent the number of heads which come up.

- find the probability distribution corresponding to the random variable X
- Construct a probability graph.

Solution

- the sample space is

$$S = \{HHH, THH, HTH, HHT, HTT, THT, TTH, TTT\}$$

The probability of each outcome is $\frac{1}{8}$, since all the outcomes are equally likely sample events. With each sample point we can associate a number for the random variable X , as shown in the table below:

Sample point	HHH	THH	HTH	HHT	HTT	THT	TTH	TTT
Number of Head	3	2	2	2	1	1	1	0
$p(X = x)$	$1/8$	$1/8$	$1/8$	$1/8$	$1/8$	$1/8$	$1/8$	$1/8$

The table above shows that the random variable can take the values 0, 1, 2, 3. The next task is to compute the probability distribution $p(x_i)$ of X . Thus,

$$p(3) = P(X = 3) = P(HHH) = \frac{1}{8}$$

$$\begin{aligned} p(2) &= P(X = 2) = P(\{HHT\} \cup \{HTH\} \cup \{THH\}) \\ &= P(\{HHT\}) + P(\{HTH\}) + P(\{THH\}) \\ &= \frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{3}{8} \end{aligned}$$

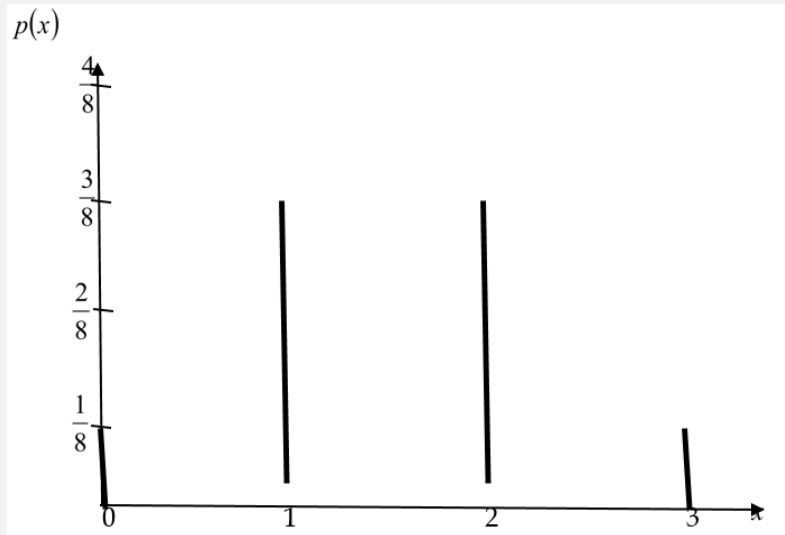
$$\begin{aligned}
 p(1) &= P(X = 1) = P(\{TTH\} \cup \{THT\} \cup \{HTT\}) \\
 &= P(\{TTH\}) + P(\{THT\}) + P(\{HTT\}) \\
 &= \frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{3}{8}
 \end{aligned}$$

$$p(0) = P(X = 0) = P(TTT) = \frac{1}{8}$$

Thus, the probability distribution of X is tabulated as:

x_i	0	1	2	3
$p(x_i)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

(b). The diagram which follows graphically describes the above distribution.



Example 26

A shipment of 8 similar microcomputers to a retail outlet contains 3 that are defective. If a school makes random purchase of 2 of these computers, find the probability distribution for the number of defectives

Solution

Let X be a random variable whose values x are the possible numbers of defective computers purchased by the school. Then x can be any of the numbers 0, 1 and 2. Now

$$f(0) = P(X = 0) = \frac{\binom{3}{0}\binom{5}{2}}{\binom{8}{2}} = \frac{10}{28}$$

$$f(1) = P(X = 1) = \frac{\binom{3}{1}\binom{5}{1}}{\binom{8}{2}} = \frac{15}{28}$$

$$f(2) = P(X = 2) = \frac{\binom{3}{2}\binom{5}{0}}{\binom{8}{2}} = \frac{3}{28}$$

Thus, the probability distribution of X is:

x	0	1	2
$f(x)$	$\frac{10}{28}$	$\frac{15}{28}$	$\frac{3}{28}$

As a check, we can see whether the three probabilities we found will sum up to one because as usual, the total probability associated with a number of mutually exclusive, exhaustive events must be one.

Note: A probability distribution is a display of all possible outcomes of an experiment along with the probabilities of each outcome. In fact, it is a list of all possible outcomes of some experiment and the probability associated with each outcome.

Example 27

Verify that the following probability distribution functions are probability mass functions.

(i).

$$p(x) = \begin{cases} \frac{1}{21}(2x + 3), & x = 1, 2, 3 \\ 0 & \text{otherwise} \end{cases}$$

(ii).

$$p(x) = \begin{cases} k(x - 1), & x = 3, 4, 5 \\ 0 & \text{otherwise} \end{cases}$$

Solution

(i). For the probability distribution function to be a probability mass function the following must satisfy:

$p(x) > 0$ for all x and

$$\begin{aligned} \sum_{x=1}^3 p(x) &= \frac{1}{21} \sum_{x=1}^3 (2x + 3) \\ &= \frac{1}{21} [\{2(1) + 3\} + \{2(2) + 3\} + \{2(3) + 3\}] \\ &= \frac{1}{21} [5 + 7 + 9] \\ &= 1 \end{aligned}$$

(ii). The value of k is determined by assuming that $p(x)$ is a probability function. Thus,

$$\begin{aligned} \sum_{x=3}^5 p(x) &= \sum_{x=3}^5 k(x - 1) = 1 \\ \Rightarrow k[(3 - 1) + (4 - 1) + (5 - 1)] &= 1 \\ \Rightarrow k[2 + 3 + 4] &= 1 \\ \Rightarrow 9k &= 1 \\ \Rightarrow k &= \frac{1}{9} \end{aligned}$$

3.3 Continuous Random Variable

If X is a continuous random variable, the probability that X takes on any one particular value is generally zero. Therefore, we cannot define a continuous random variable in the same way as for a discrete random variable. In order to arrive at a probability distribution for a continuous random variable we note that the probability that X lies between two different values is meaningful. Thus, a continuous random variable is the type whose spaces are not composed of a countable number of points but takes on values in some interval or a union of intervals of the real line.

Definition: If the set of all possible values of a random variable X , takes on an uncountably infinite number of values or values in some interval or a union of intervals of the real line, it is called a continuous random variable if there exists a function f , called probability density function of X such that

(i) $f(x_i) \geq 0 \forall x_i$ (Non-negative)

(ii)

$$\int_{-\infty}^{\infty} f(x_i) dx = 1$$

(iii)

$$P(a \leq X \leq b) = \int_a^b f(x) dx \quad -\infty \leq a \leq b \leq \infty$$

3.2.1 Probability Distribution of Continuous Random Variable

Continuous random variable can be considered as one that takes on values in some interval of the real line. The probability distribution for a continuous random variable is more often called a probability density function (p.d.f) or simply density function and is denoted by $f(x)$.

Definition: The probability density function, (p.d.f.) of a random variable X of the continuous type, with space R that is an interval or union of intervals, defined on the set of real numbers is an integrable function $f(x)$ satisfying the following conditions.

(i) $f(x_i) \geq 0 \forall x_i$ (Non-negative)

(ii)

$$\int_{-\infty}^{\infty} f(x_i) dx = 1$$

(iii) The probability of the event $X \in A$ is

$$P(X \in A) = \int_A f(x) dx$$

A typical probability function of the continuous type is as shown on the sketch below

$$f(x)$$

0 a b X

fig. 2.1 An example of a Probability Density Function

In this figure, the total area under the curve must be equal to 1 and the value of the $P(a \leq x \leq b)$ is equal to the area of the shaded region.

Example 28

Suppose that the error in the reaction temperature, in $^{\circ}C$ for a controlled laboratory experiment

is a continuous random variable X having the probability density function:

$$f(x) = \begin{cases} \frac{x^2}{3}, & -1 < x < 2 \\ 0 & \text{elsewhere} \end{cases}$$

Verify (a). $\int_{-\infty}^{\infty} f(x) dx$ (b). $P(0 < x \leq 1)$

Solution

(a).

$$\begin{aligned} \int_{-\infty}^{\infty} f(x) dx &= \int_{-1}^2 \frac{x^2}{3} dx \\ &= \frac{x^3}{9} \Big|_{-1}^2 = \frac{8}{9} + \frac{1}{9} = 1 \end{aligned}$$

(b).

$$\begin{aligned} P(0 < x \leq 1) &= \int_0^1 \frac{x^2}{3} dx \\ &= \frac{x^3}{9} \Big|_0^1 = \frac{1}{9} - 0 = \frac{1}{9} \end{aligned}$$

Example 29

(a). Find the constant C such that the function below is a probability density function:

$$f(x) = \begin{cases} cx^2, & 0 < x < 3 \\ 0 & \text{otherwise} \end{cases}$$

(b). Compute $P(1 < x < 2)$

Solution

(a).

$$\begin{aligned} \int_{-\infty}^{\infty} f(x) dx &= \int_0^3 cx^2 dx \\ &= \frac{cx^3}{3} \Big|_0^3 = 9c \end{aligned}$$

But since

$$\int_{-\infty}^{\infty} f(x) dx = 1, \quad 9c = 1 \quad \therefore c = \frac{1}{9}$$

(b).

$$\begin{aligned} P(1 < x \leq 2) &= \int_1^2 \frac{x^2}{9} dx \\ &= \frac{x^3}{27} \Big|_1^2 = \frac{8}{27} - \frac{1}{27} = \frac{7}{27} \end{aligned}$$

Example 30

A machine produced copper wire, and occasionally there is a flaw at some point along the wire. The length of wire (in meters) produced between successive flaws is a continuous random variable X with p.d.f of the form

$$f(x) = \begin{cases} c(1+x)^{-3} & x > 0 \\ 0 & x \leq 0 \end{cases}$$

where c is a constant

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$\int_0^{\infty} c(1+x)^{-3} dx = 1$$

Let $U = (1 + x)$ and $du = dx$ apply power rule for integral

$$c \int_0^{\infty} U^{-3} du = 1$$

$$c \left[\frac{U^{-2}}{-2} \right]_0^{\infty} = 1$$

$$c \left[\frac{1}{2} \right] = 1$$

$$c = 2$$

Example 31

For each of the following functions, find the constant c so that $f(x)$ is a p.d.f of a random variable X .

$$(i) f(x) = 4xc, 0 \leq x \leq 1$$

$$(ii) f(x) = c\sqrt{x}, 0 \leq x \leq 4$$

$$(iii) f(x) = c/x^{3/4}, 0 < x < 1$$

Solution

$$(i) f(x) = 4x^c$$

$$\int_0^1 f(x) dx = 1$$

$$\int_0^1 4x^c dx = 1$$

$$4 \int_0^1 x^c dx = 1$$

$$4 \left[\frac{x^{c+1}}{c+1} \right]_0^1 = 1$$

$$4 \left[\frac{1}{c+1} \right] = 1$$

$$c+1 = 4$$

$$c = 3$$

$$(ii) f(x) = c\sqrt{x}$$

(iii)

$$c \int_0^4 x^{1/2} dx = 1$$

$$c \left[\frac{x^{1/2+1}}{1/2+1} \right]_0^4 = 1$$

$$c \left[\frac{4^{3/2}}{3/2} \right] = 1$$

$$c = 3/16$$

$$f(x) = \frac{c}{x^{3/4}}$$

$$c \int_0^1 x^{-3/4} dx = 1$$

$$c \left[\frac{x^{-3/4+1}}{-3/4+1} \right]_0^1 = 1$$

$$\frac{c}{1/4} = 1$$

$$c = \frac{1}{4}$$

Post-Test

1. Define the following terms;
2. discrete random variable
3. continuous random variable
4. Let $f(x) = \frac{x}{15}$, $x = 1, 2, 3, 4, 5$, zero elsewhere be the p.d.f. of X. Find;
5. $\Pr[1 \text{ or } 2]$

$$\Pr[1 < X < 3]$$

$$(i) \Pr\left[\frac{1}{2} \leq X \leq \frac{5}{2}\right]$$

- (ii) For each of the following functions, find the constant c so that $f(x)$, satisfies the conditions of being a p.d.f. of a random variable X.

$$(i) f(x) = cx^3, 0 < x < 1$$

(ii)

$$f(x) = \frac{c}{x^4}$$

,

$$0 < x < \infty$$

3.4 Cumulative Distribution Function

The cumulative distribution function (c.d.f.) or simply the distribution function is the most universal characteristic of a random variable. It exists for all random variables whether they are

discrete or continuous.

Definition: Let X be a random variable and x any real number. The cumulative distribution function of X is a function $F(x)$ defined as the probability that the random variable X takes a value less than or equal to x such that

$$F(x) = P(X \leq x)$$

The function $F(x)$ in the above definition may also be written as

$$F(x) = P(-\infty < X \leq x)$$

3.3.1 Cumulative distribution function of Discrete Random Variables

The statement of the theorem below for a distribution function of a discrete random variable follows trivially from the definition of cumulative distribution function.

Theorem (Distribution function of a discrete random variable): Let X be a discrete random variable with probability $p(x_i)$, then the cumulative distribution function is defined by

$$F(x) = \sum_{x_i \leq x} p(x_i)$$

For the discrete case, the random variable X takes on only a finite number of values $x_1, x_2, x_3, \dots, x_n$. In this case, the distribution function $F(x)$ is given by

$$F(x) = \begin{cases} 0 & -\infty \leq x < x_1 \\ p(x_1) & x_1 \leq x < x_2 \\ p(x_1) + p(x_2) & x_2 \leq x < x_3 \\ \vdots & \vdots \\ p(x_1) + p(x_2) + \dots + p(x_n) & x_n \leq x < +\infty \end{cases}$$

The figure below depicts the graph $F(x)$ which is discontinuous at at the possible values $x_1, x_2, x_3, \dots, x_n$. At these points $F(x)$ is continuous from the right but discontinuous from the left. Because of the appearance of its graph, the cumulative distribution function for a discrete random variable is also called a staircase function or a step function, having jump discontinuities at all possible values $X = x_i$ with a step at the x_i of height $p(x_i)$. The graph increases only through these jumps at $x_1, x_2, x_3, \dots, x_n$. Everywhere else in the interval $[x_i, x_{i+1}]$, the cumulative distribution function $F(x_i)$ is constant.

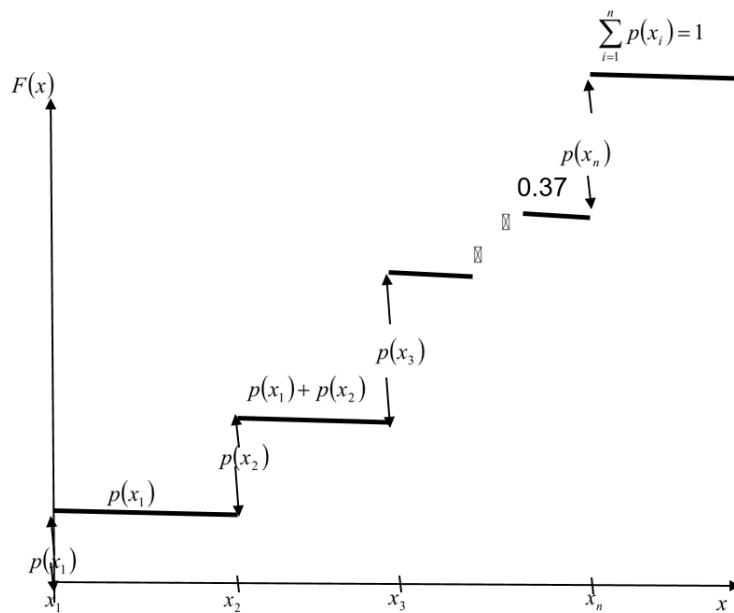


Figure 3.2: Figure 2 Cumulative Distribution Function (Discrete Case)

Example 32

A fair coin is tossed three times. Let X represent the number of heads which come up

- find the cumulative distribution function and
- Sketch the graph.

Solution

(a). To obtain the cumulative distribution function, we need the following steps:

Step 1:

Find the probability distribution of the random variable X . The probability distribution of this example has been found in an earlier example and produces the results here for convenience.

x_i	0	1	2	3
$p(x_i)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

Step 2:

Find the cumulative distribution function:

$$\begin{aligned}
 F(0) &= p(X \leq 0) = p(0 \leq X < 1) \\
 &= p(X < 0) + p(X = 0) \\
 &= 0 + p(0) \\
 &= \frac{1}{8}
 \end{aligned}$$

$$\begin{aligned}
 F(1) &= p(X \leq 1) = p(0 \leq X < 2) \\
 &= p(X < 0) + p(X = 0) + p(X = 1) \\
 &= 0 + p(0) + p(1) \\
 &= 0 + \frac{1}{8} + \frac{3}{8} \\
 &= \frac{4}{8}
 \end{aligned}$$

$$\begin{aligned}
 F(2) &= p(X \leq 2) = p(0 \leq X < 3) \\
 &= p(X < 0) + p(X = 0) + p(X = 1) + p(X = 2) \\
 &= 0 + p(0) + p(1) + p(2) \\
 &= 0 + \frac{1}{8} + \frac{3}{8} + \frac{3}{8} \\
 &= \frac{7}{8}
 \end{aligned}$$

$$\begin{aligned}
 F(3) &= p(X \leq 3) = p(0 \leq X \leq 3) \\
 &= p(X < 0) + p(X = 0) + p(X = 1) + p(X = 2) + p(X = 3) \\
 &= 0 + p(0) + p(1) + p(2) + p(3)
 \end{aligned}$$

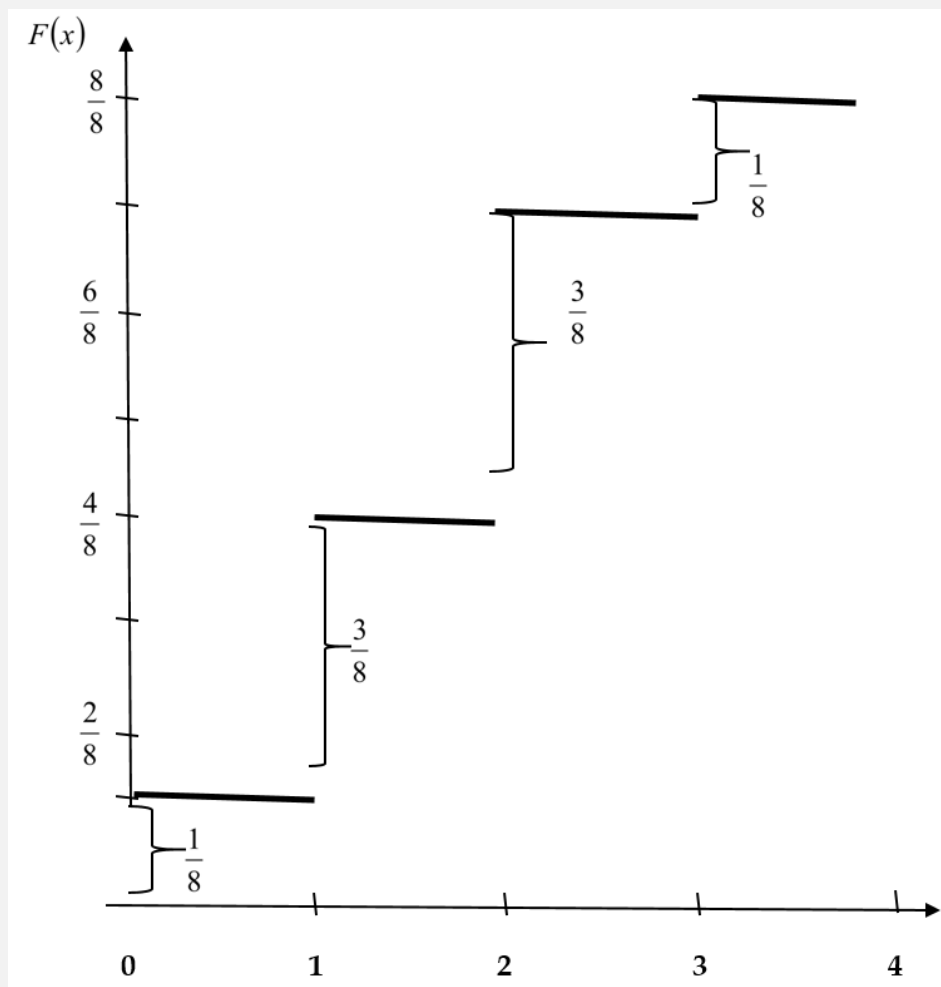
$$= 0 + \frac{1}{8} + \frac{3}{8} + \frac{3}{8} + \frac{1}{8}$$

$$= 1$$

Hence the cumulative distribution function is

$$F(x) = \begin{cases} 0 & x < 0 \\ \frac{1}{8} & 0 \leq x < 1 \\ \frac{4}{8} & 1 \leq x < 2 \\ \frac{7}{8} & 2 \leq x < 3 \\ 1 & x \geq 3 \end{cases}$$

(b), The graph of $F(x)$ is shown in the figure below:



Finding the probability distribution from the cumulative distribution function is a straightforward situation. If $F(x)$ is the cumulative distribution function of a discrete random variable X , we then find the points at which the cumulative distribution function jumps, and the jump sizes. The probability function has masses exactly at those jump points, with the probability masses being equal in magnitude to the respective jump sizes. It is for this reason that it is called probability mass function.

Example 33

Suppose you are given the cumulative distribution function below:

$$F(x) = \begin{cases} 0 & x < 0 \\ \frac{1}{8} & 0 \leq x < 1 \\ \frac{4}{8} & 1 \leq x < 2 \\ \frac{7}{8} & 2 \leq x < 3 \\ 1 & x \geq 3 \end{cases}$$

Find its probability distribution.

Solution

It would be noted from the graph of the cumulative distribution function that the magnitudes or the heights (that is, $p(x_i)$) of the jumps (steps) at 0, 1, 2, 3 are $\frac{1}{8}, \frac{3}{8}, \frac{3}{8}, \frac{1}{8}$ respectively, hence

$$p(x) = \begin{cases} \frac{1}{8} & x = 0 \\ \frac{4}{8} & x = 1 \\ \frac{3}{8} & x = 2 \\ \frac{1}{8} & x = 3 \end{cases}$$

Note:

We can obtain this result without the graph by finding the difference in the adjacent values of $F(x)$.

3.3.2 Cumulative distribution function of Continuous Random Variables

Again, the statement of the theorem below for a distribution function of a continuous random variable follows trivially from the definition of cumulative distribution function.

Theorem (Distribution function of a continuous random variable): Let X be a continuous random variable with probability distribution function $f(x)$, Then the cumulative distribution function $F(x)$ is defined by

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt \quad \text{for } -\infty < x < \infty$$

The typical graph of the cumulative distribution function of the continuous random variable is as shown below.

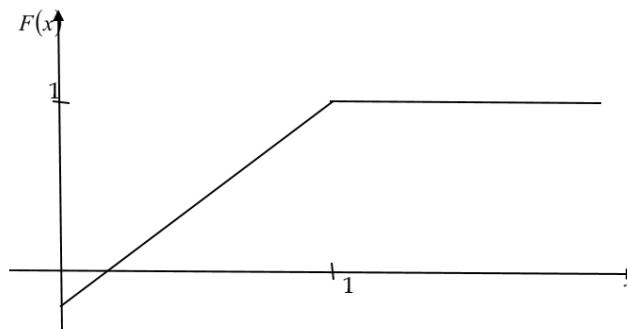


Figure 3.3: Figure 2 Cumulative Distribution Function (Continuous Case)

The graph of $F(x)$ is continuous. Its slope need not be everywhere continuous, but where this is so, it is equal to the probability density function. Thus, $f(x) = \frac{d}{dx}F(x)$ if the derivative exists and $P(a < X < b) = F(b) - F(a)$.

Example 34

The probability density function of a continuous random variable X is given by

$$f(x) = \begin{cases} 0 & x < 0 \\ \frac{x}{2} & 0 \leq x \leq 2 \\ 0 & x > 2 \end{cases}$$

Find the cumulative distribution function and sketch its graph.

Solution

If $x < 0$, then

$$F(x) = \int_{-\infty}^x f(t) dt = 0$$

If $0 < x < 2$, then

$$\begin{aligned} F(x) &= \int_{-\infty}^0 f(t) dt + \int_0^x f(t) dt \\ &= 0 + \int_0^x \frac{t}{2} dt = 0 + \int_0^x \frac{t}{2} dt \\ &= 0 + \left. \frac{t^2}{4} \right|_0^x = \frac{x^2}{4} \end{aligned}$$

If $x > 2$, then

$$\begin{aligned} F(x) &= \int_{-\infty}^0 f(t) dt + \int_0^2 f(t) dt + \int_2^x f(t) dt \\ &= 0 + \int_0^2 \frac{t}{2} dt + \int_2^x f(t) dt \\ &= 0 + \int_0^2 \frac{t}{2} dt + \int_2^x f(t) dt \\ &= 0 + \left. \frac{t^2}{4} \right|_0^2 + 0 = 1 \end{aligned}$$

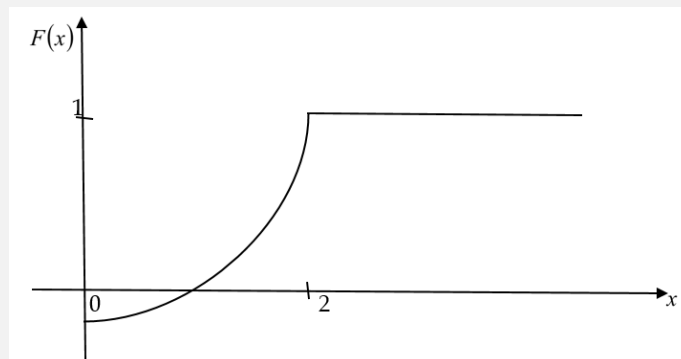


Figure 2 Cumulative Distribution Function (Continuous Case)

Chapter 4

Special Probability Distributions

Learning Objectives

- Understand the assumptions for each of the discrete and continuous probability distributions presented.
- Select an appropriate discrete and continuous probability distribution to calculate probabilities in specific applications.
- Calculate probabilities, determine means and variances for each of the discrete and continuous probability distributions presented.

4.0 Special Probability Distributions

4.1 Introduction

In general, it is useful to present the probability distribution of a random variable by a model. Probability calculations can then be made convenient by substituting appropriate values into the algebraic model.

In this section, we shall *briefly* define certain special distributions that are widely used in applications of probability and statistics. Distributions that would be considered include both discrete and continuous univariate distributions.

4.2 Discrete Probability Distribution

4.1.1 Bernoulli Distribution:

A single trial of an experiment may result in one of the two mutually exclusive outcomes such as defective and non-defective, dead or alive, yes or no, male or female, etc. Such a trial is called Bernoulli trial and a sequence of these trials form a Bernoulli process, satisfying the following conditions:

- (i). Each trial results in one of the two mutually exclusive outcomes, *success* and *failure*.
- (ii). The probability of a success, p remains constant, from trial to trial. The probability of failure is denoted by $q = 1 - p$.
- (iii). The trials are independent. That is, the outcome of any particular trial is not affected by the outcome of any other trial.

Definition:

A random variable, is said to have a Bernoulli distribution if it assumes the values 0 and 1 for the two outcomes. The probability distribution for the success in the trial is defined by

$$P(x) = p^x(1 - p)^{1-x}, \quad x = 0 \text{ or } 1 \text{ and } 0 < p < 1$$

where the mean and variance of the distribution are as follows:

$$\mu = E(X) = p;$$

$$\sigma^2 = \text{Var}(X) = p(1 - p)$$

An important distribution arising from counting the number of successes in a fixed number of independent Bernoulli trials is the Binomial distribution.

Example 35

An urn contains 5 red and 15 green balls. Draw one ball at random from the urn. Let $X=1$ if the ball drawn is red, and $X=0$ if a green ball is drawn. Obtain

- (i) the p.d.f. of X ,
- (ii) mean of X and
- (iii) variance of X .

Solution

The p.d.f. of a Bernoulli distribution is $f(x) = p^x q^{1-x}$, $x = 0, 1$ where

$$p = \frac{5}{20}$$

and

$$q = \frac{15}{20}$$

$$f(x) = \left(\frac{5}{20}\right)^x \left(\frac{15}{20}\right)^{1-x}, \quad x = 0, 1$$

Mean of

$$\bar{X} = E(X) = \sum_{x=0}^1 x \left(\frac{5}{20}\right)^x \left(\frac{15}{20}\right)^{1-x} = (0) \left(\frac{5}{20}\right)^0 \left(\frac{15}{20}\right)^1 + (1) \left(\frac{5}{20}\right)^1 \left(\frac{15}{20}\right)^0$$

$$= \left(\frac{5}{20}\right)$$

Variance of X =

$$V(X) = \sum_0 x^2 f(x) - [E(X)]^2 = (1) \left(\frac{5}{20}\right)^1 \left(\frac{15}{20}\right)^0 - \left(\frac{5}{20}\right)^2$$

$$= \left(\frac{5}{20}\right) - \left(\frac{5}{20}\right)^2 = \left(\frac{3}{16}\right)$$

4.1.2 The Binomial Distribution

The binomial distribution is a discrete probability distribution, where the experiment is repeated n times under identical conditions and each of the n trials is independent of each other which results in one of the two outcomes.

Thus, in the event of independent trials (often called Bernoulli trials) let p be the probability that an event will happen (success) and $q = 1 - p$ the probability that the event will fail in any single trial. Such experiments are called Binomial experiments and the probability that the event will happen exactly x times in n trials is given by the probability function:

$$f(x) = \Pr(X = x) = \binom{n}{x} p^x q^{n-x} = \frac{n!}{x!(n-x)!} p^x q^{n-x}$$

where the random variable denotes the number of success in n trials and $x = 0, 1, 2, 3, \dots, n$.

The shape of the distribution depends on the two parameters n and p .

- (i) when $p < 0.5$ and n is small, the distribution will be skewed to the right.
- (ii) when $p > 0.5$ and n is small, the distribution will be skewed to the left
- (iii) when $p = 0.5$ the distribution will be symmetric.
- (iv) In all cases, as n gets larger the distribution gets closer to being a symmetric, bell-shaped distribution.

Properties

Mean $\mu = np$

Variance $\sigma^2 = npq$

Standard Deviation $\sigma = \sqrt{npq}$

Example 36

If 20% of the bolts produced by a machine are bad. Determine the probability that out of 4 bolts chosen at random.

- (i) one is defective
- (ii) none is defective
- (iii) at most 2 bolts will be defective.

Solution

$$n = 4, p = 0.2, q = 0.8$$

$$P[X = 1] = f(1) = \binom{4}{1} 0.2^1 0.8^3 = 0.4096$$

$$P[X = 0] = f(0) = \binom{4}{0} 0.2^0 0.8^4 = 0.4096$$

$$P[X \leq 2] = P[X = 0, 1, 2] = P[X = 0] + P[X = 1] + P[X = 2]$$

$$0.4096 + 0.4096 + 0.1536 = 0.9728$$

$$\text{or } = 1 - P[X > 2] = 1 - P(X = 3) - P(X = 4)$$

$$\text{or } 1 - P[X \geq 3] =$$

$$1 - \binom{4}{3} 0.2^3 0.8^1 - \binom{4}{4} 0.2^4 0.8^0$$

$$= 1 - 0.0256 - 0.0016$$

$$= 0.9728$$

Example 37

(a) Suppose that it is known that 30% of a certain population is immune to some disease. If a random sample of 10 is selected from this population. What is the probability that it will contain exactly 4 immune persons?

$$n = 10, p = 0.3, x = 4$$

$$f(4) = \binom{10}{4} (0.3)^4 (0.7)^6$$

$$= 0.2$$

(b) In a certain population 10% of the population is color-blind. If a random sample of 25 people is drawn from this population (use table). Find the probability that

i. $P(X \geq 5) = 1 - P(X < 5) = 0.0980$

ii. $P(X \leq 4) = 0.902$ or $1 - P(X \geq 5) = 1 - 0.0980 = 0.902$

iii. $P(6 \leq X \leq 10) = p(6) + p(7) + p(8) + \dots + p(10) = 0.0333$

$$= 0.0334$$

Example 38

From the experiment “toss four coins and count the number of tails” what is the variance of X?

$n = 4$, $p =$, $q =$

$V(X) = npq = 4 \times \dots = 1$

Example 39

Roll a fair 6 – sided die 20 times and count the number of times that 6 shows up. What is the standard development of your random variable?

$n = 20$, $p =$

$$\frac{1}{6}$$

$q =$

$$\frac{5}{6}$$

$$V(X) = npq$$

$=$

$$20 \times \frac{1}{6} \times \frac{5}{6} = \frac{100}{36}$$

$\sigma =$

$$\sqrt{V(X)} = \sqrt{\frac{100}{36}} = \frac{10}{6}$$

Example 40

The following data are the number of seeds germinating out of 10 on damp filter paper for 80 sets of seeds. Fit a binomial distribution to these data.

x	0	1	2	3	4	5	6	7	8	9	10	Total
f	6	20	28	12	8	6	0	0	0	0	0	80

Solution

Here $n = 10$, $N = 80$ and $\sum f_i = 80$

Arithmetic mean =

$$\frac{\sum f_i x_i}{\sum f_i} = \frac{174}{80}$$

$np =$

$$\frac{174}{80} \therefore p = \frac{174}{80n} = \frac{174}{800} = 0.2175, q = 1 - p = 0.7825$$

Hence the binomial distribution to be fitted is $b(x; 10, 0.2175)$. These are approximately,

x	0	1	2	3	4	5	6	7	8	9	10	Total
f	6.89	19.14	23.94	17.74	8.63	2.88	0.67	0.1	0.01	0.00	0.00	80

4.1.3 Negative Binomial Distribution (Pascal's Distribution)

Let us consider an experiment in which the properties are the same as those listed for a binomial experiment with the exception that the trials will be repeated until a fixed number of successes occur. Therefore, instead of finding the probability of x successes in n trials, where n is fixed, we are now interested in the probability that the... k^{th} success occurs on the x^{th} trial. Experiments of this kind are called 'negative binomial experiments'. (Walpole and Myres, 1993). The number X of trials to produce k successes in a negative binomial experiment is called a "negative binomial random variable" and its probability distribution is called the "negative binomial distribution".

Since its probabilities depend on the number of successes desired and the probability of success on a given trial, we shall denote them by the symbol $b^*(x; k, p)$. For the general formula $b^*(x; k, p)$, consider the probability of a success on the trial preceded by $k-1$ successes and $x-k$ failures in some specified order. The probability for the specified order ending in success is $p^{k-1}q^{x-k}p = p^kq^{x-k}$. The total number of sample points in the experiment ending in success, after the occurrence of $k-1$ successes and $x-k$ failures in any order is equal to the number of partitions of $x-1$ trials into two groups with $k-1$ successes corresponding to one group and $x-k$ failures corresponding to the other group. This number is given by the term $\binom{x-1}{k-1}$, each mutually exclusive and occurring with equal probability p^kq^{x-k} . We obtain the general formula by multiplying p^kq^{x-k} by $\binom{x-1}{k-1}$.

In other words:

$$b^*(x; k, p) = \binom{x-1}{k-1} p^{k-1} q^{x-k} p = \binom{x-1}{k-1} p^k q^{x-k} \quad x = k, k+1, \dots$$

p = probability of success

$q = (1 - p)$ = probability of failure

x = total number of trials on which the k^{th} success occurs.

Areas of application of negative binomial distribution include many biological situations such as death of insects, number of insect bites per fruit (e.g. mango).

Example 41

Consider an exploration company that is determined to discover two new fields in a virgin basin it is prospecting, and will drill as many holes as required to achieve its goal. We can investigate the probability that it will require 2, 3, 4, ..., n exploratory holes before two discoveries are made. The same conditions that govern the binomial distribution may be assumed, except that the number of trials is not fixed.

Solution

x dry holes will be drilled before r discoveries are made.

$$\begin{aligned} P &= \binom{r+x-1}{x} (1-p)^x p^r \\ &= \frac{(r+x-1)!}{(r-1)! x!} (1-p)^x p^r \end{aligned}$$

If the regional success ratio is assumed to be 10% then the probability that a two-hole program

will meet the company's goal of two discoveries will be:

$$\begin{aligned} P &= \frac{(2+0-1)!}{(2-1)!0!}(1-0.1)^0(0.1)^2 \\ &= \frac{1!}{1!0!} \times 0.9^0 \times 0.1^2 \\ &= 1 \times 1 \times 0.01 = 0.01 \end{aligned}$$

The probability that five holes will be required to achieve two successes is:

$$p = \frac{(2+3-1)!}{(2-1)!3!}(1-0.1)^3(0.1)^2$$

=

$$\frac{24}{1 \times 6} \times 0.729 \times 0.01 = 0.029$$

or

$$p \binom{x-1}{k-1} p^k q^{x-k} = \binom{4}{1} (0.1)^2 (0.9)^3 = \frac{4!}{(4-1)!1!} 0.01 \times 0.729 = 0.029$$

Example 42

Find the probability that a person tossing three coins will get either all heads or all tails for the second time in the fifth toss?

Solution

$$x = 5, k = 2, p = \left(\frac{1}{8} + \frac{1}{8} \right) = \frac{1}{4}$$

$$b^*(5; 2, \frac{1}{4}) = \binom{4}{1} \left(\frac{1}{4} \right)^2 \left(\frac{3}{4} \right)^3 = \frac{27}{256}$$

The negative binomial distribution derives its name from the fact that each term in the expansion of $p^k(1-p)^{x-k}$ corresponds to the values of $b^*(x; k, p)$ for $x = k, k+1, k+2, \dots$

4.1.4 Geometric Distribution

The geometric distribution is a special case of the negative binomial distribution for which $k = 1$. This is the probability distribution for the number of trials required for a single success.

Thus:

$$B^*(x; 1, p) = p q^{x-1}$$

The geometric distribution is denoted by:

$$g(x, p) = p q^{x-1}$$

Example 43

In a certain theodolite manufacturing process, it is known that on the average, 1 in every 100 is defective. What is the probability that the fifth item inspected is the first defective theodolite found?

Solution

Using the geometric distribution with $x = 5$; $p = 0.01$, we have:

$$g(5, 0.01) = p(1 - p)^{5-1} = 0.01 \times 0.99^4 = 0.0096$$

For geometric distribution:

$$\mu = \frac{1}{p} = 100; \quad \sigma^2 = \frac{1-p}{p^2} = 9900$$

Example 44

Consider the case in **Example 4.4**, what is the probability that the first discovery will be the fifth exploration hole?

Solution

$$n = 5; p = 0.1$$

$$\therefore g(5, 0.1) = p(1 - 0.1)^{5-1} = 0.1 \times 0.9^4 = 0.0656$$

4.1.5 Poisson Distribution**Introduction:**

Experiments yielding numerical values of a random variable (x), the number of successes occurring during a given time interval or in a specified region, are often called poisson experiments. The given time interval may be of any length, such as a minute, a day, a week, a month or even a year. Hence, a poisson experiment might generate observations for the random variable representing the number of telephone calls per hour received by an office, the number of days school is closed due to snow during the winter, or the number of postponed games due to rain during a basketball season. The specified region could be a line segment, an area, a volume or perhaps a material. In this case, might represent the number of field mice per acre, the number of bacteria in a given culture, or the number of typing errors per page.

The Poisson process:

A poisson experiment is derived from the Poisson process and possesses the following properties:

- The number of successes occurring in one time interval or specified region are independent of those occurring in any other disjoint time interval or region of space.
- The probability of a single success occurring during a very short time interval or in a small region is proportional to the length of the time interval or the size of the region and does not depend on the number of successes occurring outside this time interval or region.
- The probability of more than one success occurring in such a short time interval or falling in such a small region is negligible.

The probability distribution of the Poisson random variable is called the Poisson distribution and is denoted by $P(x; \mu)$ since its values depend only on μ , the average number of successes occurring in the given time interval or specified region. This formula is given by the definition below:

Definition: The probability distribution of the Poisson random variable, representing the number of successes occurring in a given time interval or specified region is given by:

$$P(x; \mu) = \frac{e^{-\mu} \mu^x}{x!}; \quad x = 0, 1, 2, 3, \dots,$$

Where μ is the average number of successes occurring in the given time interval or specified region and $e = 2.71828$.

Theorem: The mean and variance of the Poisson distribution $P(x; \mu)$ both have the value λ .

Example 45

Suppose that an urn contains 100,000 marbles and 120 are red. If a random sample of 1000 is drawn what are the probabilities that 0, 1, 2, 3, and 4 respectively will be red.

$n = 1000$,

$$p = \frac{120}{100000}$$

$= 0.0012$, $q = 0.9988$

Solution

Binomial =

$$\binom{1000}{x} 0.0012^x \times 0.9988^{1000-x}$$

For $x=3$,

$$\binom{1000}{3} 0.0012^3 \times 0.9988^{997} = 0.0867$$

i.e $166167000 \times 1.728^{-09} \times 0.30206$

Using the Poisson method,

$$\lambda = np = 1000 \times 0.0012 = 1.2$$

$$e - 1.2 = 0.3012$$

$$f(3) = \frac{e^{-1.2} 1.2^3}{3!} = 0.0867$$

$$\begin{aligned} P(X > 5) &= 1 - p(X \leq 4) \\ &= 1 - 0.9985 = 0.0015 \end{aligned}$$

Example 46

Let X have a Poisson distribution with a mean of $\lambda = 5$. Find

$$P(X < 6)$$

$$P(X > 5)$$

$$P(X = 6)$$

$$P(X > 4)$$

Solution

$$(i) P(X \leq 6) = \sum_{x=0}^6 \frac{5^x e^{-5}}{x!} = 0.762$$

$$(ii) P(X > 5) = 1 - P(X \leq 5) = 1 - 0.616 = 0.384$$

$$(iii) P(X = 6) = P(X \leq 6) - P(X \leq 5) = 0.762 - 0.616 = 0.146$$

$$(iv) P(X \geq 4) = 1 - P(X < 4)$$

Example 47

A hospital administrator, who has been studying daily emergency admissions over a period of several years, has come to the conclusion that they are distributed according to the Poisson law. Hospital records reveal that emergency admissions have averaged three per day during this period. If the administrator is correct in assuming a Poisson distribution. Find the probability that

- (i) exactly two emergency admissions will occur on a given day.
- (ii) No emergency admissions will occur on a particular day.
- (iii) Either 3 or 4 emergency cases will be admitted on a particular day.

Solution

(i)

$$p(X = 2) = \frac{e^{-\lambda} \lambda^x}{x!}$$

$$\lambda = 3$$

=

$$\frac{e^{-3} 3^2}{2!} = \frac{0.05(9)}{2} = 0.225$$

$$(i) \ p(X = 0) = \frac{e^{-3} 3^0}{0!} = 0.05$$

$$(ii) \ p(X = 3) + p(X = 4) = \frac{e^{-3} 3^3}{3!} + \frac{e^{-3} 3^4}{4!}$$

=

$$e^{-3} \left[\frac{27}{6} + \frac{81}{24} \right]$$

=

$$0.05 \left[\frac{9}{2} + \frac{27}{8} \right]$$

$$= 0.05(7.875)$$

$$= 0.394$$

Example 48

Fit a Poisson distribution to the following data which gives the number of yeast cells per square for 400 squares.

No. of cells per square (x)	0	1	2	3	4	5	6	7	8	9	10	Total
No. of squares (f)	103	143	98	42	8	4	2	0	0	0	0	400

Solution

The expected theoretical frequency for r successes is $Ne^{-m}m^r/r!$, but m is not given in this example. The mean of the Poisson distribution is m . Hence

$$m = \frac{\sum fx}{\sum f} = \frac{529}{400} = 1.32$$

$$P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!} = \frac{(1.32)^x e^{-1.32}}{x!}$$

thus,

$$f = 400 \frac{e^{-1.32}(1.32)^x}{x!}$$

, therefore,

No. of cells per square (x)	0	1	2	3	4	5	6	7	8	9	10	Total
No. of squares (f)	107	141	93	41	14	4	0	0	0	0	0	400

4.1.5 Relationship between the Binomial and Poisson distributions

In the binomial distribution, if n is large while the probability p of occurrence of an event is close to zero, so that $q = 1 - p$ is close to one, the event is called a '**rare event**'. In practice, we shall consider an event as rare if the number of trials is at least 50 ($n > 50$) while np is less than 5. For such cases the Poisson distribution very closely approximates the binomial distribution $f(x) = \frac{\lambda^x e^{-\lambda}}{x!}$ with $\lambda = np$ (Spiegel, 1980). Thus the Poisson distribution is a limiting case of the Binomial distribution when $n \rightarrow \infty$, $p \rightarrow 0$ and np ($n = \mu$) remain constant (Davis, 1986).

Example 49

In a manufacturing process in which glass is being produced, defects or bubbles occur, occasionally rendering the pieces undesirable for marketing. If it is known that on the average 1 in every 1000 of these items produced have one or more bubbles. What is the probability that a random sample of 8000 will yield fewer than 7 items possessing bubbles?

Solution

This is essentially a binomial experiment with $n = 8000$ and $p = 0.001$. Since p is very close to zero and n is quite large, we shall approximate with the Poisson distribution, using $\lambda = (8000)(0.001) = 8$. Hence, if X represents the number of bubbles, we have:

$$\begin{aligned} P(X < 7) &= \sum_{x=0}^6 b(x; 8000, 0.001) \\ &= \sum_{x=0}^6 p(x; 8) \\ &= 0.3134 \end{aligned}$$

Post-Test

1. Suppose that 24% of a certain population have blood group B, for a sample of size 20 drawn

from this population, find the probability that

2. Exactly 3 persons with blood group B will be found.
3. Three or more persons $\equiv$ the characteristics of interest will be found
4. Fewer than three will be found.
5. Exactly five will be found.
6. In a large population, 16% of the members are left-handed. In a random sample of size 10, find
7. The probability that exactly 2 will be left-handed $p(X = 2)$

$$P(X \geq 2)$$

$$P(X < 2)$$

$$P(1 \leq X \leq 4)$$

1. Suppose mortality rate of a certain disease is 0.1, suppose 10 people in a community contract the disease, what is the probability that
2. None will survive
3. 50% will be
4. At least 3 will die
5. Exactly 3 will die
6. Suppose it is known that the probability of recovery from a certain disease is 0.4. If 15 people are stricken with the disease what is the probability that
7. or more will recover?
8. 4 or more will recover?
9. at least 5 will recover?
10. fewer than three recover?
11. In the study of a certain aquatic organism, a large number of samples were taken from a pond, and the number of organisms in each sample was counted. The average number of organisms per sample was found to be two. Assuming the number of organisms to be Poisson distributed. Find the probability that:
12. The next sample taken will contain one or more organisms.
13. The next sample taken will contain exactly three organisms.
14. The next sample taken will contain fewer than five organisms.
15. It has been observed that the number of particles emitted by a radioactive substance, which reach a given portion of space during time t , follows closely the Poisson distribution with parameter $\lambda=100$. Calculate the probability that:
16. No particles reach the portion of space under consideration during time t ;
17. Exactly 120 particles do so;
18. At least 50 particles do so.

19. The phone calls arriving at a given telephone exchange within one minute follow the Poisson distribution with parameter value equal to ten. What is the probability that in a given minute:
20. No calls arrive?
21. Exactly 10 calls arrive?
22. At least 10 calls arrive

4.3 Continuous Probability Distribution

4.2.1 The Normal Distribution:

The graph of the normal distribution which is a bell-shaped smooth curve approximately describes many phenomena that occur in nature, industry and research. In addition, errors in scientific measurements are extremely well approximated by a normal distribution. Thus, the normal distribution is one of the most widely used probability distributions for modelling random experiments. It provides a good model for continuous random variables involving measurements such as time, heights/weights of persons, marks scored in an examination, amount of rainfall, growth rate and many other scientific measurements.

Definition:

The probability density function for the normal random variable X which is simply called normal distribution is defined by:

$$f(x) = \begin{cases} \frac{1}{\sqrt{2\pi}\sigma^2} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, & -\infty < x < \infty \\ 0, & \text{elsewhere} \end{cases}$$

Where $\sigma > 0$, $-\infty < x < \infty$ and the mean and variance of the measurements x , are $E(x) = \mu$ and $\text{Var}(x) = \sigma^2$.

If the random variable is modelled by the normal distribution with mean, μ and variance, σ^2 then it is simply denoted as $x \sim N(\mu, \sigma)$.

This fact considerably simplifies the calculations of probabilities concerning normally distributed variables, as seen in the following illustration:

Suppose that X is $N(\mu, \sigma^2)$, let $c_1 < c_2$, and since $P(X = c_1) = 0$, then

$$\begin{aligned} P(c_1 < X < c_2) &= P(X < c_2) - P(X < c_1) \\ &= P\left(\frac{X - \mu}{\sigma} < \frac{c_2 - \mu}{\sigma}\right) - P\left(\frac{X - \mu}{\sigma} < \frac{c_1 - \mu}{\sigma}\right) \\ &= \Phi\left(\frac{c_2 - \mu}{\sigma}\right) - \Phi\left(\frac{c_1 - \mu}{\sigma}\right) \end{aligned}$$

Note that $\Phi(-x) = 1 - \Phi(x)$.

The normal distribution possesses the following properties.

Properties

Mean $E(X) = \mu$

Variance $\text{Var}(X) = \sigma^2$

Standard Deviation = σ

Example 50

1. If Z is $N(0, 1)$, find;
2. $P(0.53 < Z < 2.06)$
3. $P(Z > 2.89)$
4. If X is $N(75, 100)$, find
5. $P(X < 60)$
6. $P(6 \leq X \leq 12)$
7. If X is normally distributed with a mean of 6 and a variance 25, find $P(6 \leq X \leq 12)$

Solution

1. (i) $P(0.53 < Z < 2.06) = \Phi(2.06) - \Phi(0.53) = 0.9803 - 0.7019 = 0.2784$
- (ii) $P(Z > 2.89) = 1 - \Phi(2.89) = 1 - 0.9981 = 0.0019$
- (2)

$$P(X < 60) = P\left(\frac{X - 75}{10} < \frac{60 - 75}{10}\right) = P(Z < -1.5) = 0.0668$$

$$P(6 \leq X \leq 12) = P\left(\frac{6 - 6}{5} \leq Z \leq \frac{12 - 6}{5}\right) = P(0 \leq Z \leq 1.2)$$

$$= \Phi(1.2) - \Phi(0) = 0.8849 - 0.5000 = 0.3849$$

4.2.2 Normal approximation to the Binomial

Whenever the number of trials in a binomial experiment is small it is easy to find the probabilities associated with the experiment quickly as calculations become laborious. The normal distribution is often a good approximation to a discrete distribution when the latter takes on a symmetrical bell shape.

If X is a binomial random variable with mean $\mu = np$ and variance $\sigma^2 = npq$, then the limiting form of the distribution of:

$$Z = \frac{X - np}{\sqrt{npq}} \quad \text{as } n \rightarrow \infty$$

is the standard normal distribution $n(Z; 0, 1)$.

It turns out that the normal distribution with $\mu = np$ and $\sigma^2 = npq = np(1 - p)$ not only provides a very accurate approximation to the binomial distribution when n is large and p is not extremely close to 0 or 1, but also provides a fairly good approximation even when n is small and p is reasonably close to.

In fact,

$$\frac{X - np}{\sqrt{npq}}$$

is asymptotically normal.

Example 51

Consider a binomial distribution with p.d.f. of $b(x; 15, 0.4)$. Calculate the probability that x assumes values from 7 to 9 inclusive.

Solution

The exact probability is given by:

$$\begin{aligned}
 P(7 \leq X \leq 9) &= \sum_{x=7}^9 b(x; 15, 0.4) \\
 &= \sum_{x=0}^9 b(x; 15, 0.4) - \sum_{x=0}^6 b(x; 15, 0.4) \\
 &= 0.9662 - 0.6098 \text{ (from tables)} \\
 &= 0.3564
 \end{aligned}$$

Using normal approximation, corresponding values under a histogram:

$$\mu = np = 15 * 0.4 = 6$$

$$\sigma^2 = npq = 15 * 0.4 * 0.6 = 3.6$$

$$\therefore \sigma = \sqrt{3.6} = 1.897$$

(The 0.5 used here is called ‘Continuity Correction Factor’) $z_1 = \frac{6.5-6}{1.897} = 0.26$

$$z_2 = \frac{9.5 - 6}{1.897} = 1.85$$

Now,

$$\begin{aligned}
 P(7 \leq x \leq 9) &= p(0.26 < z < 1.85) \\
 &= p(z < 1.85) - p(z < 0.26) \\
 &= 0.9678 - 0.6026 = 0.3652
 \end{aligned}$$

Note:

The 0.5 that was added to or subtracted from the 9 and 7 respectively is called “continuity correction factor”. It corrects for the fact that we are using a continuous distribution to approximate a discrete distribution. Generally, the degree of accuracy, which depends on how well the curve fits the histogram, will increase as n increases. If both np and nq are greater than or equal to 5, the approximation will be good.

Fig.4.4 shows the areas of the normal distribution representing probabilities within some defined standard deviations.

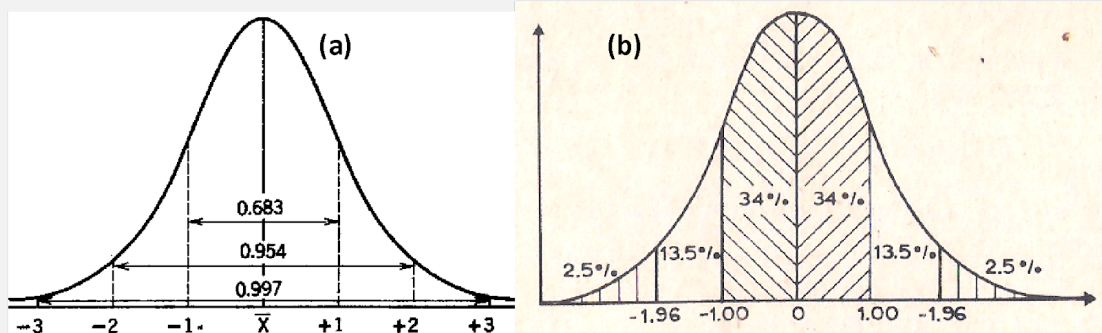


Fig. 4.4 Areas enclosed by successive standard deviation of the standard normal distribution.

4.2.3 Uniform Distribution

Suppose that a continuous random variable X can assume values in a bounded interval only, say the open interval (a, b) , and suppose the p.d.f. of X is given as

$$f(x; a, b) = f(x) = \frac{1}{b - a}, \quad a < x < b$$

= 0, elsewhere.

This distribution is referred to as the Uniform or Rectangular Distribution on the interval (a, b) and is simply written as $X \sim U(a, b)$, where 'a' and 'b' are the parameters of the distribution. It provides a probability model for selecting a point at random from the interval (a, b) .

Properties

Mean

$$\mu = \frac{a + b}{2}$$

Variance

$$\sigma^2 = \frac{(b - a)^2}{12}$$

Standard Deviation $\sigma =$

$$\sqrt{\frac{(b - a)^2}{12}}$$

Example 52

:

The hardness of a certain alloy (measured on Rockwell scale) is a random variable X . Assume that $X \sim U[50, 75]$.

- (a) Find $P[60 < X < 70]$
- (b) Find $E(X)$
- (c) Find $\text{Var}(X)$

Solution

(a)

$$P[60 < X < 70]$$

=

$$\int_{60}^{70} \frac{1}{b - a} dx$$

=

$$\frac{1}{75 - 50} [x]_{60}^{70}$$

=

$$\frac{2}{5}$$

(a) $E(X) =$

$$\frac{1}{b - a} \int_{50}^{75} x dx$$

=

$$\frac{125}{2}$$

Or

 $E(X) =$

$$\frac{b+a}{2} = \frac{75+50}{2} = \frac{125}{2}$$

(a)

$$\text{Var}(X) = E(X^2) - E^2(X) = \frac{1}{25} \int_{50}^{75} x^2 dx - \left(\frac{125}{2}\right)^2 = \frac{625}{12}$$

Or

$$\text{Var}(X) = \frac{(b-a)^2}{12} = \frac{625}{12}$$

4.2.4 The Gamma Distribution:

The Gamma distribution arises in the study of waiting times, for example, in the lifetime of devices. It is also useful in modeling many nonnegative continuous variables. The gamma distribution requires the knowledge of the gamma function.

Definition: The gamma function is defined by

$$\Gamma(\alpha) = \int_0^{\infty} x^{\alpha-1} e^{-x} dx$$

for

$$\alpha > 0$$

and for all $x > 0$

$\Gamma(\alpha)$ is read as “gamma function of α ”

Integrating by parts with $u = x^{\alpha-1}$ and $dv = e^{-x} dx$, we obtain

$$\begin{aligned} \Gamma(\alpha) &= -e^{-x} x^{\alpha-1} \Big|_0^{\infty} + \int_0^{\infty} (\alpha-1) x^{\alpha-2} e^{-x} dx \\ &= (\alpha-1) \int_0^{\infty} x^{\alpha-2} e^{-x} dx \end{aligned}$$

Which yields the recursion formula

$$\Gamma(\alpha) = (\alpha-1)\Gamma(\alpha-1).$$

Repeated application of the recursion formula gives

$$\begin{aligned} \Gamma(\alpha) &= (\alpha-1)(\alpha-2)\Gamma(\alpha-2) \\ &= (\alpha-1)(\alpha-2)(\alpha-3)\Gamma(\alpha-3) \end{aligned}$$

and so forth. Note that when $\alpha = n$, where n is a positive integer,

$$\Gamma(n) = (n-1)(n-2)(n-3) \dots \Gamma(1) = (n-1)!$$

General Gamma Distribution:

The continuous random variable has a gamma distribution, with parameters α and β if its density function is given by:

$$f(x) = \begin{cases} \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-\frac{x}{\beta}} & x > 0 \\ 0 & \text{elsewhere} \end{cases}$$

where $\alpha > 0$ and $\beta > 0$.

Property of Gamma distribution

- (a) $E(X) = \alpha\beta$
- (b) $\text{Var}(X) = \alpha\beta^2$

4.2.5 Exponential Distribution

An exponential distribution is a special gamma distribution for which $\alpha = 1$. It has many applications in the field of statistics, particularly in the areas of reliability theory and waiting times or queueing problems. It is a continuous distribution that can be related to the Poisson distribution in the discrete sense.

Definition: A continuous random variable has an exponential distribution with parameter β if its density function is given by:

$$f(x) = \begin{cases} \frac{1}{\beta} e^{-\frac{x}{\beta}} & x > 0 \\ 0 & \text{elsewhere} \end{cases}$$

where $\beta > 0$.

Properties

Mean $E(X) = \theta$

Variance $\text{Var}(X) = \theta^2$

Standard Deviation $\sigma = \theta$

So if λ is the mean of changes in the unit interval, then $\theta = \frac{1}{\lambda}$ is the mean waiting time for the first change.

Example 53

Let the p.d.f. of X be $f(x) = \left(\frac{1}{2}\right) e^{-x/2}$, $0 \leq x < \infty$.

- (i) What is the mean and variance of X ?
- (ii) Calculate $P(X > 3)$
- (iii) Calculate $P(X > 5 | X > 2)$
- (iv) Calculate $P(X < 2)$

Solution

- (i) $E(X) = \theta = 2$ and $\text{Var}(X) = \theta^2 = 4$

- (ii)

$$P(X > 3) = \frac{1}{2} \int_3^\infty e^{-\frac{x}{2}} dx = e^{-\frac{3}{2}} = 0.2231$$

(iii)

$$P(X > 5 | X > 2) = \frac{\frac{1}{2} \int_5^\infty e^{-\frac{x}{2}}}{\frac{1}{2} \int_2^\infty e^{-\frac{x}{2}}} = \frac{e^{-\frac{5}{2}}}{e^{-\frac{2}{2}}} = e^{-\frac{3}{2}} = 0.2231$$

(iv)

$$P(X < 2) = \frac{1}{2} \int_0^2 e^{-\frac{x}{2}} = 1 - e^{-\frac{2}{2}} = 1 - e^{-1} = 0.6321$$

NB. Discuss other discrete and continuous distributions with students by stating their properties respectively.

Post-Test

1. Let X have an exponential distribution with a mean of $\theta = 20$. Compute
2. $P(10 < X < 30)$
3. $P(0 < X < 30)$
4. $P(X > 30)$
5. $P(X > 40 | X > 10)$
6. Telephone calls enter a college switchboard according to a Poisson process on the

average of two every 3 minutes. Let X denote the waiting time until the first call that arrives after 10 A.M.

- (i) What is the p.d.f. of X?
- (ii) Find $P(X > 2)$
- (iii) Customers arrive randomly at a bank teller's window. Given that one customer arrived during a particular 10-minute period, let X equal the time within the 10 minutes that the customer arrived. If X is $U(0, 10)$, find
- (iv) The p.d.f. of X.
- (v) $P(X \geq 8)$
- (vi) $P(2 \leq X < 8)$
- (vii) $E(X)$
- (viii) $\text{Var}(X)$
- (ix) Explain the relationship that exists between the Poisson and the Exponential distributions.
- (x) If X is $N(75, 100)$, find $P(X < 35)$ and $P(70 < X < 100)$.
- (xi) If Z is $N(0, 1)$, find values of c such that
- (xii) $P(Z \geq c) = 0.025$
- (xiii) $P(|Z| \leq c) = 0.95$
- (xiv) $P(Z > c) = 0.05$
- (xv) Let X be $N(\mu, \sigma^2)$, so that $P(X < 89) = 0.90$ and $P(X < 94) = 0.95$. Find μ and σ^2 .
- (xvi) Show that the random variable $Z = (X - \mu)/\sigma$ is distributed $N(0, 1)$.
- (xvii) Suppose that $Z \sim N(0, 1)$. Find the following probabilities:
- (xviii) $P(Z \leq 1.53)$

(xix) $P(Z > -0.48)$

(xx) $P(0.35 < Z < 2.01)$

(xxi) $P(|Z| > 1.28)$

(xxii) Find the value of 'a' and 'b' such that

(xxiii) $P(Z \leq a) = 0.648$

(xxiv) $P(|Z| \leq b) = 0.95$

Chapter 5

Mathematical Expectation

Learning Objectives

- Understand and compute the expected values of random variables using engineering data.

5.0 Mathematical Expectation

5.1 Introduction

A very important concept in probability and statistics is that of mathematical expectation, expected value or briefly expectation of a random variable. The expectation of X is very often called the mean of X and is denoted by μ_x or simply μ when a particular random variable is understood. This expected value of x gives a simple value, which acts as a representative, or average of the value of x and for this reason it is often called a measure of central tendency.

Consider that the random variable x has the values x_1, x_2, x_n and $f(x_1), f(x_2), \dots, f(x_n)$ as the probabilities of occurrence. The mean or expected value of x is:

$$\mu = E(X) = \sum x f(x) \quad \text{if } X \text{ is discrete}$$

and

$$\mu = E(X) = \int_{-\infty}^{\infty} x f(x) dx \quad \text{if } X \text{ is continuous}$$

It is said that the expectation $E(X)$ exist for a continuous distribution if and only if the integral is absolutely convergent, that is if and only if $\int_{-\infty}^{\infty} |x| f(x) dx < \infty$

Example 54 (Spiegel, 1980)

Suppose that a game is played with a single die assumed fair. In this game a player wins \$20 if a 2 turns up, \$40 if a 4 turns up; loses \$30 if a 6 turns up; while he neither wins nor loses if any other face turns up. Find the expected sum of money won.

Solution

Table 5.1

I	1	2	3	4	5	6
X_i	0	20	0	40	0	-30
$f(x_i)$	1/6	1/6	1/6	1/6	1/6	1/6

The probability function of x is displayed Table 7.1 Thus the expected value is:

$$\sum x_i f(x_i) = (0) \left(\frac{1}{6} \right) + (20) \left(\frac{1}{6} \right) + (0) \left(\frac{1}{6} \right) + (40) \left(\frac{1}{6} \right) + (-30) \left(\frac{1}{6} \right) = 5$$

It follows that the player can expect to win 5. In a fair game, therefore he should be expected to pay \$5 in order to play the game.

Example 55

Suppose that the pdf of a random variable X with a continuous contribution is $f(x)=$

$$\begin{cases} 2x & \text{for } 0 < x < 1 \\ 0 & \text{Otherwise} \end{cases}$$

Then

$E(X)=$

$$\int_0^1 xf(x) dx = \int_0^1 x(2x) dx = \left. \frac{2}{3}x^3 \right|_0^1 = \frac{2}{3}$$

5.2 The Expectation of a Function

Let us consider a new random variable $g(x)$, which depends on X , that is, each value of $g(x)$ is determined by knowing the values of x . For instance, $g(x)$ might be x^2 or $3x - 1$, so that whenever X assumes the value 2, $g(X)$ assumes the value $g(2)$. In particular, if X is a discrete random variable with probability distribution $f(x)$, $x=-1, 0, 1, 2$ and $g(X) = X^2$ then

$$P[g(X) = 0] = P(x = 0) = f(0)$$

$$P[g(X) = 1] = p(x = -1) + p(x = 1) = f(-1) + f(1)$$

$$P[g(X) = 4] = p(x = 2) = f(2)$$

So that the probability distribution of $g(x)$ may be written

$g(x)$	0	1	4
$P[g(X)=g(x)]$	$f(0)$	$f(-1)+f(1)$	$f(2)$

By definition of an expected value of a random variable, we obtain

$$\mu_{g(x)} = E[g(x)]$$

$$g(x) = 0f(0) + 1[f(-1) + f(1)] + 4f(2)$$

$$= (-1)^2f(-1) + (0)^2f(0) + (1)^2f(1) + (2)^2f(2)$$

$$= \sum_x g(x)f(x)$$

In the general form, for a random variable X with a probability distribution $f(x)$, the expectation of the random variable $g(X)$ is:

$$\mu_{g(x)} = E[g(X)] = \sum g(x)f(x)$$

If X is discrete, and

$$\mu_{g(x)} = E[g(X)] = \int_{-\infty}^{\infty} g(x)f(x) dx$$

If X is continuous.

Example 56

Suppose that the pdf of a random variable X with a continuous distribution is

$$f(X) = \begin{cases} 2x & \text{for } 0 < x < 1 \\ 0 & \text{Otherwise} \end{cases}$$

Determine the expectations of Y when $Y = X^{1/2}$

Solution

$$E[Y] = \int_{x=0}^1 g(x)f(x) dx = \int_{x=0}^1 x^{\frac{1}{2}}(2x) dx = \frac{4}{5}x^{\frac{5}{2}} \Big|_0^1 = \frac{4}{5}$$

Example 57

Suppose that the number of cars, X_1 that pass through a car wash between 4pm and 5pm on any sunny Friday has the following probability distribution:

X	4	5	6	7	8	9
f(x)	1/12	1/12	1/4	1/4	1/6	1/6

Let $g(x) = 2X-1$ represents the amount of money in the dollars paid to the attendants expected earnings for the particular time period. How much does the attendant expect to earn on any sunny Friday?

Solution

The attendant can expect to receive

$$\begin{aligned} E[g(X)] &= E(2x - 1) = \sum_{x=4}^9 (2x - 1)f(x) \\ &= 7 \left(\frac{1}{12} \right) + 9 \left(\frac{1}{12} \right) + 11 \left(\frac{1}{4} \right) + 13 \left(\frac{1}{4} \right) + 15 \left(\frac{1}{6} \right) + 17 \left(\frac{1}{6} \right) \\ &= \$12.67 \end{aligned}$$

Example 58

Let X be a continuous random variable with density functions f

$$f(x) = \begin{cases} \frac{x^2}{3} & -1 < x < 2 \\ 0 & \text{elsewhere} \end{cases}$$

Find the expected value of $g(X) = 4x + 3$

Solution

$$\begin{aligned} E[g(x)] &= \int g(x)f(x) dx \\ E(4x + 3) &= \int_{-1}^2 \frac{(4x + 3)x^2}{3} dx \\ &= \frac{1}{3} \int_{-1}^2 (4x^3 + 3x^2) dx \end{aligned}$$

$$\begin{aligned}
&= \left. \frac{x^4 + x^3}{3} \right|_{-1}^2 \\
&= \frac{24}{3} = 8
\end{aligned}$$

5.3 Properties of Expectations

Basic Theorems

Suppose that X is a random variable for which the expectation $E(X)$ exists. We shall present several results pertaining to the basic properties of expectations.

Theorem 1

If $Y = ax + b$ where a and b are constants, then $E(Y) = aE(x) + b$.

Proof: First we shall assume, for convenience, that X has a continuous distribution for which the pdf is $f(x)$. Then,

$$E(Y) = E(ax + b) =$$

$$\begin{aligned}
&\int_{-\infty}^{\infty} (ax + b)f(x) dx \\
&= a \int_{-\infty}^{\infty} xf(x) dx + b \int_{-\infty}^{\infty} f(x) dx \\
&= aE(x) + b
\end{aligned}$$

A similar proof can be given for a discrete distribution or for a more general type of distributions

Example 59

Suppose that $E(x) = 5$ then

$$(i) E(3x - 5) = 3E(X) - 5 = 3(5) - 5 = 10$$

$$(ii) E(-3x + 15) = -3E(x) + 15 = -3(5) + 15 = 0$$

Theorem 2

If there exists a constant a such that $P(x \geq a) = 1$, then $E(x) \geq a$. If there exists a constant b such that $P(x \leq b) = 1$, then

$$E(X) \leq b$$

Theorem 3

If $X_1, X_2, \dots, X_n$ are n random variables such that each expectation $E(X_i)$ exists ($i = 1, 2, \dots, n$), then

$$E(X_1 + \dots + X_n) = E(X_1) + \dots + E(X_n)$$

This theorem means that the expectations of the sum of several random variables must be equal to the sum of their individual expectations, regardless of whether or not the random variables are independent.

Follow Theorems 1 and 3:

$$E(a_1x_1 + \dots + a_nx_n + b) = a_1E(x_1) + \dots + a_nE(X_n) + b$$

Theorem 4

$E(\prod_{i=1}^n x_i) = \prod_{i=1}^n E(x_i)$

$$E\left(\prod_{i=1}^n x_i\right) = \prod_{i=1}^n E(x_i)$$

It must be emphasised that the expectations of the product of a group of random variables are NOT ALWAYS equal to the product of their individual expectations. If the random variables are independent, then they are equal.

Example 60

Suppose that X_1 , X_2 , and X_3 are independent random variables such $E(XE(X_i)) = 0$ and $EE(X_i^2) = 1$ for $i = 1, 2, 3$, Determine $EE[x_i^2(x_2 - 4x_3)^2]$

Solution

Since X_1 , X_2 and X_3 are independent, it follows that the two random variables X_1^2 and $(X_2 - 4X_3)^2$ are also independent. Therefore

$$\begin{aligned} EE[X_1^2(X_2 - 4X_3)^2] &= E(X_1^2) E[(X_2 - 4X_3)^2] \\ &= EE(X_2^2 - 8X_2X_3 + 16X_3^2) \\ &= EE(X_2^2) - 8E(X_2X_3) + 16E(X_3^2) \text{ Since } E(X_i^2) = 1 \end{aligned}$$

$$= 1 - 8E(X_2)E(X_3) + 16$$

$$= 1 - 0 + 16 = 17$$

5.4 Expectation of Joint Probability Function

Let X and Y be random variables with joint probability distribution $f(x, y)$. The mean or expected value of the random variable $g(X, Y)$ is

$$\mu_{g(X, Y)} = E[g(X, Y)] = \sum \sum g(x, y) f(x, y) \quad \text{if } X \text{ and } Y \text{ are discrete, and}$$

$$\mu_{g(X, Y)} = E[g(X, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f(x, y) dx dy \quad \text{if } X \text{ and } Y \text{ are continuous.}$$

5.3.1 Marginal Distribution

The marginal distribution of X alone and of Y alone are given by:

$$g(x) = \sum_y f(x, y); \quad h(y) = \sum_x f(x, y) \quad \text{for discrete variables}$$

$$g(x) = \int_y f(x, y) dy; \quad h(y) = \int_x f(x, y) dx \quad \text{for continuous variables}$$

5.3.2 Conditional Distribution

Let X and Y be two random variables, discrete or continuous. The ‘Conditional distribution’ of the random variable X , given that $Y = y$ is given by:

$$f(x|y) = \frac{f(x, y)}{h(y)} \quad h > 0$$

For the joint density function:

$$P(a < X < b | Y = y) =$$

$$\sum_x f(x | y) \quad \text{for discrete variable}$$

=

$$\int_a^b f(x | y) dx \quad \text{for continuous variable}$$

Example 61

The joint density for the random variable (x,y) is given by:

$$f(x, y) = \begin{cases} 10xy^2 & 0 < x < y < 1 \\ 0 & \text{Otherwise} \end{cases}$$

Find the marginal densities g(x) and h(y) and the conditional density f(y|x).

Solution

$$g(x) = \int_{-\infty}^{\infty} f(x, y) dy = \int_x^1 10xy^2 dy = \frac{10xy^3}{3} \Big|_{y=x}^1 = \frac{10}{3}x(1 - x^3)$$

for $0 < x < 1$

$$h(y) = \int_{-\infty}^{\infty} f(x, y) dx = \int_0^y 10xy^2 dx = 5x^2y^2 \Big|_0^y = 5y^4 \quad \text{for } 0 < y < 1$$

Now,

$$f(y|x) = \frac{f(x, y)}{g(x)} = \frac{10xy^2}{\frac{10}{3}x(1 - x^3)} = \frac{3y^2}{(1 - x^3)}, \quad 0 < x < y < 1$$

Example 62

Let X and Y be the random variables with joint probability distribution given by the table below. Find the expected value of $g(X, Y) = XY$

		X			
	f(x,y)	0	1	2	Row total
Y	0		$\frac{9}{28}$		$\frac{15}{28}$
	1	$\frac{3}{14}$	$\frac{3}{14}$		$\frac{3}{7}$
	2	$\frac{1}{28}$			$\frac{1}{28}$
Column total		$\frac{5}{14}$	$\frac{15}{28}$		1

Solution

(a)

$$\begin{aligned}
E(X, Y) &= \sum_{x=0}^2 \sum_{y=0}^2 xyf(x, y) \\
&= (0)(0)f(0, 0) + (0)(1)f(0, 1) + (0)(2)f(0, 2) \\
&\quad + (1)(0)f(1, 0) + (1)(1)f(1, 1) + (1)(2)f(1, 2) \\
&\quad + (2)(0)f(2, 0) + (2)(1)f(2, 1) + (2)(2)f(2, 2) \\
&= f(1, 1) = \frac{3}{14}
\end{aligned}$$

Example 63

Find

$$E\left(\frac{Y}{X}\right)$$

for the density function

$$f(x, y) = \begin{cases} \frac{x(1+3y^2)}{4}, & 0 < x < 2, 0 < y < 1 \\ 0 & \text{Otherwise} \end{cases}$$

Solution

$$\begin{aligned}
E\left(\frac{Y}{X}\right) &= \int_0^1 \int_0^2 \frac{y(1+3y^2)}{4} dx dy = \frac{1}{4} [yx + 3y^3x]_0^2 dy \\
&= \frac{1}{4} \int_{y=0}^1 [2y + 6y^3] dy \\
&= \frac{1}{4} \left[y^2 + \frac{6}{4}y^4 \right]_0^1 = \frac{1}{4} \left[1 + \frac{6}{4} \right] = \frac{1}{4} \times \frac{10}{4} = \frac{5}{8}
\end{aligned}$$

5.5 Variance and covariance

The expected value of a random variable X is of special importance in statistics because it describes where the probability distribution is centred. By itself, however, the mean does not give adequate description of the shape of the distribution. We need to characterise the variability in the distribution.

5.4.1 Variance

Suppose that X is a random variable with mean, $\mu = E(X)$. The variance of X denoted by $\text{var}(X)$ is:

$$\text{Var}(X) = E[(X - \mu)^2] \text{ also } \text{Var}(X) = E(X^2) - \mu^2$$

It must be kept in mind that this may or may not be finite. If the expectation is not finite, it is said that $\text{Var}(X)$ does not exist. However, if the possible values of X are bounded, the $\text{Var}(X)$ must exist.

For a random variable with probability distribution $f(x)$ and mean μ , the variance of X is

$$\sigma^2 = \text{Var}(X) = E[(x - \mu)^2] = \sum (x - \mu)^2 f(x)$$

if X is discrete, and

$$\sigma^2 = \text{Var}(x) = E[(X - \mu)^2] = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx$$

If X is continuous

Example 64

Let the random variable X represent the number of automobiles that are used for official business purposes on any given workday. The probability distributions for company A is given by:

X	1	2	3
f(x)	0.3	0.4	0.3

And for company B by:

X	0	1	2	3	4
f(x)	0.2	0.1	0.3	0.3	0.1

Show that the variances of the probability distribution for company B is greater than that of company A.

Solution

For company A

$$\mu = E(x) = \sum xf(x)$$

$$= (1)(0.3) + (2)(0.4) + (3)(0.3) = 2.0$$

and then

$$\sigma_A^2 = \sum_{x=1}^3 (x - \mu)^2 f(x) = \sum (x - 2)^2 f(x)$$

$$= (1 - 2)^2(0.3) + (2 - 2)^2(0.4) + (3 - 2)^2(0.3) = 0.6$$

For company B

$$\mu = E(x) = (0)(0.2) + (1)(0.1) + (2)(0.3) + (3)(0.3) + (4)(0.1) = 2.0$$

$$\sigma_B^2 = \sum_{x=0}^4 (x - 2)^2 f(x)$$

$$= (0 - 2)^2(0.2) + (1 - 2)^2(0.1) + (2 - 2)^2(0.3) + (3 - 2)^2(0.3) + (4 - 2)^2(0.1) = 1.6$$

$$\therefore \sigma_B^2 > \sigma_A^2$$

Properties of Variance

1. For any constants a and b :

$$\text{Var}(ax + b) = a^2 \text{Var}(x)$$

- (2) For any random variable X :

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

5.4.2 Covariance

When we consider the joint distribution of two random variables, the mean, the median and the variance provide useful information about their marginal distribution. However, these values do not provide any information about the relationship between the two variables or about their tendency to vary together rather than independently. The quantity covariance enables us to measure the association between two random variables, to determine the variance of the sum of any number of dependent random variables and to predict the value of one random variable by using the observed value of some other related variable.

Let X and Y be random variables having a specified joint distribution and Let $E(X) = \mu_x$, $E(Y) = \mu_y$, $\text{Var}(X) = \sigma_x^2$ and $\text{Var}(Y) = \sigma_y^2$. The covariance of X and Y , (denoted by $\text{COV}(X, Y)$), is defined as:

$$\begin{aligned} \text{Cov}(X, Y) &= \sigma_{xy} = E[(X - \mu_x)(Y - \mu_y)] \\ &= \sum \sum (X - \mu_x)(Y - \mu_y) f(x, y) \text{ if } X \text{ and } Y \text{ are discrete, and} \end{aligned}$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \mu_x)(y - \mu_y) f(x, y) dx dy \quad \text{if } X \text{ and } Y \text{ are continuous.}$$

It can be shown (DeGroot, 1989) that if $\sigma_x^2 < \infty$ and $\sigma_y^2 < \infty$, then the expectation above will exist and $\text{cov}(x, y)$ can be finite. However, the value of $\text{cov}(x, y)$ can be positive, negative or zero. When X and Y are independent covariance is zero. The converse however, is not generally true. Two variables may have zero covariance and still not be statistically independent. To dependent variables can be unrelated.

Some properties of covariance

1. For any random variables X and Y such

That $\sigma_x^2 < \infty$ and $\sigma_y^2 < \infty$,

$$\text{Cov}(X - Y) = E(XY) - E(X)E(Y).$$

Example 65

Two variables X and Y have a joint distribution as:

$$f(x, y) = \begin{cases} \frac{2}{5}(2x + 3y) & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{Elsewhere} \end{cases}$$

Find the covariance of X and Y

Solution

$$\sigma_{x,y} = E(X, Y) - \mu_x \mu_y$$

But $\mu_x = E(X)$ and $\mu_y = E(Y)$

Hence we first must compute the marginal density functions But for a joint distribution $f(x, y)$ marginal distributions of X and Y alone are given by:

$$g(x) = \sum_y f(x, y) \text{ and } h(y) = \sum_x f(x, y) \text{ for discrete variable}$$

$$g(x) = \int_y f(x, y) dy \text{ and } h(y) = \int_x f(x, y) dx \text{ for continuous variable}$$

Thus:

$$\begin{aligned} g(x) &= \int_{-\infty}^{\infty} f(x, y) dy = \int_0^1 \frac{2}{5} (2x + 3y) dy \\ &= \frac{4xy}{5} + \frac{6y^2}{10} \Big|_0^1 = \frac{4x + 3}{5} \end{aligned}$$

$$h(y) = \int_{-\infty}^{\infty} f(x, y) dx = \int_0^1 \frac{2}{5} (2x + 3y) dx = \frac{2(1 + 3y)}{5}$$

$$E(X) = \int_0^1 \frac{x(4x + 3)}{5} dx = \frac{1}{5} \int_0^1 4x^2 + 3x = \frac{1}{5} \left(\frac{4}{3}x^3 + \frac{3}{2}x^2 \right) \Big|_0^1$$

=

$$\frac{1}{5} \left(\frac{4}{3} + \frac{3}{2} \right) = \frac{17}{30} \quad \dots\dots (1)$$

It could have been solved from

$$\int_{x=0}^1 \int_{y=0}^1 xf(xy) dx dy$$

$$E(y) = \int \frac{2}{5} (1 + 3y) dy = \frac{2}{5} \int_0^1 (y + 3y^2) dy$$

$$\begin{aligned} &= \frac{2}{5} \left(\frac{1}{2}y^2 + y^3 \right) \Big|_0^1 \\ &= \frac{2}{5} \left(\frac{1}{2} + 1 \right) = \frac{2}{5} \times \frac{3}{2} = \frac{3}{5} \quad \dots\dots (2) \end{aligned}$$

It also could have been solved from

$$\int_{x=0}^1 \int_{y=0}^1 yf(xy) dx dy$$

$$\begin{aligned} E(XY) &= \int_0^1 \int_0^1 xy \frac{2}{5} (2x + 3y) dy dx \\ &= \frac{2}{5} \int_{y=0}^1 \int_{x=0}^1 (2x^2y + 3xy^2) dx dy \end{aligned}$$

$$\begin{aligned} &= \frac{2}{5} \int \left(\frac{2}{3}x^3y + \frac{3}{2}x^2y^2 \right) \Big|_0^1 dy \\ &= \frac{2}{5} \int \frac{2}{3}y + \frac{3}{2}y^2 dy \end{aligned}$$

$$\begin{aligned} &= \frac{2}{5} \left(\frac{1}{3}y^2 + \frac{1}{2}y^3 \right) \Big|_0^1 \\ &= \frac{2}{5} \left(\frac{1}{3} + \frac{1}{2} \right) = \left(\frac{2}{5} \times \frac{5}{6} \right) = \frac{1}{3} \quad \dots\dots (3) \end{aligned}$$

$$\begin{aligned}\therefore \sigma_{xy} &= E(XY) - E(X)E(Y) \\ &= (3) - (1)(2) \\ &= \frac{1}{3} - \left(\frac{17}{30} \times \frac{3}{5}\right) \\ &= 0.333 - 0.34 \\ &\approx -0.01\end{aligned}$$

Chapter 6

Introduction to Estimation

Learning Objectives

- Understand point and interval estimates and how it can be applied to engineering data.

6.0 Estimation

6.1 Introduction

The basic reasons for the need to estimate population parameters from sample information is that it is ordinarily too expensive or simply infeasible to enumerate complete populations to obtain the required information. The cost of complete censuses may be prohibitive in finite populations while complete enumerations are impossible in the case of infinite populations. Hence, estimation procedures are useful in providing the means of obtaining estimates of population parameters with desired degree of precision. We now consider *estimation*, the first of the two general areas of *statistical inference*. The second general area is *hypothesis testing* which will be examined later.

The subject of *estimation* is concerned with the methods by which population characteristics are measured from sample information. The objectives are to present:

1. properties for judging how well a given sample statistic estimates the parent population parameter.
2. several methods for estimating these parameters.

There are basically two types of estimation: *point estimation* and *interval estimation*. In *point estimation*, a single sample statistic, such as $\bar{X}$, s or $\bar{p}$ is calculated from the sample to provide a best estimate of the true value of the corresponding population parameter such as μ , σ or p . Such a statistic is termed a *point estimator*.

The function or rule that is used to estimate the value of a parameter is called an *estimator*. An *estimate* is a particular value calculated from a particular sample of observations.

On the other hand, an *interval estimate* consists of two numerical values defining an interval which, with varying degrees of confidence, we feel includes the parameter being estimated.

6.2 Properties of a Point Estimator

Unbiasedness: If the expected value or mean of all possible values of a statistic over all possible samples is equal to the population parameter being estimated, the sample statistic is said to be unbiased. That is, if the expected value of an estimator is equal to the corresponding population parameter, the estimator is unbiased.

Example 66

The sample mean is an unbiased estimator of the population mean.

$$E(\bar{X}) = E\left[\frac{1}{n} \sum_{i=1}^n x_i\right] = \mu$$

Efficiency: The most efficient estimator among a group of unbiased estimators is the one with the smallest variance. This concept refers to the sampling variability of an estimator.

Example 67

Let $\hat{\theta}_1, \hat{\theta}_2$ and $\hat{\theta}_3$ be unbiased estimators for θ . Let $\hat{\theta}_4$ be a biased estimator among this group of estimators. Assume that $\text{Var}(\hat{\theta}_4) < \text{Var}(\hat{\theta}_2) < \text{Var}(\hat{\theta}_1) < \text{Var}(\hat{\theta}_3)$. Which of these estimators is most efficient?

Consistency: An estimator is consistent if as the sample size increases, the probability increases that the estimator will approach the true value of the population parameter. Alternatively, an estimator is consistent if it satisfies the following conditions:

1. $\text{Var}(\hat{\theta}) \rightarrow 0$ as $n \rightarrow \infty$
2. $\hat{\theta}$ becomes unbiased as $n \rightarrow \infty$

6.3 Interval Estimation

For most practical purposes, it would not suffice to have merely a single value estimate of a population parameter. Any single point estimate will be either right or wrong. Therefore, instead of obtaining only a single estimate of a population parameter, it would certainly seem to be extremely useful and perhaps necessary to obtain two estimators, say $\bar{X}_1$ and $\bar{X}_2$, and say with some confidence that the interval between $\bar{X}_1$ and $\bar{X}_2$ includes the true mean μ .

Thus, an interval estimate of a population parameter is a statement of two values between which it is estimated that the parameter lies.

We shall be discussing the construction of confidence intervals as a means of interval estimation.

The confidence we have that a population parameter, θ , will fall within some confidence interval will equal $(1 - \alpha)$, where α is the probability that the interval does not contain θ (i.e. the probability α , is an allowance for error).

To construct a 95% confidence interval $\Rightarrow \alpha = 0.05$. That is, the probability is 0.05 that the value θ will not lie within the interval.

Note that

$$\alpha + (\text{Confidence limit level}) = 1.0$$

The larger the confidence interval, the smaller the probability of error (α) for the interval estimator.

Confidence Interval For μ (σ known)

A confidence interval is constructed on the basis of sample information. It also depends on the size of α . Assume the population variance σ^2 is known and the population is normal, then the $100(1 - \alpha)\%$ percent C.I for μ is given by

$$\bar{X} - Z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right) \leq \mu \leq \bar{X} + Z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right)$$

Simply written as

$$\bar{X} \pm Z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right)$$

where $Z_{\alpha/2}$ is the Z value representing an area $\alpha/2$ to the right and left tails of the standard normal probability distribution.

Example 68

The yield of a chemical process is being studied. From previous experience yield is known to be normally distributed and $\sigma = 3$. The past five days of plant operation have resulted in the following percent yields: 91.6, 88.75, 90.8, 89.95, and 91.3. Find a 95% two-sided confidence interval on the true mean yield.

Solution

$$n=5, \sigma = 3, \bar{x} = 90.48, Z_{\alpha/2} = Z_{0.025} = 1.96$$

$$90.48 - Z_{0.025} \left(\frac{3}{\sqrt{5}} \right) \leq \mu \leq 90.48 + Z_{0.025} \left(\frac{3}{\sqrt{5}} \right)$$

$$90.48 - 1.96(1.3416) \leq \mu \leq 90.48 + 1.96(1.3416)$$

$$87.8505 \leq \mu \leq 93.1095$$

Example 69

A manufacturer produces piston rings for an automobile engine. It is known that ring diameter is normally distributed with $\sigma = 0.001$ millimeters. A random sample of 15 rings has a mean diameter of $\bar{x} = 74.036$ millimeters.

- Construct a 99% two-sided confidence interval on the mean piston ring diameter.
- Construct a 95% confidence interval on the mean piston ring diameter.

Solution

$$n=15, \sigma = 0.001, \bar{x} = 74.036, Z_{\alpha/2} = Z_{0.005} = 2.575, Z_{\alpha/2} = Z_{0.025} = 1.96$$

(a)

$$74.036 - Z_{0.005} \left(\frac{0.001}{\sqrt{15}} \right) \leq \mu \leq 74.036 + Z_{0.005} \left(\frac{0.001}{\sqrt{15}} \right)$$

$$74.036 - 2.575(0.000258) \leq \mu \leq 74.036 + 2.575(0.000258)$$

$$74.036 - 0.000665 \leq \mu \leq 74.036 + 0.000665$$

$$74.0353 \leq \mu \leq 74.0367$$

(b)

$$74.036 - Z_{0.025} \left(\frac{0.001}{\sqrt{15}} \right) \leq \mu \leq 74.036 + Z_{0.025} \left(\frac{0.001}{\sqrt{15}} \right)$$

$$74.036 - 0.00051 \leq \mu \leq 74.036 + 0.00051$$

$$74.0355 \leq \mu \leq 74.0365$$

Example 70

ASTM Standard E23 defines standard test methods for notched bar impact testing of metallic materials. The Charpy V-notch (CVN) technique measures impact energy and is often used to determine whether or not a material experiences a ductile-to-brittle transition with decreasing temperature. Ten measurements of impact energy (J) on specimens of A238 steel cut at 60C

are as follows: 64.1, 64.7, 64.5, 64.6, 64.5, 64.3, 64.6, 64.8, 64.2, and 64.3. Assume that impact energy is normally distributed with $\sigma = 1J$. We want to find a 95% CI for μ , the mean impact energy. The resulting 95% CI is

$$\begin{aligned}\bar{x} - Z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right) &\leq \mu \leq \bar{x} + Z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right) \\ 64.46 - Z_{0.025} \left(\frac{1}{\sqrt{10}} \right) &\leq \mu \leq 64.46 + Z_{0.025} \left(\frac{1}{\sqrt{10}} \right) \\ 64.46 - 1.96 \left(\frac{1}{\sqrt{10}} \right) &\leq \mu \leq 64.46 + 1.96 \left(\frac{1}{\sqrt{10}} \right) \\ 64.46 - 0.6198 &\leq \mu \leq 64.46 + 0.6198 \\ 63.84 &\leq \mu \leq 65.08\end{aligned}$$

That is, based on the sample data, a range of highly plausible values for mean impact energy for A238 steel at 60°C is $63.84J \leq \mu \leq 65.08J$.

Exercises:

1. A confidence interval estimate is desired for the gain in a circuit on a semiconductor device. Assume that gain is normally distributed with standard deviation $\sigma = 20$.
 - (a) Find a 95% CI for μ when $n = 10$ and $\bar{x} = 1000$.
 - (b) Find a 95% CI for μ when $n = 25$ and $\bar{x} = 1000$.
 - (c) Find a 99% CI for μ when $n = 10$ and $\bar{x} = 1000$.
 - (d) Find a 99% CI for μ when $n = 25$ and $\bar{x} = 1000$.
2. A civil engineer is analyzing the compressive strength of concrete. Compressive strength is normally distributed with $\sigma^2 = 1000$ (psi)². A random sample of 12 specimens has a mean compressive strength of $\bar{x} = 3250$ psi.
 - (a) Construct a 95% two-sided confidence interval on mean compressive strength.
 - (b) Construct a 99% two-sided confidence interval on mean compressive strength. Compare the width of this confidence interval with the width of the one found in part (a).

Confidence Interval For μ (σ unknown/ $n \geq 30$)

In practice, the standard deviation of a population σ , is not likely to be known. When σ is unknown and n is 30 or more, we proceed as before and estimate σ with the sample standard deviation s . the resulting $1 - \alpha$ large sample confidence interval for μ becomes

$$\bar{X} - Z_{\alpha/2} \left(\frac{s}{\sqrt{n}} \right) \leq \mu \leq \bar{X} + Z_{\alpha/2} \left(\frac{s}{\sqrt{n}} \right)$$

Example 71

1. A sample of 40 ten-year-old girls gave a mean weight of 71.5 and standard deviation of 12 pounds respectively. Assuming normality, find the
 2. 90% confidence interval for μ .
 3. 95% confidence interval for μ
 4. 99% confidence interval for μ .
5. A hospital administrator took a sample of 45 overdue accounts from which he computed a mean of \$250 and a standard deviation of \$75. Assuming that the amounts of all overdue accounts are normally distributed. Find the
 6. 90% confidence interval for μ .

7. 95% confidence interval for μ .
8. The 99% confidence interval for μ .

Confidence Interval For μ (σ unknown/ $n < 30$)

When the σ is not known and the sample size is small, the procedure for interval estimation of population mean is based on a probability distribution known as the *student t-distribution*. When the population variance is unknown, and the sample size is small, the correct distribution for constructing a confidence interval for μ is the *t-distribution*. Here, an estimate s must be calculated from the sample to substitute for the unknown population standard deviation. The *t-distribution* is used such that

$$t = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

, where

$$s = \sqrt{\frac{\sum (X_i - \bar{X})^2}{(n - 1)}}$$

The *t-distribution* is based on the assumption that the population is normal. A $100(1 - \alpha)\%$ CI for the population mean, μ with the population normal and σ unknown is given by

$$\bar{x} - t_{\alpha/2, \nu} \left(\frac{s}{\sqrt{n}} \right) \leq \mu \leq \bar{x} + t_{\alpha/2, \nu} \left(\frac{s}{\sqrt{n}} \right)$$

where

$$\nu = n - 1$$

Notice that a requirement for the valid use of the *t-distribution* is that the sample must be drawn from a normal distribution.

Example 72

A sample of 25 ten-year-old boys yielded a mean weight and standard deviation of 73 and 10 pounds respectively. Assuming a normally distributed population, find 90, 95 and 99 percent confidence intervals for the mean of the population from which the sample came.

Solution

$n = 25$, $\bar{x} = 73$, $s = 10$

$$\bar{x} \pm t_{\alpha/2} \left(\frac{10}{\sqrt{25}} \right)$$

$$73 \pm t_{0.05}(2)$$

$$73 \pm 1.711(2)$$

$$(69.578, 76.422)$$

Summary:

Confidence Interval for μ

Population Mean μ	Confidence Interval
Sample size	
Large	
→ σ assumed known	$\bar{X} \pm Z_{\alpha/2} (\sigma/\sqrt{n})$
→ σ unknown (<i>estimated by s</i>)	$\bar{X} \pm Z_{\alpha/2} (s/\sqrt{n})$
Small	
→ σ assumed known	$\bar{X} \pm Z_{\alpha/2} (\sigma/\sqrt{n})$
→ σ unknown (<i>estimated by s</i>)	$\bar{X} \pm t_{\alpha/2} (s/\sqrt{n})$

6.3 Confidence Interval For A Population Proportion

It is often necessary to construct confidence intervals on a population proportion. For example, suppose that a random sample of size n has been taken from a large (possibly infinite) population and that X ($\leq n$) observations in this sample belong to a class of interest. Then $\bar{P} = X/n$ is a point estimator of the proportion of the population p that belongs to this class. Note that n and p are the parameters of a binomial distribution. Furthermore, we know that the sampling distribution of $\bar{P}$ is approximately normal with mean p and variance $p(1-p)/n$ if p is not too close to either 0 or 1 and if n is relatively large. Typically, to apply this approximation we require that np and $n(1-p)$ be greater than or equal to 5. We will make use of the normal approximation in this regard.

Definition: If n is large, the distribution of

$$Z = \frac{X - np}{\sqrt{np(1-p)}} = \frac{\bar{P} - p}{\sqrt{\frac{p(1-p)}{n}}}$$

is approximately standard normal.

The $100(1-\alpha)\%$ CI for p is then given by

$$\bar{P} - Z_{\alpha/2} \sqrt{\frac{p(1-p)}{n}} \leq p \leq \bar{P} + Z_{\alpha/2} \sqrt{\frac{p(1-p)}{n}}$$

Note that the quantity $\sqrt{p(1-p)/n}$ in the equation above is called the **standard error of the point estimator** $\bar{P}$. Unfortunately, the upper and lower limits of the confidence interval obtained from this equation contain the unknown parameter p . However, a satisfactory solution is to replace p by $\bar{P}$ in the standard error, which results in

$$\bar{P} - Z_{\alpha/2} \sqrt{\frac{\bar{P}(1-\bar{P})}{n}} \leq p \leq \bar{P} + Z_{\alpha/2} \sqrt{\frac{\bar{P}(1-\bar{P})}{n}}$$

This procedure depends on the adequacy of the normal approximation to the binomial. To be reasonably conservative, this requires that np and $n(1-p)$ be greater than or equal to 5. In situations where this approximation is inappropriate, particularly in cases where n is small, other methods must be used.

Example 73

1. A manufacturer of electronic calculators is interested in estimating the fraction of defective units produced. A random sample of 800 calculators contains 10 defectives. Compute a 99% confidence interval on the fraction defective.

Solution

$$n=800, x=10, \bar{p} = x/n = 0.0125, n\bar{p} = 10, n(1 - \bar{p}) = 790$$

$$\bar{p} - Z_{\alpha/2} \sqrt{\frac{\bar{p}(1 - \bar{p})}{n}} \leq p \leq \bar{p} + Z_{\alpha/2} \sqrt{\frac{\bar{p}(1 - \bar{p})}{n}}$$

$$0.0125 - Z_{0.05} \sqrt{\frac{0.0125(1 - 0.0125)}{800}} \leq p \leq 0.0125 + Z_{\alpha/2} \sqrt{\frac{0.0125(1 - 0.0125)}{800}}$$

$$0.0125 \pm 2.575(0.003928)$$

$$0.0125 \pm 0.0101 = (0.0025, 0.0226)$$

1. Of 1000 randomly selected cases of lung cancer, 823 resulted in death within 10 years. Construct a 95% confidence interval on the death rate from lung cancer.

Solution

$$n=1000, x=823, \bar{p} = x/n = 0.823, n\bar{p} = 823, n(1 - \bar{p}) = 177$$

$$\bar{p} - Z_{\alpha/2} \sqrt{\frac{\bar{p}(1 - \bar{p})}{n}} \leq p \leq \bar{p} + Z_{\alpha/2} \sqrt{\frac{\bar{p}(1 - \bar{p})}{n}}$$

$$0.823 - Z_{0.05} \sqrt{\frac{0.823(1 - 0.823)}{1000}} \leq p \leq 0.823 + Z_{\alpha/2} \sqrt{\frac{0.823(1 - 0.823)}{1000}}$$

$$0.823 \pm 1.96(0.0121)$$

$$0.823 \pm 0.0237$$

$$(0.7993, 0.8467)$$

Exercise

- i. A random sample of 50 suspension helmets used by motorcycle riders and automobile race-car drivers was subjected to an impact test, and on 18 of these helmets some damage was observed. Find a 95% confidence interval on the true proportion of helmets of this type that would show damage from this test.
- ii. In a random sample of 85 automobile engine crankshaft bearings, 10 have a surface finish that is rougher than the specifications allow. Construct the 95% confidence interval for the proportion of bearings in the population that exceeds the roughness specification.
- iii. A survey was conducted to study the dental health practices and attitudes of a certain urban adult population. Of 300 adults interviewed, 123 said that they regularly had a dental checkup twice a year. Obtain a 95% CI for p , based on these data.

Confidence Interval for the Difference Between Two Population Means (Variances Known)

There are instances where we are interested in estimating the difference between two population means. Here, from each of the populations a sample is drawn and from the data of each, the sample means $\bar{x}_1$ and $\bar{x}_2$ respectively, are computed. The estimator $\bar{x}_1 - \bar{x}_2$ yields an unbiased estimate of $\mu_1 - \mu_2$, the difference between the population means.

The quantity

$$\frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

has a $N(0, 1)$ distribution.

The $100(1 - \alpha)\%$ confidence interval for $\mu_1 - \mu_2$ is given by

$$\bar{X}_1 - \bar{X}_2 \pm Z_{\alpha/2} \left(\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \right)$$

for large sample sizes n_1 and n_2 respectively.

Example 74

- i. Tensile strength tests were performed on two different grades of aluminum spars used in manufacturing the wing of a commercial transport aircraft. From past experience with the spar manufacturing process and the testing procedure, the standard deviations of tensile strengths are assumed to be known. The data obtained are as follows: $n_1 = 10$, $\bar{x}_1 = 87.6$, $\sigma_1 = 1$, $n_2 = 12$, $\bar{x}_2 = 84.5$, and $\sigma_2 = 1.5$. If μ_1 and μ_2 denote the true mean tensile strengths for the two grades of spars, find a 90% confidence interval on the difference in mean strength $\mu_1 - \mu_2$.
- ii. Two machines are used for filling plastic bottles with a net volume of 16.0 ounces. The fill volume can be assumed normal, with standard deviation $\sigma_1 = 0.020$ and $\sigma_2 = 0.025$ ounces. A member of the quality engineering staff suspects that both machines fill to the same mean net volume, whether or not this volume is 16.0 ounces. A random sample of 10 bottles is taken from the output of each machine.

Machine 1	Machine 2		
16.03	16.01	16.02	16.03
16.04	15.96	15.97	16.04
16.05	15.98	15.96	16.02
16.05	16.02	16.01	16.01
16.02	15.99	15.99	16.00

Find a 95% confidence interval for the difference in means.

- iii. Two different formulations of an oxygenated motor fuel are being tested to study their road octane numbers. The variance of road octane number for formulation 1 is $\sigma_1^2 = 1.5$, and for formulation 2 it is $\sigma_2^2 = 1.2$. Two random samples of size $n_1 = 15$ and $n_2 = 20$ are tested, and the mean road octane numbers observed are $\bar{x}_1 = 89.6$ and $\bar{x}_2 = 92.5$. Assume normality. Construct a 95% confidence interval on the difference in mean road octane number.

Confidence Interval For The Difference Between Two Population Proportions

The magnitude of the difference between two population proportions is often of interest. An unbiased point estimator of the difference in two populations is provided by the difference in the sample proportions, $\bar{p}_1 - \bar{p}_2$. When n_1 and n_2 are large and the population proportions are not too close to 0 or 1, the central limit theorem applies

$$Z = \frac{\bar{P}_1 - \bar{P}_2 - (p_1 - p_2)}{\sqrt{\frac{p_1(1 - p_1)}{n_1} + \frac{p_2(1 - p_2)}{n_2}}}$$

where Z is a standard normal random variable and thus, normal distribution theory may be employed to obtain confidence intervals.

A $100(1 - \alpha)\%$ confidence interval for $p_1 - p_2$ is given by

$$[\bar{p}_1 - \bar{p}_2] - Z_{\alpha/2} \sqrt{\frac{\bar{p}_1(1 - \bar{p}_1)}{n_1} + \frac{\bar{p}_2(1 - \bar{p}_2)}{n_2}} \leq p_1 - p_2 \leq [\bar{p}_1 - \bar{p}_2] + Z_{\alpha/2} \sqrt{\frac{\bar{p}_1(1 - \bar{p}_1)}{n_1} + \frac{\bar{p}_2(1 - \bar{p}_2)}{n_2}}$$

Example 75

- i. Two different types of injection-molding machines are used to form plastic parts. A part is considered defective if it has excessive shrinkage or is discolored. Two random samples, each of size 300, are selected, and 15 defective parts are found in the sample from machine 1 while 8 defective parts are found in the sample from machine 2. Construct a 95% confidence interval on the difference in the two fractions defective.
- ii. Two hundred patients suffering from a certain disease were randomly divided into two equal groups. Of the first group, who received the standard treatment, 78 recovered within three days. Out of the other 100, who were treated by a new method, 90 recovered within three days. The physician wished to estimate the true difference in the proportions who would recover within three days. Find 95% CI for $p_1 - p_2$.
- iii. In a study designed to assess the side effects of two drugs, 50 animals were given Drug A, and 50 animals were given Drug B. Of the 50 receiving Drug A, 11 showed undesirable side effects, while 8 of those receiving Drug B reacted similarly. Find the 90 and 95 percent confidence intervals for $p_A - p_B$.

Chapter 7

Tests of Hypotheses and Significance

Learning Objectives

- Structure engineering decision-making problems as hypothesis tests.
- Test hypotheses on the mean of a normal distribution using either a Z -test or a t -test procedure.
- Test hypotheses on the variance or standard deviation of a normal distribution.
- Test hypotheses on a population proportion.

7.0 Tests of Hypotheses and Significance

7.1 Introduction

We now discuss the subject of hypothesis testing, which as earlier noted is one of the two basic classes of statistical inference. Testing of hypotheses involves using statistical inference to test the validity of postulated values for population parameters. If the hypothesis specifies the distribution completely it is called simple, otherwise it is called composite. For example, a demographer interested in the mean age of residents in a certain local government area might pose a simple hypothesis such as $\mu = 24$ or he might specify a composite hypothesis as $\mu < 24$ or $\mu > 24$.

A statistical test is usually structured in terms of two mutually exclusive hypotheses referred to as the null hypothesis and the alternative hypothesis denoted by H_0 and H_1 respectively.

Two types of error occur in hypothesis testing; these are type I error and type II error. Type I error occurs if H_0 is rejected when it is true. The probability of a type I error is the conditional probability, $P(\text{reject } H_0 | H_0 \text{ is true})$ and is denoted by α .

Hence,

$$\alpha = P(\text{reject } H_0 | H_0 \text{ is true}) \text{ and}$$

$$1 - \alpha = P(\text{accept } H_0 | H_0 \text{ is true})$$

Type II error if H_0 is accepted when it is false. Its probability is denoted by the symbol β , where

$$\beta = P(\text{accept } H_0 | H_0 \text{ is false}) \text{ and}$$

$$1 - \beta = P(\text{reject } H_0 | H_0 \text{ is false}) \text{ called power of the test}$$

Types I and II error can be explained as follows:

	H_0 is true	H_0 is false
Accept H_0	$1 - \alpha$ (correct decision)	β (Type II errors)
Reject H_0	α (Type I errors)	$1 - \beta$ (correct decision)

Standard format of hypothesis testing: this format involves **five** steps.

Step 1: State the null and alternative hypotheses.

Step 2: Determine the suitable test statistics. This involves choosing the appropriate random variable to use in deciding to accept or reject the null hypothesis.

Unknown Parameter	Appropriate Test Statistic
' μ ' σ known, population normal	$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$
' μ ' σ unknown, population normal	$Z = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}$ if n is 'large' usually $n \geq 30$
' μ ' σ unknown, n small, population normal	$t = \frac{\bar{X} - \mu}{s/\sqrt{n}}$, with $(n-1)$ df
' p ' population normal, n large	$Z = \frac{(x/n) - p_0}{\sqrt{\frac{\bar{p}(1-\bar{p})}{n}}}$

Step 3: Determine the critical region using the cumulative distribution table for the test statistic. The set of values that lead to the rejection of the null hypothesis is called the critical region. A statistical test may be a one-tail or two-tail test. Whether one uses a one- or two- tail test of significance depends upon how the alternative hypothesis is formulated.

Types of Hypothesis	H_0	H_1	Decision Rule Rejected H_0 if
Two-tail	$\mu = \mu_0$	$\mu \neq \mu_0$	$z > z_{\alpha/2}$ or $z < -z_{\alpha/2}$
Right-tail	$\mu \leq \mu_0$	$\mu > \mu_0$	$z > z_{\alpha/2}$
Left-tail	$\mu \geq \mu_0$	$\mu < \mu_0$	$z < -z_{\alpha/2}$

Step 4: Compute the values of the test statistic based on the sample information, e.g. Z_c , t_c , χ_c^2 .

Step 5: Make a statistical decision and interpretation. H_0 is rejected if the computed value of the test statistic falls in the critical region otherwise it is accepted.

Possible situation in testing a statistical hypothesis

	Hypothesis is correct	Hypothesis is incorrect
Hypothesis is Accepted	Correct decision	Type II error β
Hypothesis is Rejected	Type I error α	Correct decision

Type I error: We reject a hypothesis when it should be accepted. $P(\text{reject } H_0 | H_0 \text{ true})$.

Type II error: We accept a hypothesis when it should be rejected. $P(\text{accepting } H_0 | H_0 \text{ false})$.

7.2 A Single Population Mean,

We shall consider testing of hypothesis about a population mean under three different conditions:

1. When sampling is from a normally distributed population with known variance.

1. When sampling is from a normally distributed population with unknown variance.
2. When sampling is from a population that is not normally distributed.

Sampling From Normally Distributed Populations: Population Variance Known

Example 76

A researcher is interested in the mean level of some enzyme in a certain population. The data available to the researcher are the enzyme determinations made on a sample of 10 individuals from the population of interest, and the sample mean is 22. If the sample came from a population that is normally distributed with a known variance, $\sigma^2 = 45$. Can the researcher conclude that

the mean enzyme level in this population is different from 25? Take $\alpha = 0.05$.

Solution

Step 1: $H_0 : \mu_0 = 25$

$$H_1 : \mu_0 \neq 25$$

Step 2:

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$$

since μ_0 and σ are known.

Step 3:

Step 4: $\sigma^2 = 45$, $n = 10$, $\bar{x} = 22$

$$Z_c = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$$

=

$$\frac{22 - 25}{\sqrt{45}/\sqrt{10}}$$

=

$$\frac{-3}{2.1213}$$

=

$$-1.41$$

Step 5: We are unable to reject the null hypothesis, since $-1.42 > -1.96$.

Example 77

Aircrew escape systems are powered by a solid propellant. The burning rate of this propellant is an important product characteristic. Specifications require that the mean burning rate must be 50 centimeters per second. We know that the standard deviation of burning rate is $\sigma = 2$ centimeters per second. The experimenter decides to specify a type I error probability or significance level of $\alpha = 0.05$ and selects a random sample of $n = 25$ and obtains a sample average burning rate of $\bar{x} = 51.3$ centimeters per second. What conclusions should be drawn?

Solution

$$H_0 : \mu_0 = 50$$

$$H_1 : \mu_0 \neq 50$$

$$Z_c = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$$

=

$$\frac{51.3 - 50}{2/\sqrt{25}}$$

=

$$\frac{1.3}{0.4}$$

= 3.25

Conclusion: Since $z_c = 3.25 > 1.96$, we reject $H_0 : \mu_0 = 50$ at the 0.05 level of significance. We conclude that the mean burning rate differs from 50 centimeters per second, based on a sample of 25 measurements.

7.3 Tests on the Mean of a Normal Distribution: Variance Unknown

Test statistic is

$$t = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

has a t distribution with $n-1$ degrees of freedom.

Example 78

A study revealed that the upper limit of the Normal Body Temperature of males is 98.6. The body temperatures for 25 male subjects were taken and recorded as follows: 97.8, 97.2, 97.4, 97.6, 97.8, 97.9, 98.0, 98.0, 98.0, 98.1, 98.2, 98.3, 98.3, 98.4, 98.4, 98.4, 98.5, 98.6, 98.6, 98.7, 98.8, 98.8, 98.9, 98.9 and 99.0. Test the hypothesis $H_0 : \mu_0 = 98.6$ versus $H_1 : \mu_0 \neq 98.6$, using $\alpha = 0.05$.

Example 79

Nine patients suffering from the same physical handicap, but otherwise comparable were asked to perform a certain task as part of an experiment. The average time required to perform the task was seven minutes with a standard deviation of two minutes. Assuming normality, can we conclude that the true mean time required to perform the task by this type of patient is at least ten minutes?

Example 80

The increased availability of light materials with high strength has revolutionized the design and manufacture of golf clubs, particularly drivers. Clubs with hollow heads and very thin faces can result in much longer tee shots, especially for players of modest skills. This is due partly to the “spring-like effect” that the thin face imparts to the ball. Firing a golf ball at the head of the club and measuring the ratio of the outgoing velocity of the ball to the incoming velocity can quantify this spring-like effect. The ratio of velocities is called the coefficient of restitution of the club. An experiment was performed in which 15 drivers produced by a particular club maker were selected at random and their coefficients of restitution measured. In the experiment, the golf balls were fired from an air cannon so that the incoming velocity and spin rate of the ball could be precisely controlled. Determine if there is evidence (with $\alpha = 0.05$) to support a claim that the mean coefficient of restitution exceeds 0.82. The observations are:

0.8411 0.8191 0.8182 0.8125 0.8750

0.8580 0.8532 0.8483 0.8276 0.7983

0.8042 0.8730 0.8282 0.8359 0.8660

The sample mean and sample standard deviation are $\bar{x} = 0.83725$ and $s = 0.02456$.

7.4 Tests on a Population Proportion

It is often necessary to test hypotheses on a population proportion. For example, suppose that a random sample of size n has been taken from a large population and that $X(\leq n)$ observations in this sample belong to a class having a particular characteristic of interest. Then $\hat{P} = X/n$ is a point estimator of the proportion of the population p that belongs to this class. Note that n and p are the parameters of a binomial distribution. Recall that the sampling distribution of $\hat{P}$ is approximately normal with mean p and variance $p(1-p)/n$, if p is not too close to either 0 or 1 and if n is relatively large.

In many engineering problems, we are concerned with a random variable that follows the binomial distribution. For example, consider a production process that manufactures items that are classi-

fied as either acceptable or defective. It is usually reasonable to model the occurrence of defectives with the binomial distribution, where the binomial parameter p represents the proportion of defective items produced. Consequently, many engineering decision problems include hypothesis testing about p .

Consider testing

$$H_0 : p = p_0$$

$$H_1 : p \neq p_0$$

For large samples, the normal approximation to the binomial with the test statistic

$$Z = \frac{X - np_0}{\sqrt{np_0(1 - p_0)}} = \frac{X/n - p_0}{\sqrt{\frac{p_0(1 - p_0)}{n}}} = \frac{\bar{P} - p_0}{\sqrt{\frac{p_0(1 - p_0)}{n}}}$$

may be used.

This presents the test statistic in terms of the sample proportion instead of the number of items X in the sample that belongs to the class of interest.

Example 81

In a study designed to assess the relationship between a certain drug and a certain anomaly in chick embryos, 50 fertilized eggs were injected with the drug on the fourth day of incubation. On the twentieth day of incubation the embryos were examined and in 12 the presence of the abnormality was observed. Test the null hypothesis that the drug causes abnormalities in not more than 20 percent of eggs into which it is introduced. Let $\alpha = 0.05$.

Example 82

A manufacturer of intraocular lenses is qualifying a new grinding machine and will qualify the machine if the percentage of polished lenses that contain surface defects does not exceed 2%. A random sample of 250 lenses contains six defective lenses. Formulate and test an appropriate set of hypotheses to determine if the machine can be qualified. Use $\alpha = 0.05$.

Example 83

A semiconductor manufacturer produces controllers used in automobile engine applications. The customer requires that the process fallout or fraction defective at a critical manufacturing step not exceed 0.05 and that the manufacturer demonstrate process capability at this level of quality using $\alpha = 0.05$. The semiconductor manufacturer takes a random sample of 200 devices and finds that four of them are defective. Test the null hypothesis that the process fallout does not exceed 0.05.

7.5 The Difference Between two Population Means

Hypothesis testing involving the difference between two population means is most frequently employed to determine whether or not it is reasonable to conclude that the two are unequal. In such cases, one or other of the following hypotheses is tested:

$$H_0 : \mu_1 - \mu_2 = 0$$

$$H_1 : \mu_1 - \mu_2 \neq 0$$

$$H_0 : \mu_1 - \mu_2 \geq 0$$

$$H_1 : \mu_1 - \mu_2 < 0$$

$$H_0 : \mu_1 - \mu_2 \leq 0$$

$$H_1 : \mu_1 - \mu_2 > 0$$

Hypothesis Tests for a Difference in Means, Variances Known

Test statistic is

$$Z = \frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Example 84

In a large hospital for the treatment of the mentally retarded, a sample of 12 individuals with mongolism yielded a mean serum uric acid value of $\bar{x}_1 = 4.5$ mg/100 ml. In a general hospital, a sample of 15 normal individuals of the same age and sex were found to have a mean value of $\bar{x}_2 = 3.4$ mg/100 ml. If it is reasonable to assume that the two populations of values are normally distributed with variances equal to 1. Do these data provide sufficient evidence to indicate a difference in mean serum uric acid levels between normal individuals and individuals with mongolism? Let $\alpha = 0.05$.

Solution

$$n_1 = 12$$

$$n_2 = 15$$

$$\sigma_1^2 = 1$$

$$\sigma_2^2 = 1$$

$$\bar{x}_1 = 4.5$$

$$\bar{x}_2 = 3.4$$

$$H_0 : \mu_1 - \mu_2 = 0$$

$$H_1 : \mu_1 - \mu_2 \neq 0$$

$$\begin{aligned}
 Z_c &= \frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \\
 &= \frac{(4.5 - 3.4) - 0}{\sqrt{\frac{1}{12} + \frac{1}{15}}} \\
 &= \frac{1.1}{\sqrt{0.15}} = \frac{1}{0.3873} \\
 &= 2.84
 \end{aligned}$$

Reject H_0 since $2.84 > 1.96$ on the basis of these data, there is an indication that the means are not equal.

Example 85

A product developer is interested in reducing the drying time of a primer paint. Two formulations of the paint are tested; formulation 1 is the standard chemistry, and formulation 2 has a new drying ingredient that should reduce the drying time. From experience, it is known that the standard deviation of drying time is 8 minutes, and this inherent variability should be unaffected by the addition of the new ingredient. Ten specimens are painted with formulation 1, and another 10 specimens are painted with formulation 2; the 20 specimens are painted in random order. The two sample average drying times are $\bar{x}_1 = 121$ minutes and $\bar{x}_2 = 112$ minutes, respectively. What conclusions can the product developer draw about the effectiveness of the new ingredient, using $\alpha = 0.05$?

Solution

$$n_1 = 10$$

$$n_2 = 10$$

$$\sigma_1^2 = 64$$

$$\sigma_2^2 = 64$$

$$\bar{x}_1 = 121$$

$$\bar{x}_2 = 112$$

$$H_0 : \mu_1 \leq \mu_2 = 0$$

$$H_1 : \mu_1 > \mu_2$$

$$\begin{aligned}
Z_c &= \frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \\
&= \frac{(121 - 112) - 0}{\sqrt{\frac{64}{10} + \frac{64}{10}}} \\
&= \frac{9}{\sqrt{12.8}} = \frac{9}{3.5777} \\
&= 2.52
\end{aligned}$$

Conclusion: Reject H_0 .

Exercises

i. Two machines are used for filling plastic bottles with a net volume of 16.0 ounces. The fill volume can be assumed normal, with standard deviation $\sigma_1 = 0.020$ and $\sigma_2 = 0.025$ ounces. A member of the quality engineering staff suspects that both machines fill to the same mean net volume, whether or not this volume is 16.0 ounces. A random sample of 10 bottles is taken from the output of each machine.

Machine 1	Machine 2		
16.03	16.01	16.02	16.03
16.04	15.96	15.97	16.04
16.05	15.98	15.96	16.02
16.05	16.02	16.01	16.01
16.02	15.99	15.99	16.00

Do you think the engineer is correct? Use $\alpha = 0.05$.

ii. Two different formulations of an oxygenated motor fuel are being tested to study their road octane numbers. The variance of road octane number for formulation 1 is $\sigma_1^2 = 1.5$, and for formulation 2 it is $\sigma_2^2 = 1.2$. Two random samples of size $n_1 = 15$ and $n_2 = 20$ are tested, and the mean road octane numbers observed are $\bar{x}_1 = 89.6$ and $\bar{x}_2 = 92.5$. Assume normality, and if formulation 2 produces a higher road octane number than formulation 1, the manufacturer would like to detect it. Formulate and test an appropriate hypothesis, using $\alpha = 0.05$.

Hypothesis Tests for a Difference in Means, Variances unknown but Assumed Equal.

We now extend the results of the previous lecture to the difference in means of the two distributions when the variances of both distributions σ_1^2 and σ_2^2 are unknown. If the sample sizes n_1 and n_2 exceed 30, the normal distribution procedures could be used. However, when small samples are taken, we will assume that the populations are normally distributed and base our hypothesis tests on the t distribution. This nicely parallels the case of inference on the mean of a single sample with unknown variance.

The normality assumption is required to develop the test procedure, but moderate departures from normality do not adversely affect the procedure. Two different situations must be treated. In the first case, we assume that the variances of the two normal distributions are unknown but equal; that is, $\sigma_1^2 = \sigma_2^2 = \sigma^2$. In the second, we assume that they are unknown and not necessarily equal.

The test statistic is

$$t_c = \frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{S_p^2}{n_1} + \frac{S_p^2}{n_2}}} = \frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

with

$$n_1 + n_2 - 2$$

df

The two sample variances are combined to form an estimator of σ^2 . The **pooled estimator** of σ^2 is defined as follows.

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$$

Example 86

The diameter of steel rods manufactured on two different extrusion machines is being investigated. Two random samples of sizes $n_1 = 15$ and $n_2 = 17$ are selected, and the sample means and sample variances are $\bar{x}_1 = 8.73$, $s_1^2 = 0.35$, $\bar{x}_2 = 8.68$, and $s_2^2 = 0.40$, respectively. Assume that $\sigma_1^2 = \sigma_2^2$ and that the data are drawn from a normal distribution. Is there evidence to support the claim that the two machines produce rods with different mean diameters? Use $\alpha = 0.05$ in arriving at this conclusion.

Example 87

Two catalysts are being analyzed to determine how they affect the mean yield of a chemical process. Specifically, catalyst 1 is currently in use, but catalyst 2 is acceptable. Since catalyst 2 is cheaper, it should be adopted, providing it does not change the process yield. A test is run in the pilot plant and results in the data shown in the following table. Is there any difference between the mean yields? Use $\alpha = 0.05$, and assume equal variances.

Observation Number	Catalyst 1	Catalyst 2
1	91.50	89.19
2	94.18	90.95
3	92.18	90.46
4	95.39	93.21
5	91.79	97.19
6	89.07	97.04
7	94.72	91.07
8	89.21	92.75
	$\bar{x}_1 = 92.255$	$\bar{x}_2 = 92.733$
	$s_1 = 2.39$	$s_2 = 2.98$

Example 88

Serum amylase determinations were made on a sample of 15 apparently normal subjects. The sample yielded a mean of 96 units/100ml and a standard deviation 35 units/100ml. Serum amylase determinations were also made on 22 hospitalized subjects. The mean and standard deviation from this second group are 120 and 40 units/100ml., respectively. Would we be justified in calculating that the implied population means are different? Let $\alpha = 0.05$.

Hypothesis Tests for a Difference Between Two Population Proportions

Suppose that two independent random samples of sizes n_1 and n_2 are taken from two populations, and let X_1 and X_2 represent the number of observations that belong to the class of interest in

samples 1 and 2, respectively. Furthermore, suppose that the normal approximation to the binomial is applied to each population, so the estimators of the population proportions $\hat{P}_1 = X_1/n_1$ and $\hat{P}_2 = X_2/n_2$ have approximate normal distributions.

The test statistic is

$$Z = \frac{\bar{P}_1 - \bar{P}_2 - (p_1 - p_2)}{\sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}}$$

is distributed approximately as standard normal and is the basis of a test for $H_0 : p_1 = p_2$. If the null hypothesis is true, using the fact that $p_1 = p_2 = p$, the random variable

$$Z = \frac{\hat{P}_1 - \hat{P}_2}{\sqrt{p(1-p) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

is distributed approximately $N(0, 1)$. An estimator of the common parameter p is

$$\hat{P} = \frac{X_1 + X_2}{n_1 + n_2}$$

The **test statistic** for $H_0 : p_1 = p_2$ is then

$$Z = \frac{\hat{P}_1 - \hat{P}_2}{\sqrt{\hat{p}(1-\hat{p}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

Example 89

A random sample of 500 adult residents of Maricopa County found that 385 were in favor of increasing the highway speed limit to 75 mph, while another sample of 400 adult residents of Pima County found that 267 were in favor of the increased speed limit. Do these data indicate that there is a difference in the support for increasing the speed limit between the residents of the two counties? Use $\alpha = 0.05$.

Example 90

Out of a sample of 150, selected from patients admitted over a two-year period to a large hospital, 129 had some type of hospitalization insurance. In a sample of 160 similarly selected patients from a second hospital, 144 had some type of hospitalization insurance. Test the null hypothesis that $p_1 = p_2$. Let $\alpha = 0.05$.

Chapter 8

Regression and Correlation Analysis

Learning Objectives

- Use simple linear regression for building empirical models of engineering and scientific data.
- Understand how the method of least squares is used to estimate the parameters in a linear regression model.
- Test statistical hypotheses and construct confidence intervals on regression model parameters.
- Use the regression model to make a prediction of a future observation and construct an appropriate prediction interval on the future observation.
- Apply the correlation model.

8.0 Regression and Correlation Analysis

8.1 Introduction

Many problems in engineering and science involve exploring the relationships between two or more variables. **Regression analysis** is a statistical technique that is very useful for these types of problems. For example, in a chemical process, suppose that the yield of the product is related to the process-operating temperature. Regression analysis can be used to build a model to predict yield at a given temperature level. This model can also be used for process optimization, such as finding the level of temperature that maximizes yield, or for process control purposes. Other examples are, studying the relationship between blood pressure and age, the concentration of an injected drug and heart rate etc.

Regression analysis is concerned with the study of the dependence of one variable, the dependent variable, on one or more other variables, the independent or explanatory variables with a view to estimating and predicting the (population) mean or average of the former (dependent) in terms of the known or fixed (in repeated sampling) values of the latter (independent).

Very often in practice, a relationship is found to exist between two (or more) variables and one wishes to express this relationship in mathematical form by determining an equation connecting the variables.

Correlation analysis, on the other hand, is concerned with measuring the strength of the relationship between variables. When we compute measures of correlation from a set of data, we are interested in the degree of the *correlation* between variables.

8.2 The Regression Model

In the typical regression problem, the researcher has available for analysis a sample of observations from some real or hypothetical population. Based on the result of his analysis of the sample data, he is interested in reaching decisions about the population from which the sample is presumed to have been drawn. It is important that the researcher understand the nature of the population in which he is interested.

In the simple linear regression model two variables X and Y , are of interest. The variable X is usually referred to as the independent variable, while the other variable, Y is called the dependent variable; and we speak of the regression of Y on X .

The following are the assumptions underlying the simple linear regression model.

- (i) Values of the independent variable X are fixed.
- (ii) The variable X is measured without error.
- (iii) Values Y are normally distributed
- (iv) The Y values are statistically independent.
- (v) The variances of the subpopulations of Y are all equal.

$$V(Y|x) = V(\alpha + \beta x + e) = V(\alpha + \beta x) + V(e) = 0 + \sigma^2 = \sigma^2$$

- (i) The means of the subpopulations of Y all lie on the same straight line, this is the assumption of linearity.

$$E(Y|x) = \mu_{Y|x} = \alpha + \beta x \quad (1.0)$$

These assumptions may be summarized by means of the following equation which is called the simple linear regression model because it has only one independent variable or **regressor**:

$$y = \alpha + \beta x + e \quad (1.1)$$

where α and β (slope and intercept) are called population regression coefficients, e is called the **error term** with **mean zero** and variance σ^2 . The random errors corresponding to different observations are also assumed to be uncorrelated random variables.

The results of n observations of the set of random variables X and Y can be summarized by drawing a scatter diagram. A straight line passing closely to the points may be drawn. The main problem arises when the points do not all lie exactly on the straight line, but simply form a cloud of points around it. Thus, it may be possible by guess work to draw quite a number of lines each of which will appear to be able to explain the relationship between X and Y . We shall consider finding a best fit line. Such a line will then be used as a model relating the random variable Y with the random variable X .

Suppose that we have n pairs of observations $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$. The following figure shows a typical scatter plot of observed data and a candidate for the estimated regression line.

The estimates of α and β should result in a line that is (in some sense) a “best fit” to the data. The German scientist Karl Gauss proposed estimating the parameters α and β in Equation 1.1 to minimize the sum of the squares of the deviations in the diagram. We call this criterion for estimating the regression coefficients the **method of least squares**. Using Equation 1.1, we may express the n observations in the sample as

$$y_i = \alpha + \beta x_i + e_i$$

$$i = 1, 2, \dots, n$$

and the sum of the squares of the deviations of the observations from the true regression line is

$$L = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - \alpha - \beta x_i)^2$$

(1.2)

The least squares estimators of α and β , say $\hat{\alpha}$ and $\hat{\beta}$, must satisfy

$$\frac{\partial L}{\partial \alpha} = -2 \sum_{i=1}^n (y - \hat{\alpha} - \hat{\beta}x_i) = 0$$

(1.3)

$$\frac{\partial L}{\partial \beta} = -2 \sum_{i=1}^n (y - \hat{\alpha} - \hat{\beta}x_i) x_i = 0$$

Simplifying these two equations yields

$$n\hat{\alpha} + \hat{\beta} \sum_{i=1}^n x_i = \sum_{i=1}^n y_i$$

(1.4)

$$\hat{\alpha} \sum_{i=1}^n x_i + \hat{\beta} \sum_{i=1}^n x_i^2 = \sum_{i=1}^n x_i y_i$$

Equations 1.4 are called the least squares normal equations. The solution to the normal equations results in the least squares estimators $\hat{\alpha}$ and $\hat{\beta}$.

The least squares estimates of the intercept and slope in the simple linear regression model are

$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x} \quad (1.5)$$

$$\begin{aligned} \hat{\beta} &= \frac{\sum_{i=1}^n x_i y_i - \frac{\left(\sum_{i=1}^n x_i\right)\left(\sum_{i=1}^n y_i\right)}{n}}{\sum_{i=1}^n x_i^2 - \frac{\left(\sum_{i=1}^n x_i\right)^2}{n}} \\ &= \frac{n \sum_{i=1}^n x_i y_i - \left(\sum_{i=1}^n x_i\right)\left(\sum_{i=1}^n y_i\right)}{n \sum_{i=1}^n x_i^2 - \left(\sum_{i=1}^n x_i\right)^2} \end{aligned}$$

(1.6)

Equation 1.6 can also be written as

$$\hat{\beta} = \frac{\sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}}{\sum_{i=1}^n x_i^2 - n\bar{x}^2}$$

The estimated regression line is therefore

$$\hat{y} = \hat{\alpha} + \hat{\beta}x$$

Alternatively

8.3 Method of Least Squares

We shall now find a and b , the estimates of α and β so that the sum of the squares of the residuals is a minimum. The residual sum of squares is often called the Sum of Squares of Errors (SSE) about the regression line. This minimisation procedure for estimating the parameter is called the “methods of least squares”. Hence we shall find a and b so as to minimise

$$SSE = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (y_i - a - bx_i)^2$$

Differentiating SSE with respect to a and b , we have

$$\begin{aligned} \frac{\partial(SSE)}{\partial a} &= -2 \sum_{i=1}^n (Y_i - a - bx_i) \\ \frac{\partial(SSE)}{\partial b} &= -2 \sum_{i=1}^n (Y_i - a - bx_i)x_i \end{aligned}$$

Setting the partial derivative equal to zero and rearranging the terms, we obtain the equation (called the normal equations)

$$na + b \sum_{i=1}^n x_i = \sum_{i=1}^n y_i \quad \dots (1)$$

$$a \sum_{i=1}^n x_i + b \sum_{i=1}^n x_i^2 = \sum_{i=1}^n x_i y_i \quad \dots (2)$$

Solving for a and b from (1) and (2)

$$b = \frac{n \sum x_i y_i - \left(\sum_{i=1}^n x_i \right) \left(\sum_{i=1}^n y_i \right)}{n \sum x_i^2 - \left(\sum_{i=1}^n x_i \right)^2} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{SS_{xy}}{SS_{xx}}$$

$$a = \frac{\sum_{i=1}^n y_i - b \sum_{i=1}^n x_i}{n} = \bar{y} - b\bar{x}$$

Equations (1) and (2) can also be solved using matrices as:

$$\begin{bmatrix} n & \sum x \\ \sum x & \sum x^2 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} \sum y \\ \sum xy \end{bmatrix}$$

$$\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} n & \sum x \\ \sum x & \sum x^2 \end{bmatrix}^{-1} \begin{bmatrix} \sum y \\ \sum xy \end{bmatrix}$$

Example 91

Assuming we have the following quantities $n = 8$, $\sum x = 140$, $\sum y = 382$, $\sum xy = 3870$, and $\sum x^2 = 3500$.

Solution

Using the above equation:

$$\begin{bmatrix} 8 & 140 \\ 140 & 3500 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 382 \\ 3870 \end{bmatrix}$$

Solving gives $a = 94.67$ and $b = -2.68$

Thus, the estimated regression line is given by:

$$= 94.67 - 2.68x$$

It may be noted that the least-squares line passes through the point $(\bar{x}, \bar{y})$ called the ‘Centroid’ or centre of gravity of the data. The slope b of the regression line is independent of the origin of coordinates. It is therefore said that b is invariable under the translation of axes.

Besides assuming that the regression of y and x is a linear function having the form $E(Y|X) = \alpha + \beta x$ we have made three further assumptions which may be summarised as follows:

1. **Normality:** We have assumed that each variable y_i has a normal distribution.
2. **Independence:** We have assumed that the variables $y_1, \dots, y_n$ are independent.
 1. **Homoscedasticity:-** We have assumed that the variable $y_1, \dots, y_n$ have the same variance σ^2 . This assumption is called the assumption of homoscedasticity. In general, it is said that random variables having the same variance are homoscedastic, and random variables having different variance are heteroscedastic.

Example 92

A team of professional mental health workers in a long-stay psychiatric hospital wished to measure the level of response of withdrawn patients to a program of remotivation therapy. A standard test was available for this purpose, but it was expensive and time-consuming to administer. To overcome this obstacle, the team developed a test that was much easier to administer. To test the usefulness of the new instrument for measuring the level of patient response, the team decided to examine the relationship between scores made on the new test and scores made on the standardized test. The objective was to use the new test if it could be shown that it was a good predictor of a patient’s score on the standardized test. The results are shown in the table below:

Patients' Scores on Standardized Test and New Test		
Patient Number	Score on New Test (X)	Score on Standardized Test (Y)
1	50	61
2	55	61
3	60	59
4	65	71
5	70	80
6	75	76
7	80	90
8	85	106
9	90	98
10	95	100
11	100	114

Obtain the estimates of the regression coefficients.

Example 93

A research on “Near Surface Characteristics of Concrete: Intrinsic Permeability”, presented data on compressive strength x and intrinsic permeability y of various concrete mixes and cures. Summary quantities are $n = 14$, $\sum x_i = 43$, $\sum y_i = 572$, $\sum x_i^2 = 157.42$, $\sum y_i^2 = 23,530$, and $\sum x_i y_i = 1697.80$. Assume that the two variables are related according to the simple linear regression model.

- Calculate the least squares estimates of the slope and intercept.
- Use the equation of the fitted line to predict what permeability would be observed when the compressive strength is $\bar{x} = 4.3$.
- Give a point estimate of the mean permeability when compressive strength is $\bar{y} = 3.7$.

Example 94

The following data were obtained from a study investigating the relationship between noise exposure and hypertension.

Y	1	0	1	2	5	1	4	6	2	3	5	4	6	8	4	5	7	9	7	6
X	60	63	65	70	70	70	80	90	80	80	85	89	90	90	90	90	94	100	100	100

- Fit the simple linear regression model using least squares.
- Find the predicted mean rise in blood pressure level associated with a sound pressure level of 85 decibels.

8.5 Correlation Analysis

Closely related but conceptually very much different from regression analysis is **correlation analysis**, where the primary objective is to measure the *strength or degree of linear association* between two variables. The **correlation coefficient** measures this strength of (linear) association. For example, we may be interested in finding the correlation between smoking and lung cancer; between scores on mathematics and fluid mechanics examinations, between high school grades and

college grades etc. In regression analysis, as already noted, we are not primarily interested in such a measure. Instead, we try to estimate the average value of one variable on the basis of the fixed values of another variable.

The population correlation coefficient between two random variables, X and Y is defined as

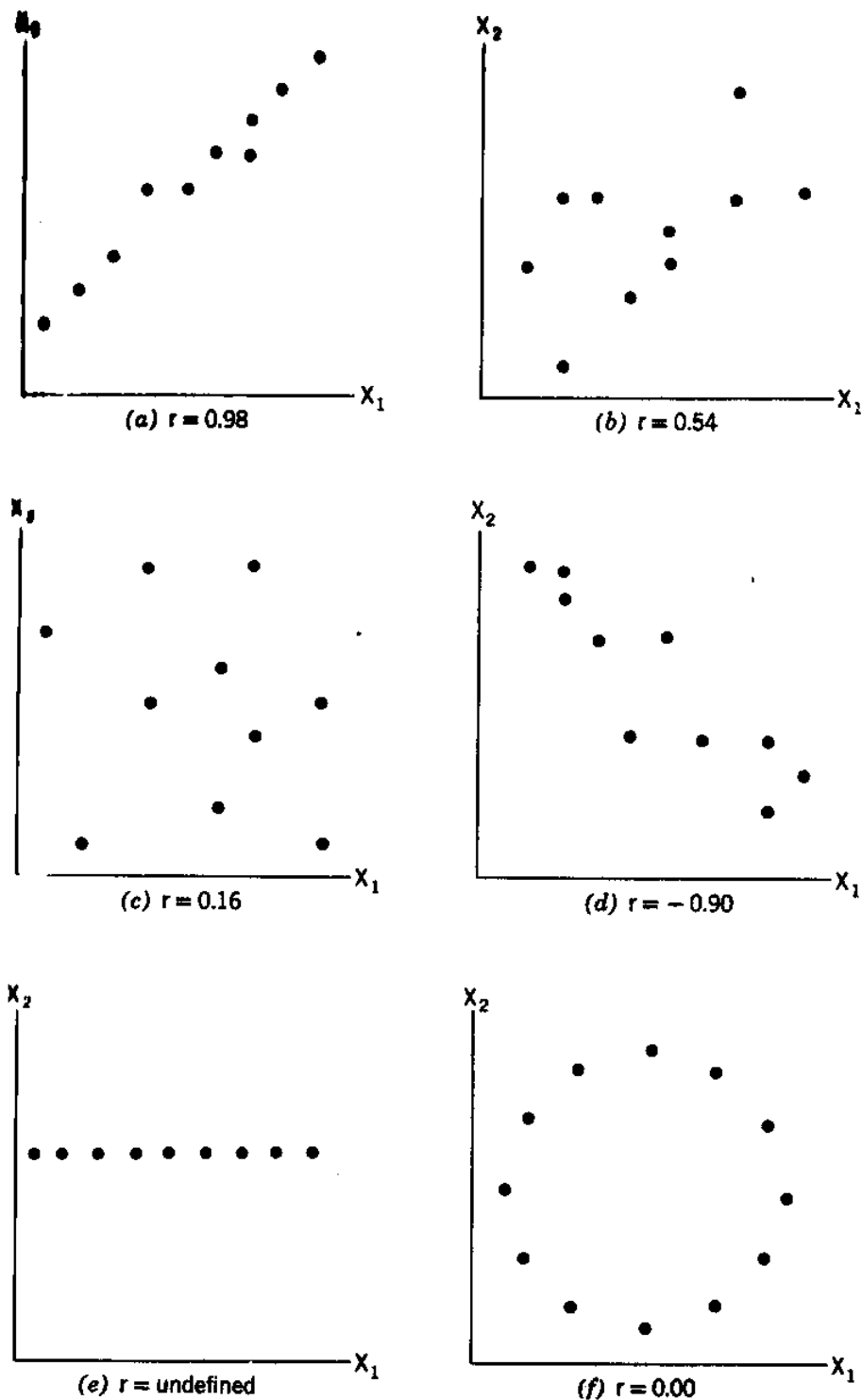
$$\rho = \frac{E[X - E(X)][Y - E(Y)]}{\sqrt{\text{Var}(X)\text{Var}(Y)}} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}} = \frac{\sigma_{XY}}{\sigma_X\sigma_Y}$$

where σ_{XY} is the covariance between variables X and Y , σ_X and σ_Y are the standard deviations of X and Y respectively. It is possible to draw inferences about the correlation coefficient ρ using its estimator, the sample correlation coefficient, r . “ r ” is the correlation coefficient between “ n ” pairs of observations whose values are (X_i, Y_i) and is given by

$$\begin{aligned} r &= \frac{\sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}}{\sqrt{(\sum_{i=1}^n x_i^2 - n\bar{x}^2)(\sum_{i=1}^n y_i^2 - n\bar{y}^2)}} \\ &= \frac{n \sum_{i=1}^n x_i y_i - (\sum_{i=1}^n x_i)(\sum_{i=1}^n y_i)}{\sqrt{(n \sum_{i=1}^n x_i^2 - (\sum_{i=1}^n x_i)^2)(n \sum_{i=1}^n y_i^2 - (\sum_{i=1}^n y_i)^2)}} \end{aligned}$$

Properties of r :

1. It is symmetrical in nature (the two variables are treated symmetrically). That is, there is no distinction between the dependent and independent variables.
2. Both variables are assumed to be random.
3. It can be positive or negative, the sign depending on the sign of the term in the numerator which measures the sample co variation of the two variables.
4. It lies between the limits of -1 and +1; that is, $-1 \leq r \leq +1$.
5. If X and Y are independent, the correlation coefficient between them is zero but if $r=0$ it does not mean that the two variables are independent.

Figure 8.1: Fig. 6.2 Scatter plots with various r values**Testing hypothesis about the correlation coefficient.**

A test of the special hypothesis $\rho = 0$ versus an appropriate alternative is equivalent to testing $\beta = 0$ for the simple linear regression model. In doing this the t -distribution with $n-2$ degrees of freedom may be needed.

t

$$= \frac{b}{\sqrt{SS_{xx}}}$$

which can also be written as to test:

$$H_0 : \rho = 0$$

$$H_1 : \rho \neq 0$$

Example 95

Using the following data, test the hypothesis that there is no linear correlation among the variables that generated them; at 5% level of significance:

$$SS_{xx} = 0.11273 \quad SS_{yy} = 11,807,324,786 \quad SS_{xy} = 34,422,75972$$

Solution

$$r = \frac{SS_{xy}}{\sqrt{SS_{xx}SS_{yy}}} = \frac{34422.75972}{\sqrt{(0.11273)(11807324786)}} = 0.9435$$

$$H_0 : \rho = 0$$

$$H_1 : \rho \neq 0$$

$$\alpha = 0.05 \quad df = (n - 2)$$

critical region: $t < -2.052$ and $t > 2.052$

$$t = \frac{0.9435(\sqrt{29-2})}{\sqrt{1-(0.9435)^2}} = 14.79$$

$$P < 0.0001$$

Decision:

Since $t > t_{0.025}(27)$ reject the hypothesis of no linear correlation.

More generally, if X and Y follow the bivariate normal distribution, it can be shown that quantity.

is a random variable that follows approximately the normal distribution with mean. and variance equal to $1/(n-3)$. There the procedure is to compute

$$\begin{aligned} z &= \frac{\sqrt{n-3}}{2} \left[\ln \left(\frac{1+r}{1-r} \right) - \ln \left(\frac{1+\rho_0}{1-\rho_0} \right) \right] \\ &= \frac{\sqrt{n-3}}{2} \ln \left[\frac{(1+r)(1-\rho_0)}{(1-r)(1+\rho_0)} \right] \end{aligned}$$

Example 96

Consider the immediate preceding example data, test the null hypothesis that $\rho = 0.9$ against the alternative that $\rho > 0.9$ at 5% level of significance.

Solution

$$H_0 : \rho = 0.9$$

$$H_1 : \rho > 0.9$$

Critical region: $Z > 1.645$

Decision: Since $Z < Z_{0.05}$ there is no evidence that the correlation coefficient is not equal to 0.9

In ordinary usage of this method, it is not necessary to use the formula for Z that corresponds to r values between 0.0 and 0.99. Tables containing Fisher - Z values Z_f are available. In this case to test $H_0: \rho = \rho_0$ vs $H_1: \rho \neq \rho_0$

$Z =$ we have $Z =$

Critical region is $Z \leq -Z_{\alpha/2}$ and $Z \geq Z_{\alpha/2}$ where Z_f and μ_f are the Fisher - Z values for r and ρ_0 respectively.

Example 97

The following data gave X = the water content of snow on April 1 and Y = the yield from April to July (in inches) on the Snake River watershed in Wyoming for 17 years.

x	y	X	y
23.1	10.5	37.9	22.8
32.8	16.7	30.5	14.1
31.8	18.2	25.1	12.9
32.0	17.0	12.4	8.8
30.4	16.3	35.1	17.4
24.0	10.5	31.5	14.9
39.5	23.1	21.1	10.5
24.2	12.4	27.6	16.1
52.5	24.9		

Estimate the correlation between X and Y .

Example 98

Two methods of measuring cardiac output were compared in 10 experimental animals with the following results

<i>Cardiac Output (l./min)</i>	
<i>Method I</i> x	<i>Method I</i> Y
0.8	0.5
1.0	1.2
1.3	1.1
1.4	1.3
1.5	1.1
1.4	1.8
2.0	1.6
2.4	2.0
2.7	2.4
3.0	2.8

Compute the sample correlation coefficient.

Example 99

A group of eight athletes ran a 400 metres race twice. The times in seconds were recorded as follows for each athlete.

<i>Runner</i>	
<i>1st Trial</i> x	<i>2nd Trial</i> Y
48.4	48.0
51.2	54.3
48.6	49.4
49.5	48.4
51.6	54.0
49.3	47.2
50.8	51.8
49.7	50.3

Calculate the correlation coefficient between these two trials.

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